

Academic Booklet
Academic Year 2025–26

Master of Computer Applications (MCA) / Master of Science in Information Technology (MScIT) / MSc in Data Science Semester-2

Faculty of IT & Computer Science

Parul Institute of Computer Application

Parul University

Our Management Team



Registrar	Provost	Pro Vice Chancellor	Campus Director
			
Prof. Manish Pandya	Dr. K.N. Madhusoodanan	Dr. Kunjal Sinha	Dr. Babita Chaube

FITCS Team

Photo	Faculty Details
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About the University

Parul University, Vadodara, stands today as one of Gujarat's leading private universities - a vibrant hub of higher education, healthcare, research, and innovation. Its journey began with the Parul Arogya Seva Mandal Trust, which first made its mark in the healthcare sector before establishing Gujarat's first self-financed Homeopathic Medical College in Ahmedabad in 1993. This marked the beginning of a legacy committed to quality and holistic learning.

Over time, the vision grew into the Parul Group of Institutions, offering programs in Engineering, Technology, Ayurveda, Nursing, Physiotherapy, Pharmacy, Management, Architecture, and more. The Government of Gujarat established Parul University through the Gujarat Private Universities (Second Amendment) Act, 2015, recognizing its growth and impact.

Starting with 16 institutes and about 50 programs, the University has grown into a multidisciplinary ecosystem of 38 constituent colleges and 21 faculties offering Diploma, UG, PG, and Ph.D. programs, including seven constituent colleges that function as teaching hospitals. Today, it is home to over 65,000 students and 8,000 staff, including 4,500+ international students from 75+ countries, on a 125-acre eco-friendly campus equipped with modern classrooms, laboratories, hostels, sports arenas, and cultural spaces.

Parul University has earned NAAC accreditation with the highest A++ grade (3.55 CGPA) in its very first assessment cycle, UGC Category-1 status with Graded Autonomy, and the prestigious Centre of Excellence status by the State Government. In NIRF 2025, it ranked among India's top 150 universities, 41st in Pharmacy, and within the top 50 for Innovation. It has achieved 5-star ratings in Pharmacy, Engineering, Management, and Applied Sciences, and 4-star ratings in University and Medical categories in GSIRF-2024; along with Diamond Ratings in QS I-GAUGE, with Platinum in Medicine, Engineering, and Pharmacy. It also made a global debut in the Times Higher Education Impact Rankings 2025, securing ranks among India's top 50 for Quality Education (SDG 4), Gender Equality (SDG 5), Good Health & Well-being (SDG 3), and Partnerships for the Goals (SDG 17). The hospitals hold NABH accreditation with Platinum Level Certification for Digital Health Standards, along with the NABL-accredited Molecular Laboratory at Parul Sevashram Hospital.

True to its mission, Parul University Parul University delivers holistic education, fosters innovation, and advances sustainable development. With its achievements, global collaborations, and commitment to quality, it stands as Gujarat's leading private university, setting new benchmarks in higher education.

Vision

To make successful academic quests through entrepreneurship, research, modernization and partnerships, thus making PU the finest educational destination.

Mission

Bridging the gap between academia and career, by paying emphasis on development programs for both students and staff. Promoting healthy relationships between PU's existing students, alumni, teachers and staff. Forming associations with other universities and corporate firms of

the nation and the world. Presenting state of art infrastructure with high quality and energized work ethics.

Quality Policy

To strive towards attaining the status of global educational university by setting higher benchmarks in quality education to deliver excellence in academics, research, innovation and extension activities through the implementation of best practices adopted by renowned academic institutes in teaching and learning processes by continuously monitoring the effectiveness of the University's practices, fostering a quality learning ecosystem through state-of- the-art facilities to enable the beneficiaries to enhance their skillsets and knowledge, with enhanced emphasis on comprehensive development.

About the Institute

Parul Institute of Computer Application (PICA) is not just an educational institution; it's a gateway to a world of innovation and opportunity. With a diverse range of programs including BCA, BSc-IT, IMCA, MCA, and MSc-IT, PICA offers students the chance to explore their passions and carve their path in the dynamic realm of technology. Our curriculum, enriched by state-of-the-art facilities and expert faculty, ensures that every student's journey is one of holistic growth and discovery. Through a blend of theoretical learning and practical application, we empower our graduates to excel in their chosen field, armed with the skills, knowledge, and confidence to make a meaningful impact

Vision

To make students contribute in the area of Computer Applications by developing innovative Computing professionals to achieve excellent standards of quality education by keeping Pace with rapidly changing technologies, with a promising future to bring a new level of Inspiration among the students.

Mission

To generate competent future IT leaders by providing excellent technology-driven academic, research and diverse cultural exposure.

Quality

We are committed to implementing the principles of the National Education Policy to enhance the quality and relevance of education. We pledge to provide a nurturing environment that fosters intellectual curiosity, critical thinking, and lifelong learning. Through continuous improvement and innovation in teaching methodologies and curriculum design, we strive to meet the evolving needs of our students and society.

Code of Conduct for Students of Parul University

1. All students of Parul University shall compulsorily display their University ID cards by wearing them around their neck. If any student is found without an ID card on any day, he/she will be marked absent for that day.
2. The University expects all students to behave in a manner befitting a prudent person.
3. Students shall be dressed in a presentable manner which does not invite criticism from any quarter.
4. Students shall strictly adhere to class timings and be punctual in attending all classes.
5. Students shall display cordial, genial, and respectful behaviour towards their teachers.
6. Students should be polite, cooperative, and respectful in dealing with the employees of the University.
7. Students shall maintain the highest order of cleanliness in the classroom as well as in the college premises.
8. Students should not indulge in boisterous behaviour at any place on the university campus.
9. Students shall follow the directions issued in accessing common places such as the library, canteen, sports fields, auditorium, gymnasium, swimming pool, etc.
10. Students shall strictly follow the schedules given by the class teacher regarding assignments, class tests, examinations, and practicals, and shall complete the assigned work within the duration specified by their teachers.
11. Students shall follow the instructions given by the teacher during practical sessions in relation to the use of laboratories, workshops, implements, and equipment.
12. Whenever students have queries regarding their performance, either from the class teacher or from any office in the College/University, they should follow the laid-down procedures and approach the concerned authority with utmost respect.
13. Students shall pay all prescribed fees at the stipulated times and avoid being penalized for non-payment.
14. Students shall not indulge in unfair means during the conduct of class tests or internal and external examinations.
15. Students shall not indulge in unlawful assembly at any place on the campus.
16. Any problem encountered by students should be brought to the notice of the authorities immediately available in the College/University.
17. Students should never take law into their own hands and must report any matter of lawlessness or harassment to the College Authorities immediately, who will initiate suitable action.
18. Students shall participate in all national events such as Independence Day and Republic Day organized by the University.

19. Students should not indulge in any activities that adversely affect the reputation of the University.
20. Students shall not consume prohibited substances such as alcohol, narcotics, marijuana, heroin, cocaine, etc., and shall not keep in their custody or hostel premises any illegal objects or materials such as firearms, missiles, bombs, narcotics, alcohol, or other intoxicants.
21. Smoking and chewing of tobacco are strictly prohibited on campus.
22. The UGC has directed all universities to strictly implement anti-ragging measures. It is the responsibility of the institutions within the University to ensure the safety of newcomers and protect them from any incidents that may harm their physical or mental well-being. Any student found involved in ragging will face strict disciplinary action as per the *UGC Regulations on Curbing the Menace of Ragging in Higher Educational Institutions, 2009*.
23. Any violation of the provisions mentioned above will be viewed as an act of misconduct. The University, after conducting a thorough probe into such incidents, shall initiate strict disciplinary action against the delinquent students.

Code of Conduct for Department

As we embark on this journey together, it is essential that we maintain a positive and respectful learning environment for everyone. To ensure a harmonious and productive academic community, we have established a Code of Conduct that outlines the expected behavior of all FITCS Students.

1. Wearing an I-Card is Mandatory.
2. Treat fellow students, faculty, and staff with respect and courtesy.
3. Do not engage in any type of misconduct or academic cheating.
4. Attend classes regularly and on time; 75% attendance is Mandatory, else the student will be detained and must repeat the whole Semester again.
5. Fill in the leave form (Available with Office Assistants) before taking leave and get it signed by your MFT/Mentor and submit the duly signed leave form to the Office Assistant for seeking Attendance.
6. Note: only 5 days leave will be sanctioned per Semester.
7. If a student is caught using Mobile during Lab or Lecture Hours for the first time, the student's mobile will be taken in custody for 3 days with an apology.
8. For second offense of using mobile during Lab or Lecture Hours, the mobile will be taken in custody for 6 days and Parents will be called to meet the Dean/Principal/HOD.
9. If found shooting or sharing any unofficial video/photo, student shall be suspended for 1 week, phone in custody for 15 days, and Parents will be called.
10. Cheating in any Exam will lead to disqualification from the remaining Subject papers.

For questions or concerns regarding the code of conduct, contact HOD (BCA).

Code of Discipline

- **Academic Integrity:** Be honest in all academic work; avoid plagiarism and cheating.
- **Respectful Behavior:** Treat everyone with respect; no discrimination, harassment, or bullying.
- **Attendance:** Attend classes regularly and participate actively.
- **Professionalism:** Exhibit professionalism in behavior, attire, and communication.
- **Resource Use:** Use university resources responsibly.
- **Community Engagement:** Participate in community service.
- **Respect:** Treat everyone with respect and dignity.
- **Compliance:** Follow all laws and university policies.
- **Integrity:** Maintain honesty in all activities.
- **Responsibility:** Be accountable for one's actions.

Unaccepted Behaviors

- **Dishonesty:** Plagiarism, cheating, or any form of academic fraud.
- **Disrespect:** Discrimination, harassment, bullying, or any form of disrespect.
- **Absenteeism:** Frequent unexcused absences.
- **Unprofessionalism:** Inappropriate attire, unpunctuality, irresponsible communication.
- **Resource Misuse:** Unauthorized or unethical use of university resources.
- **Disruption:** Actions that disrupt educational environment.
- **Substance Abuse:** Use or possession of illegal drugs or alcohol on campus.
- **Violence:** Any form of physical violence or threats.
- **Theft:** Stealing or damaging university or personal property.
- **Cyber Misconduct:** Unauthorized access or misuse of digital resources.
- **Insubordination:** Defying university authority.
- **Unauthorized Gatherings:** Unapproved gatherings or protests.
- **Unethical Behavior:** Actions compromising university ethics.

Disciplinary Measures

- **Warnings:** Verbal or written warnings for minor infractions.
- **Probation:** Specific conditions for improvement.
- **Detention from Academics:** Temporary restriction from attending.
- **Suspension:** Temporary suspension from university services.
- **Expulsion:** Permanent removal for serious violations.

About the Programme

Details about NEP-2020

The National Education Policy 2020 (NEP-2020) heralds a transformative vision for the Indian education landscape, addressing the evolving needs of the 21st century. This comprehensive framework emphasizes a holistic and multidisciplinary approach to education, integrating science, arts, humanities, and sports to nurture creativity and critical thinking. NEP-2020 advocates for flexibility in course choices, enabling students to pursue subjects across streams and fostering an interdisciplinary learning environment. It places a strong emphasis on early childhood care and education (ECCE), introducing a new structural framework that encompasses the foundational, preparatory, middle, and secondary stages of learning. The policy also prioritizes teacher training and professional development, assessment reforms, and the integration of technology in education to enhance access and quality. NEP-2020 aims to restructure higher education, with a focus on increasing enrollment, promoting research and innovation, and granting autonomy to educational institutions. Furthermore, it underscores the importance of equity and inclusion, striving to bridge the gap in access to education and promote the preservation and promotion of Indian languages. Overall, NEP-2020 represents a paradigm shift towards a more inclusive, flexible, and forward-thinking education system that is aligned with the aspirations of a rapidly changing world.

Key Features of NEP 2020

- Flexibility in Learning
- Early Childhood Care and Education (ECCE)
- Teacher Training and Professional Development
- Assessment Reforms
- Higher Education Restructuring
- Technology Integration
- Equity and Inclusion
- Autonomy to Educational Institutions
- Promotion of Indian Language

Program Educational Objectives (PEOs)

- The statements below indicate the career and professional achievements that the **MCA/MSc IT/MSc Data Science curriculum** enables graduates to attain.

PEO Number	Program Educational Objective Description
PEO1	To prepare graduates who will be effective professionals in identifying, analyzing and developing ICT applications for various sectors of society with their proficient technical skills.
PEO2	To prepare socially and ethically responsible graduates who can work in a team or as an individual with proven abilities and expertise.
PEO3	To prepare graduates with life-long learning attitude and strong communication skills in tandem with analytical and conceptual expertise.

Program Outcomes (PLOs)

- Program outcomes are statements conveying the intent of a program of study.

PLO Number	Program Outcome Description
PLO1	**Computational Knowledge** – Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements. Utilize the fundamental knowledge of computer, information, and technology for fulfilling social and industrial needs.
PLO2	**Problem Analysis & Design/Development of Solutions** – Identify, formulate, research literature and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines. Design and evaluate solutions for complex computing problems and systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. Compare, classify, and analyze the data for creating and developing different algorithms.
PLO3	**Project Management and Finance** – Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to man- age projects and multidisciplinary environments.
PLO4	**Communication Efficacy** – Communicate effectively with the computing community and with society at large about computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
PLO5	**Professional Ethics** – Implement and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practice.
PLO6	**Innovation and Entrepreneurship** – Identify a timely opportunity and use innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large. Enable students to become entrepreneurs by integrating innovation in science and technology. Cultivate the skills for entrepreneurship by integrating innovation in science and technology and become lifelong learners.

Program Specific Learning Outcomes (PSOs)

- Program Specific Learning Outcomes

PSO Number	Program Specific Outcome Description
PSO1	Demonstrate proficiency in designing, developing, testing, and deploying software applications using various programming languages, development frameworks, and software engineering methodologies.
PSO2	Successfully complete a series of project-based assignments that integrate knowledge, skills, and competencies acquired throughout the program to solve real-world IT challenges or address specific industry needs.

Academic Calender

PARUL UNIVERSITY
FACULTY OF IT & COMPUTER SCIENCE
MCA, M.Sc. IT & MSc-Data Science
Academic Calendar 2025-26 (EVEN)

Mon	Tue	Wed	Thu	Fri	Sat	Sun	Week
19-Jan	20	21	22	23	24	25	1
26	27	28	29	30	31	01-Feb	2
2	3	4	5	6	7	8	3
9	10	11	12	13	14	15	4
16	17	18	19	20	21	22	5
23	24	25	26	27	28	01-Mar	6
2	3	4	5	6	7	8	7
9	10	11	12	13	14	15	8
16	17	18	19	20	21	22	9
23	24	25	26	27	28	29	10
30	31	01-Apr	2	3	4	5	11
6	7	8	9	10	11	12	12
13	14	15	16	17	18	19	13
20	21	22	23	24	25	26	14
27	28	29	30	01-May	2	3	15
4	5	6	7	8	9		16

Semester	Commencement Date
2	19-01-2026

Holidays

26-Jan - Republic Day
16-02-2026 - Maha Shivratri
04-03-2026 - Dhuleti
05-04-2026 - Good Friday
14-04-2026 - Dr. Ambedkar Jayanti

Orientation Programme - 19-01-2026
Mid Term Exam
Term End
Remedial Examination
Term Work Submission
Parent Teacher Meet : 23-March-2024

Events / Activities	Planned Month
AppliFest	Feb/March-2026
TechFuze	Feb-2026
Industrial Visit	Feb-2026
Expert Sessions Week	March-2026
International Conference	29th March, 2026
FDP/STTP	May/June-2026
CT - CLASS TEST	

Exam	Start	End	Result
Mid Term Exam	16-03-2026	27-03-2026	30-03-2026
Remedial Exam	30-03-2026	07-03-2026	07-03-2026
End Semester Exam (Theory)	18-05-2026	30-05-2026	-
End Semester Exam (Practical)	11-05-2026	11-05-2026	-
Supplementary Exam (Theory)	01-06-2026	26-06-2026	-
Supplementary Exam (Practical)	18-05-2026	30-05-2026	-
Assignment Submission (Current Sem)	14-04-2026	20-04-2026	-
Assignment Submission (Previous Sem)	23-02-2026	23-02-2026	-

Expert Talks / Seminars / Workshops on

1) Advanced Java Programming - 04-03-2026
2) Data Communication and Networking - 05-03-2026
3) Artificial Intelligence - 06-03-2026
4) Big Data Analytics - 06-03-2026
5) Cloud Computing - 06-03-2026
6) Cyber Security & Forensics - 06-03-2026
7) Full Stack Web Development - 06-03-2026
8) Computer Graphics - 07-03-2026

Prepared By	Approved By
Head of Department	Director & Dean-FITCS

Important date of the academic calendar

Activities	**Even Semester (2025–26)**
Start of the Semester	19/01/2026
Class Test (from)	16/02/2026
Assignment Submission (Previous Semester)	23/02/2026
Remedial Mid-Term Exam of Previous Semester (from)	02/03/2026
Result of Remedial Mid-Term Exam of Previous Semester	07/03/2026
Mid-Term Exam (from)	16/03/2026
Result of Mid-Term Exam	30/03/2026
Assignment Submission (Current Semester)	20/04/2026
Last date for submission of marks of Continuous Evaluation / Internal Evaluation	02/05/2026
End of the Semester	09/05/2026
End Semester Examination (Practical)	11/05/2026 to 16/05/2026
End Semester Examination (Theory) and Supplementary Examination (Practical)	18/05/2026 to 30/05/2026
Supplementary Exam Previous Semester (Theory)	01/06/2026 to 06/06/2026

List of Holidays

Semester -2 (MCA/MSc IT/MSc Data Science)

Date	**Holiday**
14/01/2026 to 16/01/2026	Uttrayan
26/01/2026	Republic Day
04/03/2026	Holi-Dhuleti
26/03/2026	Shree Ram Navami
14/04/2026	Dr. BabaSahab Ambedkar Jayanti
Every 1st & 3rd Saturday	Regular Monthly Holiday

Class Test Schedule

Date	**MCA/ MSc IT**	**MSc in Data Science**
16/02/2026	Class Test 1 (Advanced Java Programming)	Class Test 1 (Data Analysis Statistical Modelling)
23/02/2026	Class Test 1 (Data Comm. & N/w (MCA), Comp. N/w & N/w Security (MScIT))	Class Test 1 (Data Mining and Machine Learning)
02/03/2026	Class Test 1 (Specialization-I)	Class Test 1(Database Management Systems)
09/03/2026	Class Test 1 (Graphics programming)	Class Test 1(Python and R Programming)
01/04/2026	Class Test 1 (S/w Engg. with Agile)	Class Test 1(Elective-I)
CDC Dept Subject	Class Test 1 (English and Mathematical Aptitude)	

Mid Term Exam Schedule

Exam	Start Date	End Date	Result Date
Remedial Exam of Previous Sem	02/03/2026	07/03/2026	07/03/2026
Mid-Term Exam (Current Sem)	16/03/2026	27/03/2026	30/03/2026

Term End Exam Schedule

Exam Type	Start Date	End Date
End Semester Exam (Practical)	11/05/2026	16/05/2026
End Semester Exam (Theory) & Supplementary (Practical)	18/05/2026	30/05/2026
Supplementary (Theory) & Exam Previous Semester	01/06/2026	06/06/2026
Assignment Submission (Current Sem)	14/04/2026	20/04/2026
Assignment Submission (Previous Sem)	23/02/2026	-

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MCA/MSc IT **DIVISION:** A

Effective from: 19-01-2026

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	DCN - Prof. Satyamani Gayathri Devi Classroom - 301	Comprehnsive Assesment	Comprehnsive Assesment	AJP- Prof. Prof. Honey P Classroom - 301	DCN - Prof. Satyamani Gayathri Devi , NF 1 LAB - 601,602	
08:25 AM to 09:20 AM	AJP- Prof. Honey P Classroom - 301	AJP- Prof. Prof. Honey P Classroom - 301	AJP- Prof. Prof. Honey P Classroom - 301	EMA Dr. Bindi Bhatt Classroom 301		
09:20 AM to 09:30 AM Short Break						
09:30 AM to 10:25 AM	Specialization - AI Prof. Vijya Tulsani,Dr Praveen Tomar (Lab no :612,613)	Computer Graphics - Prof. Megha T. New Classroom - 208	PULMS Self Study	Library	AJP Lab- Prof. Honey P., Prof. Rinku P Lab no 613,712	Specialization AI Prof. Vijya Tulsani Classroom 607
10:25AM to 11:20AM		Specialization - AI Prof. Vijya Tulsani New Classroom 208	Specialization - AI Prof. Vijya Tulsani Classroom - 605	Specialization - AI Prof. Vijya Tulsani Classroom 303		Comprehnsive Assesment
11:20 AM to 12:20 AM Lunch Break						
12:20PM to 01:15 PM	PULMS Self Study	Library	AJP Lab- Prof. Honey Prof Hardik P Lab no 605,606	Seminar	Computer Graphics Prof Preeti Sharma,Prof Ashutosh S Lab 609,610	PULMS Self Study
01:15PM to 02:10 PM	Computer Graphics - Prof Megha T Classroom - 304	Computer Graphics - Prof. Megha T NEW Classroom - 207				

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MCA/MSc IT **DIVISION:** B

Effective from: 19-01-2026

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	AJP- Prof. Honey P Classroom - 302	AJP- Prof. Honey P Classroom - 302	AJP- Prof. Honey P Classroom - 302	DCN - Prof. Satyamani Gayathri Devi, NF 1 LAB - 603,604	AJP- Prof. Honey P Classroom - 302	
08:25 AM to 09:20 AM	DCN - Prof. Satyamani Gayathri Devi Classroom - 302	Seminar	Seminar		MCA Sem 2 EMA Classroom 302 Dr. Bindi Bhatt	
09:20 AM to 09:30 AM Short Break						
09:30 AM to 10:25 AM	Specialization - AI Prof. Vijya Tulsani,Dr Praveen Tomar (Lab no :612,613)	Computer Graphics - Prof. Megha T. New Classroom - 208	PULMS Self Study	Library	AJP Lab- Prof. Honey P., Prof. Rinku P Lab no 613,712	Specialization AI Prof. Vijya Tulsani Classroom 607
10:25AM to 11:20AM			Specialization - AI Prof. Vijya Tulsani Classroom - 605		Library	Comprehnsive Assesment
11:20 AM to 12:20 AM Lunch Break						
12:20PM to 01:15 PM	PULMS Self Study	Library	AJP Lab- Prof. Honey Prof Hardik P Lab no 605,606	Seminar	Computer Graphics Prof Preeti Sharma,Prof Ashutosh S Lab 609,610	PULMS Self Study
01:15PM to 02:10 PM	Computer Graphics - Prof Megha T Classroom - 304		Specialization AI Prof. Vijya Tulsani New Classroom - 307	Comprehensive Assesment		

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MCA/MSc IT **DIVISION:** C

Effective from: 19-01-2026

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	EMA Dr. Bindi Bhatt Classroom 305	Computer Graphics - Prof. Preeti Sharma Lab- 601,602	Computer Graphics - Prof. Preeti Sharma Classroom - 303	Comprehensive Assesment	DCN - Prof Aniket P,NF 1 Lab No 610,611	
08:25 AM to 09:20 AM	Computer Graphics - Prof. Preeti Sharma Classroom - 305		DCN - Prof. Aniket P Classroom - 303	Computer Graphics - Prof. Preeti Sharma Classroom - 303		
09:20 AM to 09:30 AM Short Break						
09:30 AM to 10:25 AM	Specialization - AI Prof. Vijya Tulsani,Dr Praveen Tomar (Lab no :612,613)	Computer Graphics - Prof. Megha T. New Classroom - 208	PULMS Self Study	Library	AJP Lab- Prof. Honey P., Prof. Rinku P Lab no 613,712	Specialization AI Prof. Vijya Tulsani Classroom 607
10:25AM to 11:20AM				Specialization - AI Prof. Vijya Tulsani Classroom 303	SE using Agile [M Sc [IT]] Prof Satyamani Gayathri Classroom No 602 Library [MCA]	Comprehnsive Assesment
11:20 AM to 12:20 AM Lunch Break						
12:20PM to 01:15 PM	PULMS Self Study	Library	AJP Lab- Prof. Honey Prof Hardik P Lab no 605,606	Seminar	Computer Graphics Prof Preeti Sharma,Prof Ashutosh S Lab 609,610	PULMS Self Study
12:20PM to 01:15 PM	Computer Graphics - Prof Megha T Classroom - 304	Computer Graphics - Prof. Megha T NEW Classroom - 207	Specialization AI Prof. Mehul Dalwadi New Classroom - 208		Specialization AI Prof. Mehul Dalwadi Classroom 605	PULMS Self Study

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MCA/MSc IT **DIVISION:** D

Effective from: 19-01-2026

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	Computer Graphics - Prof. Preeti Sharma Classroom - 307	EMA Dr. Bindi Bhatt Classroom 306	SEMINAR	Computer Graphics - Prof. Preeti Sharma Classroom - 307	AJP Lab- Prof Chandan Kumar Singh Lab no 612,613	
08:25 AM to 09:20 AM	DCN - Prof. Aniket Paul Classroom - 307	Library	Computer Graphics - Prof. Preeti Sharma Classroom - 307	Comprehensive Assesment		
09:20 AM to 09:30 AM Short Break						
09:30 AM to 10:25 AM	Specialization - AI Prof. Vijya Tulsani,Dr Praveen Tomar (Lab no :612,613)	Computer Graphics - Prof. Megha T. New Classroom - 208	PULMS Self Study	Library	AJP Lab- Prof. Honey P., Prof. Rinku P Lab no 613,712	Specialization AI Prof. Vijya Tulsani Classroom 607
10:25AM to 11:20AM				Specialization - AI Prof. Vijya Tulsani Classroom 303	AJP- Prof. Chandan Kumar Singh Classroom - 605	
11:20 AM to 12:20 AM Lunch Break						
12:20PM to 01:15 PM	PULMS Self Study	Library	AJP Lab- Prof. Honey Prof Hardik P Lab no 605,606	Seminar	Computer Graphics Prof Preeti Sharma,Prof Ashutosh S Lab 609,610	PULMS Self Study
01:15PM to 02:10 PM	Computer Graphics - Prof Megha T Classroom - 304	Computer Graphics - Prof. Megha T NEW Classroom - 207	AJP- Prof. Chandan Kumar Singh New Classroom - 308	PULMS Self Study	CSF Mr Shreyash Lab - 607 608	AJP- Prof. Chandan Kumar Singh ClassRoom - 601

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MCA/MSc IT **DIVISION:** E

Effective from: 19-01-2025

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	DCN - Prof Anktit Paul Classroom - 304	CSF Prof. Lakshman Classroom 305	CSF Prof. Lakshman Classroom 304	Comprehensive Assesment	Computer Graphics LAB - Prof. Preeti Sharma 601,602	
08:25 AM to 09:20 AM	Comprehensive Assesment	EMA Dr. Bindi Bhatt Classroom - 305	CSF Prof. Lakshman Class - 304			
09:20 AM to 09:30 AM Short Break						
09:30 AM to 10:25 AM	Specialization - AI Prof. Vijya Tulsani,Dr Praveen Tomar (Lab no :612,613)	Computer Graphics - Prof. Megha T. New Classroom - 208	PULMS Self Study	Library	AJP Lab- Prof. Honey P., Prof. Rinku P Lab no 613,712	Specialization AI Prof. Vijya Tulsani Classroom 607
10:25AM to 11:20AM	AJP - Prof. Jigar B Classroom no 602	Specialization - AI Prof. Vijya Tulsani New Classroom 208		Specialization - AI Prof. Vijya Tulsani Classroom 303	SE using Agile [M Sc [IT]] Prof Satyamani Gayathri Classroom No 602 Library [MCA]	Comprehnsive Assesment
11:20 AM to 12:20 AM Lunch Break						
12:20PM to 01:15 PM	PULMS Self Study	Library	AJP Lab- Prof. Honey Prof Hardik P Lab no 605,606	SEMINAR	Computer Graphics Prof Preeti Sharma,Prof Ashutosh S Lab 609,610	PULMS Self Study
01:15PM to 02:10 PM			SEMINAR			AJP - Prof. JB Classroom no 606

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MCA/MSc IT **DIVISION:** F

Effective from: 19-01-2026

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	Comprehensive Assesment	Seminar	FSW Prof. Shivani class - 308	AJP Lab- Prof. Niharika A NF 1 Lab no 605,606	AJP Lab- Prof. Niharika A,NF 1 Lab no 704,705	
08:25 AM to 09:20 AM	DCN - Prof Lakshman Prasad Latua Classroom - 308	AJP - Prof. Niharika A Classroom no 308	EMA Dr. Bindi Bhatt Classroom 308			
09:20 AM to 09:30 AM **Short Break**						
09:30 AM to 10:25 AM	Specialization - AI Prof. Vijya Tulsani,Dr Praveen Tomar (Lab no :612,613)	Computer Graphics - Prof. Megha T. New Classroom - 208	PULMS Self Study	Library	AJP Lab- Prof. Honey P., Prof. Rinku P Lab no 613,712	Specialization AI Prof. Vijya Tulsani Classroom 607
10:25AM to 11:20AM		Specialization - AI Prof. Vijya Tulsani New Classroom 208	Specialization - AI Prof. Vijya Tulsani Classroom - 605		SE using Agile [M Sc [IT]] Prof Faruk Abdulla Classroom 606 Library [MCA]	**Comprehnsive Assesment**
11:20 AM to 12:20 PM **Lunch Break**						
12:20PM to 01:15 PM	Computer Graphics - Prof Megha T Classroom - 304	Library	AJP Lab- Prof. Honey Prof Hardik P Lab no 605,606	Seminar	Computer Graphics Prof Preeti Sharma,Prof Ashutosh S Lab 609,610	**PULMS Self Study**
01:15PM to 02:10 PM	Power BI LAB Dr. Sujit			PULMS SELF STUDY	Comprehensive Assesment	**SE using Agile Prof Faruk Abdulla Classroom No 602**

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MCA/MSc IT **DIVISION:** G

Effective from: 19-01-2025

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	PULMS SELF STUDY	Computer Graphics Prof Ashutosh S Lab No 608,609	EMA Classroom 605	DCN - Prof. Aniket Paul Lab No - 608,609	SEMINAR	
08:25 AM to 09:20 AM	Computer Graphics Prof Ashutosh S Classroom - 606		Computer Graphics Prof Ashutosh S Classroom 605		Computer Graphics Prof Ashutosh S Classroom 605	
09:20 AM to 09:30 AM Short Break						
09:30 AM to 10:25 AM	Specialization - AI Prof. Vijya Tulsani,Dr Praveen Tomar (Lab no :612,613)	Computer Graphics - Prof. Megha T. New Classroom - 208	PULMS Self Study	Library	AJP Lab- Prof. Honey P., Prof. Rinku P Lab no 613,712	Specialization AI Prof. Vijya Tulsani Classroom 607
10:25AM to 11:20AM		Specialization - AI Prof. Vijya Tulsani New Classroom 208			SE using Agile [M Sc [IT]] Prof Faruk Abdulla Classroom 606 Library [MCA]	Comprehnsive Assesment
11:20 AM to 12:20 AM Lunch Break						
12:20PM to 01:15 PM	PULMS Self Study	Library	AJP Lab- Prof. Honey Prof Hardik P Lab no 605,606	SEMINAR	Computer Graphics Prof Preeti Sharma,Prof Ashutosh S Lab 609,610	PULMS Self Study
01:15PM to 02:10 PM	Computer Graphics - Prof Megha T Classroom - 304		SEMINAR	AJP Prof Niharika A New Classroom 208	Specialisation BDA [Lab Batch 1] Prof Vipul G BDA [Lab Batch 2] NF 1 708,709	SE using Agile Prof Faruk Abdulla Classroom No 602 Library [MCA]

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MCA/MSc IT **DIVISION:** H

Effective from: 19-01-2026

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	Computer Graphics Prof Ashutosh S Classroom 607	SEMINAR	Computer Graphics Prof Ashutosh S Classroom 607	Computer Graphics Prof Ashutosh S Lab No 608,609	Computer Graphics Prof Ashutosh S Classroom 604	
08:25 AM to 09:20 AM	Dr Bindi Bhatt EMA Classroom 607	DCN - Prof. Lakshman Classroom 304	AJP Prof Jigar B Classroom 607		AJP Prof Jigar B Classroom 604	
09:20 AM to 09:30 AM **Short Break**						
09:30 AM to 10:25 AM	CC Dr. Ghanasyam Rathod Lab No 607 + GT Prof Ramdeepak Lab no 602	CC Dr. Ghanasyam Rathod Classroom No 605 + GT Prof Ramdeepak Lab no 712	DCN - Prof Lakshman Latua, Prof Aniket Paul Lab- 706,712	SE using Agile [M Sc[IT]] Prof Faruk Abdulla Lab No 706,712 Library [MCA]	CC Dr. Ghanasyam Rathod Classroom No 604	CC Dr. Ghanasyam Rathod Classroom 606
10:25AM to 11:20AM	Comprehensive Assesment	Spec - GT Prof Ramdeepak Lab no 712 + Spec - CC PULMS SELF STUDY		SE using Agile [M Sc[IT]] Prof Faruk Abdulla Lab No 706,712 GT - Prof Ramdeepak Lab no 605 [MCA]	SE using Agile [M Sc [IT]] Prof Faruk Abdulla Classroom 606 Library [MCA]	**AJP Prof Jigar B Classroom 606**
11:20AM to 12:20 PM Lunch Break						
12:20PM to 01:15 PM	Comprehensive Assesment	AJP Lab- Prof. Jigar B, Prof Anmol Singh Lab No 709,710	AJP Lab- Prof. Jigar Bhawsar Prof Anmol Singh Lab No 712	SEMINAR	Specialisation CC [Batch 1] Dr. Ghanshyam Rathod CC [Batch 2] Prof Mini B 602,703 + GT - Prof Ramdeepak Lab No 604	**Comprehensive Assesment**
01:15PM to 02:10 PM	SE using Agile Prof Faruk Abdulla Class No 606			AJP Prof Jigar B ClassNo 604		**SE using Agile Prof Faruk Abdulla Classroom No 602**

Time Table

PARUL UNIVERSITY- TIME TABLE

INSTITUTE NAME: PARUL INSTITUTE OF COMPUTER APPLICATION
ACADEMIC YEAR: 2025-26 **YEAR:** 1st YEAR
SEMESTER: 2nd **LEVEL:** PG
PROGRAM NAME: MSc Data Science **DIVISION:** I

Effective from: 19-01-2026

Lecture Timing	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
07:30 AM to 08:25 AM	Python and R Programming Shivaniba D. Bhadoriya Lab no 713	Python and R Programming Shivaniba D. Bhadoriya Lab no 713	PULMS SELF STUDY	Python and R Programming Shivaniba D. Bhadoriya Lab no 713	Python and R Programming hivaniba D. Bhadoriya, Lab no 713 hivaniba D. Bhadoriya, Lab no 713	
08:25 AM to 09:20 AM	Elective - II Data Analytics and Visualization Dr Ramchandran Lab no 713	Elective - II Data Analytics and Visualization Dr Ramchandran Lab no 713	Elective - II Data Analytics and Visualization Dr Ramchandran Lab no 7137	Library		
09:20 AM to 09:30 AM **Short Break**						
09:30 AM to 10:25 AM	Industry Deployable Courses (IDC) Power BI for Data Analytics Prof Jatin Lab no 713	Data Analysis and Statistical Modelling Prof Anmol Singh Lab no 713	Data Analysis and Statistical Modelling Prof Anmol Singh Lab no 713	SE using Agile [M Sc[IT]] Prof Faruk Abdulla Lab No 706,712 Library [MCA]	LIBRARY	Data Mining and Machine Learning Prof Madhav K Lab no 709
10:25AM to 11:20AM		Database Management Systems Prof Rinkal Sarvaiya Lab no 713	Database Management Systems Prof Rinkal Sarvaiya Lab no 713	Data Mining and Machine Learning Prof Madhav K Lab no 713		Data Mining and Machine Learning Prof Madhav K Lab no 709
11:20AM to 12:20 PM Lunch Break						
12:20PM to 01:15 PM	PULMS SELF STUDY	Database Management Systems Prof Rinkal Sarvaiya Lab no 713	Database Management Systems Prof Rinkal Sarvaiya Lab no 713	PULMS SELF STUDY	Industry Deployable Courses (IDC) Power BI for Data Analytics Prof Jatin Lab no 713	Python and R Programming Shivaniba D. Bhadoriya, Lab no 713
12:20PM to 01:15 PM	PULMS SELF STUDY			Database Management Systems Prof Rinkal Sarvaiya Classroom 605		

Subject Wise Faculty Details

Concerned Faculty List with Contact Details

Department Concerned with the Subject	Subject	List of Faculty Mambers	Subject Coordinator Faculty
MCA/MScIT	Advance java programming	1. Prof. Jigar Bhawsar Mob: 9229891004 Email: jigar.bhawsar24744@paruluniversity.ac.in 2. Prof. Rinku Patil [P] Mob: 9423284752 Email: rinku.patil31670@paruluniversity.ac.in 3. Prof. Hardik Parmar Mob: 9586607479 Email: hardik.parmar26611@paruluniversity.ac.in 4. Prof. Honey Parmar Mob: 9157176781 Email: honey.parmar36498@paruluniversity.ac.in 5. Prof. Chandan Kumar Mob: 9525141407 Email: chandan.singh33017@paruluniversity.ac.in 6. Prof. Niharika Mob: 9662833439 Email: niharika.agarwal36122@paruluniversity.ac.in	Prof. Honey Parmar
MCA/MScIT	Data Communications and Networking / COMPUTER NETWORKS AND NETWORK SECURITY	1. Prof. Rayavarapu Satya Mani Gayathri Devi Mob: 9392314985 Email: satyamani.rayavarapu41098@paruluniversity.ac.in 2. Daniel Kwesi Acquah-Amoah [P] Mob: 9879401859 Email: daniel.amoah36114@paruluniversity.ac.in 3. Prof. Aniket Paul Mob: 8140880956 Email: aniket.paul36074@paruluniversity.ac.in 4. Lakshman Prasad Latua Mob: 90021 67389 Email: lakshman.latua40573@paruluniversity.ac.in	Prof. Rayavarapu Satya Mani Gayathri Devi
MCA/MScIT	Comprehensive Assessment	1. Prof. Smarika Rai Mob: 9023483944	

		Email: Smarika.rai30842@paruluniversity.ac.in	Prof. Smarika Rai
MCA/MScIT	Seminar	1. Prof. Ahutosh Solanki Mob: 9157278205 Email: ashutosh.solanki43715@paruluniversity.ac.in	Prof. Ahutosh Solanki
MScIT	Software Engineering with Agile	1. Prof. Faruk Abdulla Mob: 9810470397 Email: faruk.abdulla30274@paruluniversity.ac.in 2. Prof Satyamani Gayathri Mob: 9392314985 Email: satyamani.rayavarapu41098@paruluniversity.ac.in	Prof. Faruk Abdulla
MCA/MScIT	Computer Graphics	3. Prof. Preeti Sharma Mob: 8218864370 Email: preeti.sharma32496@paruluniversity.ac.in 4. Dr. Hina Chokshi Mob: 9227873387 Email: hina.chokshi42058@paruluniversity.ac.in 5. Dr. Ghanshyam Rathod Mob: 6357979555 Email: ghanshyam.rathod36135@paruluniversity.ac.in 6. Ashutosh Solanki Mob: 9157278205 Email: ashutosh.solanki43715@paruluniversity.ac.in 7. Prof. Megha Triwedi Mob: 9106805356 Email: megha.trivedi24314@paruluniversity.ac.in	Prof. Preeti Sharma
MCA/MScIT	Artificial Intelligence - I	1. Prof. Vijya Tulsani Mob: 8866665173 Email: vijya.tulsani42087@paruluniversity.ac.in 2. Prof. Mehulkumar Dalwadi Mob: 8000131149 Email: mehulkumar.dalwadi33536@paruluniversity.ac.in	Prof. Vijya Tulsani
MCA/MScIT	Big Data Analytics	1. Prof. Vipul Gamit Mob: 8200012226 Email: vipulbhai.gamit26053@paruluniversity.ac.in	Prof. Vipul Virsinghbhai Gamit
MCA/MScIT	Cloud Computing	1. Dr. Ghanshyam Rathod Mob: 6357979555 Email: ghanshyam.rathod36135@paruluniversity.ac.in 2. Prof. Mini Bhola [P] Mob: 7203083488	Dr. Ghanshyam Rathod

		Email: miniben.bhola40198@paruluniversity.ac.in	
MCA/MScIT	Cyber Security & Forensic	1. Prof. Lakshman Prasad Latua Mob: 9002167389 Email: lakshman.latua40573@paruluniversity.ac.in 2. Mr. Shreyash	Prof. Lakshman Latua
MCA/MScIT	Full Stack Web Developer	1. Prof. Shivaniba D. Bhadoriya Mob: 9510332044 Email: shivaniba.bhadoriya36466@paruluniversity.ac.in 2. Sathwik [P] Mob: 8008469584 Email: sathwik.chebrolu40366@paruluniversity.ac.in	Prof. Shivaniba D. Bhadoriya
MCA/MScIT	Gaming Technology	1. Prof. Y Rama Deepak Mob: 9494075286 Email: rama.yadagiri35463@paruluniversity.ac.in	Prof. Y Rama Deepak
MScIT	DATA SCIENCE WITH PYTHON	1. Prof. Payal Parekh Mob: 8866573415 Email: payal.parekh40362@paruluniversity.ac.in	Not Given
MScIT	Web Application Development using ASP.NET	1. Prof. Anmol Singh Mob: 6306385299 Email: anmol.singh40298@paruluniversity.ac.in 2. Prof. Khyati Kariya Mob: 9898094186 Email: khyati.kariya33400@paruluniversity.ac.in 3. Prof. Niharika Agarwal Mob: 9662833439 Email: niharika.agarwal36122@paruluniversity.ac.in 4. Prof. Honey Parmar Mob: 9157176781 Email: honey.parmar36498@paruluniversity.ac.in	

Department/ Branch	Department Concerned with the Subject	Subject	List of Faculty Mambers	Subject Coordinator Faculty
MSC Data Science		Data Analysis & Statistical Modelling	1. Prof. Anmol Singh Mob: 6306385299 Email: anmol.singh40298@paruluniversity.ac.in	Prof. Anmol Singh
		Data Mining and Machine Learning	1. Prof. Madhav Kapadiya Mob: 9313230772 Email: madhavkumar.kapadia36168@paruluniversity.ac.in	Prof. Madhav Kapadiya
		Database Management Systems	1. Prof. Rinkal Sarvaiya Mob: 8799025175 Email: rinkal.sarvaiya31672@paruluniversity.ac.in	Prof. Rinkal Sarvaiya
		Python and R Programming	1. Prof. Shivaniba D. Bhadoriya Mob: 9510332044 Email: shivaniba.bhadoriya36466@paruluniversity.ac.in	Prof Shivani B
	Elective-1 [Only one]	Data Analytics and Visualisation Predictive Analytics/ Spatial Modelling Cloud Native Development	1. Dr. Ramachandran P Mob: 9384345662 Email: Ramchandran.p34251@paruluniversity.ac.in	Dr. Ramachandran P
	Industry Deployable Courses (IDC): (consider only 1)	Industry Deployable Courses (IDC) IDC - I - Power BI for Data Analytics	1. Prof. Jatin Morwal Mob: 8764398766 Email: jatin.morwal40859@paruluniversity.ac.in	Prof. Jatin Morwal

Syllabus of Courses

05201251 (MCA) / 05202156 (MScIT) - Advanced Java Programming

Course: MCA/MSc IT

Prerequisite: Knowledge of Core Java Programming and HTML.

Rationale : To make the students aware of the data driven web applications, web services, MVC architecture and ORM concepts of modern web application development.

Course Learning Objective:

CLOBJ 1	Understand and apply Java Database Connectivity (JDBC) concepts to establish database connections, execute SQL queries, manage transactions, and handle exceptions effectively.
CLOBJ 2	Design and develop Java Servlet–based web applications by configuring servers, managing servlet life cycle, handling client requests, implementing session tracking, and using servlet filters.
CLOBJ 3	Develop dynamic web interfaces using Java Server Pages (JSP) by applying JSP tags, implicit objects, Expression Language (EL), JavaBeans, and JSTL for clean separation of presentation and business logic.
CLOBJ 4	Build enterprise-level applications by implementing Java Web Services (SOAP and REST) and integrating Spring MVC and Hibernate frameworks for scalable, maintainable, and database-driven web applications.
CLOBJ 5	Analyze and implement RESTful web services using JAX-RS by applying HTTP methods, URI design principles, parameter injection, client APIs, and proper response and exception handling.
CLOBJ 6	Apply Spring MVC and Hibernate integration to develop layered enterprise applications by managing controllers, views, persistence, object–relational mapping, and database queries using HQL.

Course Learning Outcomes:

CLO 1	Explain the architecture and core components of Java Database Connectivity (JDBC) and develop Java applications that connect to databases, execute SQL queries, handle transactions, and process results efficiently.
CLO 2	Design and implement Java Servlets by configuring the web environment, managing the servlet life cycle, handling client requests and responses, and applying session tracking, cookies, and servlet filters.
CLO 3	Develop dynamic web applications using Java Server Pages (JSP) by applying JSP tags, implicit objects, Expression Language (EL), JavaBeans, custom tag libraries, and JSTL for effective presentation logic.
CLO 4	Analyze and build Java-based web services using SOAP and REST architectures, implement RESTful services with JAX-RS (Jersey), handle HTTP methods, parameters, responses, and exceptions, and consume services using client APIs.

CLO5	Apply Spring MVC and Hibernate frameworks to develop scalable and maintainable enterprise applications by implementing MVC architecture, object–relational mapping (ORM), and database interactions using HQL.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
4	0	4	-	6	20	20	20	60	30	150

SEE - Semester End Examination, **T** - Theory, **P** – Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Java Database Connectivity** Introduction to JDBC, JDBC architecture. Common JDBC Components - DriverManager, Driver, Connection, Statement, ResultSet, SQLException. Types of JDBC drivers, Establishing connection w Submitting queries and obtaining results, ResultSetMetadata Interface, Precompiled statements, CallableStatement, JDBC transactions.	15	9
2	**Java Servlets** Configuring Java Development Environment, Downloading and configuring Apache Tomcat web server. Introduction to servlets, Servlet life cycle, Exploring servlet API, Basic servlet structure, Handling client requests, Generating response, Working with cookies, Session tracking. Servlet filters - Introduction to filters, creating custom filters, mapping filters to servlets, mapping filters to specific URL pattern.	20	11
3	**Java Server Pages** Introduction to JSP, Need of JSP, Advantages of JSP over servlet, Life cycle of a JSP page, JSP basic tags: declaration tag, scriptlet tag, expression tag. JSP implicit objects - out, request, response, application, session, pageContext, page, exception. Introduction to java beans, Reading and writing properties of JavaBeans in JSP pages. JSP 2.0 expression language - Use of EL, Invoking Expression Language, Preventing Expression language evaluation, Accessing scoped variables. Custom tag libraries - tag handlers, tag library descriptor. JSP standard tag library (JSTL): Downloading and installing JSTL in Java web application, Core tags (c:out, c:forEach, c:if, c:choose, c:set, c:remove, c:import, c:catch).	25	14
4	**Java Web Services** Introduction to web services, Architecture of web services, Types of web services - SOAP web services, REST web services SOAP v/s REST, Introduction to REST, Configuring RESTful web service using Jersey framework in java web application, RESTful architectural principles, HTTP method and URI matching. JAX-RS injection - PathParam, QueryParam, FormParam. Introduction to Server responses and exception handling, JAX-RS client API.	20	13

5	**Introduction to Spring and Hibernate** Introduction to Spring MVC Framework, Spring MVC Architecture and Components, Advantages of Spring MVC, Modules Of Spring Framework. Spring Application Introduction to Hibernate, Architecture of hibernate, Hibernate O/R mapping, Configuring hibernate development environment, Implementing hibernate O/R Mapping, Introduction to Hibernate Query Language (HQL).	**20**	**13**
	Total	**100**	**60**

Reference Books	
1.	**Java Server Programming Java EE6 Black Book (TextBook)** Dreamtech Press
2.	**RESTful Java with JAX-RS 2.0** By Bill Burke | O'Reilly | 2nd Edition
3.	**Core Servlets and Java Server Pages Volume-1** By Mary Hall and Larry Brown | Prentice Hall | 2nd Edition
4.	**Core Servlets and Java Server Pages Volume-2** By Marty Hall, Larry Brown and Yaakov Chaikin | Prentice Hall | 2nd Edition

Course Outcome
After Learning the Course the students shall be able to:
1. design and develop data driven web applications using JDBC, servlet and JSP API. 2. develop MVC based web applications using Struts. 3. develop and integrate web services and web clients. 4. describe significance of Object Relational Mapping (ORM) in Java context.

List of Practical	
1.	**JDBC Connectivity** Develop a java application using concept of JDBC for user login. User will enter username and password. Application will match for the same in database table. If match is found then display message "Successful Login" and if not found then display message "Invalid Username and password ".
2.	**JDBC Connectivity & CRUD Operations** Develop a Menu driven java application for student information which will create the table with appropriate columns. Menu will contain the options like insert, update and delete. Based on the option, data will be inserted or updated or deleted from table based on student id (student_id will be primary key). Display appropriate message for each operation.

3.	**Data Driven GUI Application** Develop an application using GUI for user registration. In first form user will enter the data for registration (e.g. Name, Birthdate, Email_id, Phone_No). When user clicks on submit button, data will be inserted into database and user will be redirected to another form. In this form data will be selected from database and displayed in proper format.
4.	**A Data Driven Servlet Application** Develop a data driven servlet application for user authentication. Also display header values of request object and response object.
5.	**Servlet - Session Management** Develop a servlet application for implementing student grading system. In this system on first page student will enter his data (name, id and department). By clicking on next button he will be redirected to another page. On the second page he will enter marks of all 6 subjects. By clicking on "Generate result" he will be redirected to next page. On this page all information of student will be displayed along with the result (pass or fail) in appropriate color. Also display the name of student on each page he visits. (Use session management)
6.	**Servlet - Request Redirection** Develop an application for the following: User should be redirected to the URL entered in location text box.
7.	**JSP - User Authentication** Develop a data driven JSP application for user authentication.
8.	**JSP - Java Beans** Develop an application for Bank Account information using Java Beans. This application will perform various operation on bank account like withdraw, deposit and displaying the information.
9.	**JSP - Custom Tags** Develop a custom JSP tag which accepts 10 numbers from user and sorts them in specified order.
10.	**JSP - Application Context** Develop an application that will allow maximum 3 users to access the application. If number of users exceeds than limit then user will be redirected to the error page.
11.	**Simple RESTful Service** Develop a simple JAX-RS service that provides currency conversion.
12.	**Data Driven RESTful Service** Develop a RESTful service to accept student registration data sent from a web form.
13.	**RESTful Web Service - CRUD Operations** Develop a JAX-RS service that provides CRUD database operation interface for a phonebook. Call appropriate service from servlet/jsp pages to perform data oriented operations.
14.	**JAX-RS Client** Develop a JAX-RS client that consumes RESTful service developed in Program-11. Utilize the client in UI layer (JSP pages).
15.	**JAX-RS Client** Develop a RESTful service to perform authentication of the user. The JAX-RS client will be utillized in servlet and provide interface between jsp page and web service to perform authentication.

16.	**Struts2 - User Authentication** Develop a struts 2 application that accepts user name and password from user. If both values are valid then user will be redirected to home page otherwise redirected on error page.
17.	**Struts2 - User Registration and Authentication** Develop a data driven Struts 2 application that accepts registration details from user, i.e. name, contact number, address, user id and password. After successful registration, application redirects user to log in page. If user provides valid authentication details then application will redirect user to home page and will display welcome message, otherwise user will be redirected to error page.
18.	**Struts2 - Data Driven Application** Develop a phone book using struts 2. Application accepts data of contacts like name, address, contact number and email id. The application should provide features to update, delete and list data of contacts into/from database.
19.	**Struts2 - Interceptors** Create a logging service using struts 2 to monitor log in and log out timestamp of the users. (Hint: utilize interceptors)
20.	**Struts2 - Hibernate Integration** Develop an application using struts 2 that provides insert, update, delete and list operations for Student data like, name, roll number, semester, course, result (%). The data oriented operations must be performed through hibernate.

05202158 – Computer Networks and Network Security (MCA/MSc IT) (offered as MOOC-NPTEL)

Course: M.Sc. (IT)/MCA
Prerequisite: Basics of computer fundamentals**.**
Rationale : Demonstration of application layer protocols Discuss transport layer services and understand UDP and TCP protocols Explain routers, IP and Routing Algorithms in network layer Disseminate the Wireless and Mobile Networks covering IEEE 802.11 Standard**.**

Course Learning Objective:

CLOBJ 1	Understand the fundamentals of data communication, including components, transmission concepts, signal types, multiplexing techniques, and transmission errors.
CLOBJ 2	Explain standard network reference models, particularly the OSI and TCP/IP models, and compare their layers, functions, and protocols.
CLOBJ 3	Identify and analyze different types of computer networks such as LAN, MAN, WAN, wireless networks, and internetworks.
CLOBJ 4	Describe physical layer concepts, including transmission media, switching techniques, and analog and digital modulation methods.
CLOBJ 5	Understand data link and network layer mechanisms, including error control, flow control, routing algorithms, congestion control, and internetworking protocols.
CLOBJ 6	Explain transport and application layer functionalities, including TCP/UDP operations, connection management, DNS, e-mail, WWW, HTTP, and basic network security concept

Course Learning Outcomes:

CLO 1	**Describe and compare network architectures and protocols**, including OSI and TCP/IP models, for effective data communication.
CLO 2	**Analyze transmission techniques and media**, including multiplexing, modulation, and switching methods, used in wired and wireless networks.
CLO 3	**Apply data link and network layer protocols** to detect and correct errors, manage flow control, and implement routing and congestion control mechanisms.
CLO 4	**Explain transport layer services**, including TCP and UDP operations, connection management, and performance considerations.
CLO5	**Demonstrate understanding of application layer services and network security basics**, including DNS, HTTP, e-mail systems, cryptography, and compression techniques.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
4	0	2	-	5	20	20	20	60	30	150

SEE - Semester End Examination, **T** - Theory, **P** – Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Networking Fundamentals**Introduction, Data & Information, Data Communication-Characteristics of Data Communication, Components of Data Communication, Data Representation, Data Flow- Simplex, Half Duplex, Full Duplex, Computer Network- Categories of a network, Protocol-Elements of a Protocol, Networking Standards, Reference Models- OSI Model, TCP/IP Model, Comparison of OSI and TCP/IP Model.	**20**	**12**
2	**The Data Link Layer**Design Issues: Framing, Error Control, Flow Control, Protocols: FDDI, CDDI, Frame Relay, ATM, 802.11, PPP, HDLC. The Medium Access Sub-Layer Multiple Access Protocols: ALOHA, CSMA, Ethernet: Switched Ethernet, Fast Ethernet, Gigabit Ethernet, DLL Switching: Internetworking, Repeaters, Hubs, Bridges, Switches, Routers, Gateways, Virtual LANs	**25**	**12**
3	**The Network Layer**Design Issues, Routing Algorithms: Link State Routing, Distance Vector Routing, Flooding, Routing Protocols: RIP, IGRP, EIGRP, OSPF, Internetworking: Tunneling, Fragmentation, IPV4, IPV6 Basics, BGP. The Transport Layer Protocols: UDP, TCP, Headers.	**20**	**10**
4	**The Application Layer**DNS: The DNS Name Space, Name Servers-Mail: SMTP, POP3, HTTP, FTP, Telnet, Network Management: SNMP	**10**	**6**
5	**Network & System Security**Authentication Applications: Kerberos X.509, Directory Authentication Service, Electronic Mail Security, Pretty Good Privacy (PGP), S / Mime, Security: Architecture, Authentication Header, Encapsulating Security Payloads, Combining Security Associations, Key Management, Web Security: Secure Socket Layer & Transport Layer Security, Secure Electronic Transaction (Set), System Security: Intruders, Viruses, Firewall Design Principles, Trusted Systems	**25**	**14**
	Total	**100**	**54**

Reference Books	
1.	**Computer Networking: A Top-Down Approach** By James F. Kurose, Keith W. Ross.
2.	**Computer Architecture** By Parhami, Behrooz | Oxford publication, Latest Edition ISBN: 978-0-19-808407-5
3.	**Computer Networks : A Systems Approach** By Larry L. Peterson & Bruce S. Davie | Morgan Kaufmann
4.	**"Computer networks"** By Andrew Tanenbaum | Prentice Hall

List of Practical	
1.	**Study of different types of Network cables and Practically implement the cross-wired cable and straight through cable using clamping tool.**
2.	**Study of Network Devices in Detail.**
3.	**Study of network IP.**
4.	**Connect the computers in Local Area Network.**
5.	**Study of basic network command and Network configuration commands.**
6.	**Performing an Initial Switch Configuration**
7.	**Performing an Initial Router Configuration**
8.	**Configuring and Troubleshooting a Switched Network**
9.	**Connecting a Switch**

05201254 (MCA)/05202154s (MScIT) - Comprehensive Assessment

Course: MCA

Prerequisite: Basic knowledge of computer fundamentals, procedural programming, internet concepts, data structures, operating systems, databases, and problem-solving techniques.

Rationale : Student will be assessed on the basis of various courses learned till the end of respective semester.

Course Learning Objective:

CLOBJ 1	Develop a comprehensive understanding of signals, including continuous-time and discrete-time signals, analog and digital signals, and their characteristics such as amplitude, frequency, and phase.
CLOBJ 2	Analyze signals in the time domain by performing operations such as addition, multiplication, differentiation, and integration. Understand the concept of linearity and time-invariance in systems.
CLOBJ 3	Apply Fourier analysis to decompose signals into their frequency components. Perform frequency domain operations like convolution and filtering.
CLOBJ 4	Learn to transform signals from the time domain to the complex frequency domain using Laplace and Z-transforms. Analyze the stability and system behaviour in the transformed domains.
CLOBJ 5	Understand the characteristics of linear, time-invariant (LTI) systems. Compute and interpret the impulse response of systems.
CLOBJ 6	Grasp the fundamentals of sampling and the Nyquist theorem.

Course Learning Outcomes:

CLO 1	Review and summarize various subjects studied till the end of respective semester.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	-	3	-	20	20	-	60	100

SEE - Semester End Examination, **T** - Theory, **P** – Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Fundamentals of Computer Organization** Number systems, Boolean algebra, Flip-Flop, Adders, RAM, ROM, Cache memory, Buses.	10	4
2	**Fundamentals of Programming** Decision & control structure, Looping, Arrays, Functions, Pass by value, Pass by pointer, Pointers, Structure, Union, File handling.	10	5
3	**Fundamentals of Internet and Web Designing** HTML tags, Linking documents, Frames, Media elements, Image maps, CSS styles, Java script.	10	4
4	**Object Oriented Concepts and Programming** Object Oriented Programming concepts, Functions, Constructor and destructor, Operator overloading, Inheritance, Runtime polymorphism, Exception handling, IO operations.	10	4
5	**Operating Systems** Operating System concept, Process Management and threads, Memory Management, Inter-process communication and synchronization, I/O and file system, Disk scheduling.	10	5
6	**Java Programming** Inheritance, Sub classing, Package, Exception, Collection frameworks – ArrayList, Vector, Set, Map, I/O and Multi threading.	10	4
7	**Software Engineering** Software process, Requirements engineering, Data dictionary, Class diagram, Usecase diagram, Activity diagram, Sequence diagram.	10	5
8	**Advanced Database Management Systems** Object relational database, Extended relational database, PL/SQL, Cursor, Trigger, Stored procedure, Database tuning, indexing and security, Backup and recovery in database, Granting and revoking privileges.	10	5
9	**Open Source Technology using PHP** Open source software and configuration, PHP structure and syntax, Session and cookies, MySQL with PHP, working with User data and Error handling.	10	4
10	**Analysis and Design of Algorithm** Algorithm analysis, Time and Space complexity, Divide and conquer algorithms, Graph searching and traversal, Branch and bound, Dynamic programming, Backtracking, Asymptotic notations.	10	5
	Total	100	45

Reference Books

1.	**Structured Computer Organization** By A. Tanenbaum | Prentice - Hall of India Pvt. Ltd | 4th Edition

2.	**Computer Architecture and Logic Design** By Thomas C. Bartee | Tata McGraw - Hill Edition, 2010
3.	**Programming with problem solving through 'C'** By ELSEVIER
4.	**Programming in C** By E. Balagurusamy | Tata McGraw-Hill
5.	**HTML & CSS: The Complete Reference** By Thomas A. Powell | McGraw Hill
6.	**Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP** By Ivan Bayross | BPB Publication
7.	**Database System Concepts** By Silberschatz, Korth, Sudarshan | McGraw Hill Publication | 4th Edition
8.	**Database Systems, Concepts, Design and Applications** By S K Singh | Pearson Edition
9.	**An Introduction to Data Structures with Applications** By Jean-Paul Tremblay, Paul G. Sorenson | Tata McGraw-Hill | 2nd Edition, (2007)
10.	**Introduction to Algorithm** By Cormen, Leiserson, Rivest, Stein | PHI (2003) | 2nd Edition
11.	**C++: The Complete Reference** By Herbert Schildt | McGraw Hill | 5th Edition
12.	**Programming with ANSI C++** By Bhushan Trivedi | Oxford University Press | Second Edition
13.	**Operating System Concepts** By Abraham Siberschatz | Wiley | Fifth
14.	**Operating Systems Internals and Design Principles** By William Stallings | PHI | 5th Edition
15.	**Analysis and Design of Information system** By James Senn | Mc Graw Hill
16.	**System Analysis and Design** By S. Parthasarthy & B. W. Khalkar | Master Ed. Cons., Nashik. | 1st Edition
17.	**The Complete Reference Java J2SE** By Herbert Schildt | TMH Publishing Company Ltd, NewDelhi. | 5th Edition
18.	**Core Java Volume 1** By Cay Horstmann and Gary Cornell | Pearson Education | 8th Edition

19.	**Software Engineering : A Practitioner's Approach** By Pressman R.S | TMH
20.	**Software Engineering** By Sommerville | Pearson | 8th
21.	**SQL, PL/SQL – The Programming Language** By Ivan Bayross | BPB Publications
22.	**Database System Concepts** By Abraham Silberschatz, Henry Korth, S. Sudarshan | McGraw Hill International | 6th Edition
23.	**Professional LAMP: Linux, Apache, MySQL and PHP5 Web Development** By Jason Gerner, Morgan Owens, Elizabeth Naramore, Matt Warden | WROX publication
24.	**Sams Teach Yourself PHP, MySQL and Apache All in One** By Julie C Meloni | Pearson Education
25.	**Analysis and Design of Algorithm** By Ullman | TMH
26.	**Introduction to the Design & Analysis of Algorithms** By Goodman | TMH

Course Outcome

After Learning the Course the students shall be able to:

1. Explain fundamental concepts of computer organization, operating systems, and algorithm analysis including memory, processes, and complexity.
2. Apply procedural and object-oriented programming concepts using C/C++/Java for solving computational problems.
3. Design and develop web-based and database-driven applications using HTML, CSS, JavaScript, PHP, and advanced DBMS features.
4. Analyze and model software systems using software engineering principles, UML diagrams, and efficient algorithms.

05201255(MCA)/05202155(MSc IT) - Seminar

Course: MCA

Prerequisite: Basic knowledge of computer science fundamentals, familiarity with current computing technologies, ability to read and understand research papers/technical articles, and basic presentation and documentation skills.

Rationale : This course/seminar provides students with an opportunity to explore contemporary topics in Computer Science and Applications, enhance their ability to study and analyze technical literature, and develop skills in research, documentation, and professional presentation, thereby preparing them for advanced studies and industry-oriented problem solving.

Course Learning Objective:

CLOBJ 1	Develop a comprehensive understanding of signals, including continuous-time and discrete-time signals, analog and digital signals, and their characteristics such as amplitude, frequency, and phase.
CLOBJ 2	Analyze signals in the time domain by performing operations such as addition, multiplication, differentiation, and integration. Understand the concept of linearity and time-invariance in systems.
CLOBJ 3	Apply Fourier analysis to decompose signals into their frequency components. Perform frequency domain operations like convolution and filtering.
CLOBJ 4	Learn to transform signals from the time domain to the complex frequency domain using Laplace and Z-transforms. Analyze the stability and system behaviour in the transformed domains.
CLOBJ 5	Understand the characteristics of linear, time-invariant (LTI) systems. Compute and interpret the impulse response of systems.
CLOBJ 6	Grasp the fundamentals of sampling and the Nyquist theorem.

Course Learning Outcomes:

CLO 1	Learn about exploring topic on ICT domain.
CLO 2	Develop skills for identification and retrieval of relevant information.
CLO 3	Effectively communicate and present ICT topic.
CLO 4	Summarize on ICT topic.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
0	0	2	-	1	-	20	20	-	60	100

SEE - Semester End Examination, **T** - Theory, **P** – Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Seminar** Each Student is expected to choose a topic of current relevance of Computer Science/Application field. They have to refer papers / reading material from various resources including latest tools. Student should compile the material in the form of presentation / seminar report and appropriately present the topic**.**	**100**	
	Total	**100**	

Course Outcome

After Learning the Course the students shall be able to:

1. learn about exploring topic on ICT domain.
2. develop skills for identification and retrieval of relevant information.
3. effectively communicate and present ICT topic.
4. summarize on ICT topic.

05201256[MCA]/05202186[MSc IT] – Graphics Programming

Course: MCA

Prerequisite: Knowledge of linear algebra, calculus and programming skills in C/C++.

Rationale : To provide the fundamental principles and algorithms of underlying computer graphics, including line drawing algorithms, circle/ellipse drawing algorithms, 2D geometrical transformation, 3D geometric transformations, viewing in 3D (orthographic projection and perspective projection) and visible surface detection algorithms.

Course Learning Objective:

CLOBJ 1	Develop a comprehensive understanding of signals, including continuous-time and discrete-time signals, analog and digital signals, and their characteristics such as amplitude, frequency, and phase.
CLOBJ 2	Analyze signals in the time domain by performing operations such as addition, multiplication, differentiation, and integration. Understand the concept of linearity and time-invariance in systems.
CLOBJ 3	Apply Fourier analysis to decompose signals into their frequency components. Perform frequency domain operations like convolution and filtering.
CLOBJ 4	Learn to transform signals from the time domain to the complex frequency domain using Laplace and Z-transforms. Analyze the stability and system behaviour in the transf ormed domains.
CLOBJ 5	Understand the characteristics of linear, time-invariant (LTI) systems. Compute and interpret the impulse response of systems.
CLOBJ 6	Grasp the fundamentals of sampling and the Nyquist theorem.

Course Learning Outcomes:

CLO 1	Learn about exploring topic on ICT domain.
CLO 2	Develop skills for identification and retrieval of relevant information.
CLO 3	Effectively communicate and present ICT topic.
CLO 4	Summarize on ICT topic.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	0	2	-	4	20	20	20	60	30	150

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Introduction and Basic Drawing Algorithms** Computer graphics and its applications, Raster-scan system, Random-scan system, Graphics primitives. **Basic Drawing Algorithms:** Line-drawing algorithms - DDA Algorithm, Bresenham's line algorithm. **Circle Generating Algorithms:** Midpoint circle algorithm, Bresenham's circle algorithm. **Ellipse-Generating Algorithm:** Midpoint ellipse algorithm.	**20**	**8**
2	**Region Filling Algorithms and Two-Dimensional Geometric Transformations** Attributes of output primitives, Line style, Colour and intensity, Area filling algorithm, Scan line and boundary fill algorithm, Flood fill algorithm, Antialiasing technique. **Two dimensional transformations:** Translation, Scaling, Rotation, Reflection, Sheering, Composite transformation, Transformation commands, Character generation.	**20**	**12**
3	**Viewing and Clipping** Viewing coordinates, Window view port, Clipping, Window to view port transformation, Line clipping algorithm - Cohen Sutherland, Polygon clipping - Sutherland Hodgman algorithm, 3D clipping - normalized view volume, View port clipping, Clipping in homogeneous coordinates.	**15**	**7**
4	**Illumination Model** Light sources, Diffuse reflection, Colour Models like RGB, YIQ, CMY, HSV. **3D Viewing**: 3D concepts, 3D display techniques, 3D representation, 3D transformation - transition, scaling, composite transformation, rotation about arbitrary axis. **Projection**: Parallel and perspective projection, Hidden surface and line removal, Back face removal, Depth buffer and scan line methods	**15**	**7**
5	**OpenGL and Graphics ProgrammingIntroduction to OpenGL:** Basics of OpenGL, its pipeline, and setup, OpenGL coordinate system. OpenGL Programming Basics: Drawing shapes, event handling, and rendering pipeline Lighting and material properties in OpenGL. Texture Mapping & Shading: Texture mapping techniques, OpenGL texture functions, shading techniques (Flat, Gouraud, and Phong), basics of OpenGL shaders (GLSL). OpenGL Animations & Rendering: Animation concepts, double buffering, frame rate control, shadow mapping. Advanced OpenGL & Project: Introduction to modern OpenGL (VBOs, VAOs, and shaders), optimization techniques	**30**	**14**
	Total	**100**	**48**

Reference Books	
1.	**Computer Graphics (TextBook)** By Donald Hearn & M. Pauline Baker | PHI,2011 | Second Edition
2.	**Computer Graphics - a Programming approach** By S. Harrington | McGraw Hill,2014 | 2nd Edition
3.	**Principles of interactive computer graphics** By New Mann & Sprovl | McGraw Hill,2001 | 2nd edition
4.	**Procedural Elements for Computer Graphics** By David F. Rogers | Tata McGraw Hill,2001
5.	**Multimedia System Design** By Prabhat K. Andleigh & Kiran Thakur | PHI
6.	**Computer Graphics with Open GL** By Hearn Baker Carithers | Person New International Edition | 4th
7.	**Learn Open GL Graphics Programming** By Joey de Vries

Course Outcome
After Learning the Course the students shall be able to:
1. identify various components of graphics system. 2. implement algorithms for rendering basic shapes. 3. describe significance of projection, illumination models, viewing and clipping techniques. 4. implement various 2D and 3D transformation techniques and clipping algorithms.

List of Practical	
1.	**Implement and visualize a line using the Digital Differential Analyzer (DDA) algorithm.**
2.	**Implement the Bresenham's line drawing algorithm to render an efficient line.**
3.	**Draw a circle using the Midpoint Circle Algorithm.**
4.	**Implement Bresenham's circle drawing algorithm to generate a circle.**
5.	**Draw an ellipse using the Midpoint Ellipse Algorithm.**
6.	**Implement Scan-line and Boundary Fill algorithms for polygon filling.**
7.	**Implement Flood Fill Algorithm to fill a closed shape with color.**
8.	**Perform translation, scaling, rotation, reflection, and shearing on 2D objects.**
9.	**Implement the Cohen-Sutherland algorithm for line clipping within a viewport.**

10.	**Implement the Sutherland-Hodgman algorithm for polygon clipping.**
11.	**Perform 3D translation, scaling, rotation, reflection, and shearing.**
12.	**Implement Back-face culling, Depth-buffer (Z-buffer), and Scan-line algorithms for hidden surface removal.**
13.	**Implement RGB, HSV color models and anti-aliasing techniques.**
14.	**Apply shading models and textures to 3D objects.**
15.	**Create an animated scene using transformation functions.**

05202108 - Software Engineering with Agile

Course: M.Sc. (IT)

Prerequisite: Understanding of Object Oriented Design, UMl

Rationale : To analyze the requirements of end users, Understands the software development life cycle (SDLC) along with many other process models, gain understanding of test strategies and different types of testing, gain understanding of different agile processes

Course Learning Objective:

CLOBJ 1	Develop a comprehensive understanding of signals, including continuous-time and discrete-time signals, analog and digital signals, and their characteristics such as amplitude, frequency, and phase.
CLOBJ 2	Analyze signals in the time domain by performing operations such as addition, multiplication, differentiation, and integration. Understand the concept of linearity and time-invariance in systems.
CLOBJ 3	Apply Fourier analysis to decompose signals into their frequency components. Perform frequency domain operations like convolution and filtering.
CLOBJ 4	Learn to transform signals from the time domain to the complex frequency domain using Laplace and Z-transforms. Analyze the stability and system behaviour in the transformed domains.
CLOBJ 5	Understand the characteristic of linear, time-invariant (LTI) systems. Compute and interpret the impulse response of systems.
CLOBJ 6	Grasp the fundamentals of sampling and the Nyquist theorem.

Course Learning Outcomes:

CLO 1	Describe Software Development Life Cycle.
CLO 2	Explain software development process in association with its fundamental principles andmethodologies.
CLO 3	Analyze & represent end user requirements and model requirement analysis using Unified ModelingLanguage.
CLO 4	Prepare & represent software design and design software model using Unified Modeling Language.
CLO 5	Define significance of project planning, effort estimation and risk management
CLO 6	Design test cases and identify testing strategies.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	0	2	0	4	20	20	20	60	30	150

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Introduction to System and Software Development**Concept of system, Basic components of system, Information systems categories, Need of information system development. Study of different models, Software characteristics, Components, Applications, Layered technologies, Processes, Methods and tools, Generic view of software engineering, Classical Systems Development Life Cycle (SDLC) method.	**15**	**6**
2	**Requirements Analysis & Engineering**Requirement Analysis: Classical Systems Development Life Cycle (SDLC) method, Requirement determination, System Requirement Specification (SRS), Fact finding techniques. Process models - waterfall model, incremental, Evolutionary process models - prototype, spiral and concurrent development model. Requirement Engineering: Problem recognition, Requirement engineering tasks, , Use cases and functional specification, Requirements validation	**20**	**10**
3	**Structured System Analysis & Design:**System Analysis: Introduction to Structured System Analysis Development Methodology (SSADM), Tools for analysis - decision trees, decision tables, structured english, Data flow diagram, Entity Relationship (ER) diagram, Data dictionary. System Design: Design concepts, Design modeling, Software architecture, Data design, Architectural styles and patterns, Procedural design, Object oriented design.	**20**	**10**
4	**Agile Methodology:** Agile process, Extreme Programming (XP), Brief overview of other agile process models - adaptive software development, Scrum.	**20**	**7**
5	**User Interface Design**Concepts of UI, Interface design model, Internal and external design, Evaluation, Interaction and information display software.	**5**	**3**

6	**Planning a Software Project:**Management spectrum, People, Product, Process, Project, W5HHPrinciple, Importance of team management, Scope and feasibility, Effort estimation, Schedule and staffing, Quality planning, Risk management- identification, assessment, control,	**15**	**6**
7	**Case Tools and Study:**Introduction to CASE, Building blocks of CASE, Integrated CASE environment.	**5**	**3**
	Total	**100**	**45**

Reference Books

1.	**Software Engineering : A Practitioner's Approach (TextBook)** By Pressman R.S | TMH
2.	**Software Engineering** By Sommerville
3.	**Software Engineering** By Rajiv Mall | PHI
4.	**Software Engineering** By Pankaj Jalote | Wiley India
5.	**Software Engineering, An Engineering Approach** By Peters & Pedrycz | Wile-India
6.	**Software Engineering, Principles and Practice** By Jawadekar | TMH

Course Outcome

After Learning the Course the students shall be able to:

1. describe Software Development Life Cycle.
2. explain software development process in association with its fundamental principles and methodologies.
3. analyze & represent end user requirements and model requirement analysis using Unified Modeling Language.
4. prepare & represent software design and design software model using Unified Modeling Language.

05293252 – English and Mathematical Aptitude

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
2	-	-	-	2	-	100	-	-	-	100

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Numbers system** This session will prepare students on Number system which is one of the sub-topic of quantative ability section of Aptitude test of various companies.	5	2
2	**Reading comprehension** The student will be taught different ways to solve Reading Comprehension questions when it comes to Placements or different competitive exams.	5	2
3	**Time and work** This topic covers different type problems on Time and Work and also the short trick method to solve them.	5	2
4	**Sentence correction** This topic contains an important role to solve Verbal section in an Aptitude exam for any company	5	2
5	**Coding & Decoding, Series, Analogy, Odd man out and Visual reasoning** Solving different types of series, visual reasoning questions which comes in Personal Interview as well as Logical section of any Aptitude Exam.	5	2
6	**Time, speed and distance** This topic covers the upstream, downstream questions and short trick to solve it.	5	2
7	**Vocabulary** Will include to How learn new words and remember those words with different methodology, it also includes different types anagrams and puzzles .	5	2
8	**Recap on C++** Case Studies, Coding, MCQ questions from different technical exams will be discussed to prepare for technical test and PI.	8	6
9	**Recap on Data structure** Case Studies, Coding on C and C++, MCQ questions from different technical exams will be discussed to prepare for technical test and PI.	7	6
10	**Ratio and Proportion**	5	2
11	**Logical connectives, Syllogism and Venn diagrams**	5	2

12	**Worksheet on – Adjective and Adverb**	**5**	**2**
13	**Worksheet on - Collocation, Gerund and Infinitives**	**5**	**2**
14	**Worksheet on - Determiners and Articles**	**5**	**2**
15	**Worksheet on - Nouns and Pronouns**	**5**	**2**
16	**Worksheet on – PREPOSITION**	**5**	**2**
17	**Worksheet on - PRONOUN ANTECEDENT AGREEMENT**	**5**	**2**
18	**Worksheet on - Subject Verb Agreement**	**5**	**2**
19	**Worksheet on – Tenses**	**5**	**1**
	Total	**100**	**45**

Reference Books	
1.	**Handouts on communication confidence**
2.	**Norman Lewis, Word Power Made Easy, Goyal Publishers, Delhi**
3.	**Understanding and Using English Grammar** By Betty Azar & Stacy Hagen | Pearson Education
4.	**Let Us C++** By Yashavant Kanetkar
5.	**Fundamentals of Data Structure** By Ellis Horowitz

Course Outcome

After Learning the Course the students shall be able to:
1. comprehend day to day English. 2. respond to familiar issues / topics in English. 3. speak confidently on stage. 4. able to solve aptitude questions. 5. able to crack the technical exam for IT companies.

05201295 – Big Data Analytics – I

Course: MCA (A.Y.-II) 2020
Prerequisite: Basic Knowledge of Data Analytics and Computation Methods.
Rationale : The objective of this course is to provide Conceptual insight about Big Data Analytics, Installation and understanding of Hadoop Architecture and its ecosystems.

Course Learning Objective:

CLOBJ 1	To introduce the fundamental concepts, evolution, architecture, and components of Big Data systems.
CLOBJ 2	To explain the role of Big Data analytics in business decision-making and industry applications such as retail, social media, and fraud detection.
CLOBJ 3	To familiarize learners with distributed and parallel computing technologies used for large-scale data processing.
CLOBJ 4	To provide in-depth knowledge of the Hadoop ecosystem, including HDFS, MapReduce, and related tools.
CLOBJ 5	To develop understanding of Big Data storage technologies, including databases, data warehouses, and non-relational systems.
CLOBJ 6	To enable students to design and implement basic Big Data processing applications using MapReduce.

Course Learning Outcomes:

CLO 1	Explain the characteristics, structure, and lifecycle of Big Data and its importance in modern computing environments.
CLO 2	Analyze business problems and identify how Big Data analytics can be applied in domains such as social networking, retail, and fraud detection.
CLO 3	Describe and compare distributed, parallel, cloud, and in-memory computing technologies for Big Data processing.
CLO 4	Demonstrate the working of the Hadoop ecosystem, including HDFS architecture, data ingestion, and file management operations.
CLO 5	Evaluate different data storage solutions such as RDBMS, NoSQL databases, and data warehouses for Big Data applications.
CLO 6	Design and implement simple MapReduce programs to process and analyze large datasets efficiently.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/ Week	Credit	T	CE	P	T	P	
3	1	2	-	5	20	20	20	60	30	150

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Overview of Big Data** Introduction to Big Data, Evolution of Big Data, Structuring of Big Data, Fundamentals of Big Data, Big Data Analytics, Career and Future in Big Data	10	5
2	**Discovering the Use of Big Data in Business Context** Big Data in Social Networking, Big Data in Preventing Fraudulent Activities, Big Data in Detecting Fraudulent Activities in Insurance Sector, Use of Big Data in Retail Industry.	10	5
3	**Technologies for Handling Big Data** Distributed and Parallel Computing for Big Data, Introducing Hadoop, Cloud Computing and Big Data, In-Memory Computing Technology for Big Data.	8	4
4	**Understanding Hadoop Ecosystem** History of Hadoop, Hadoop Ecosystem, Analysing Data with Unix tools, Analysing Data with Hadoop, Hadoop Streaming, Hadoop Echo System, IBM Big Data Strategy, Introduction to Infosphere BigInsights and Big Sheets.	12	6
5	**HDFS(Hadoop Distributed File System)** The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O: Compression, Serialization, Avro and File-Based Data structures.	14	6
6	**MapReduce Fundamentals** Anatomy of a Map Reduce Job Run, Failures, Job Scheduling, Shuffle and Sort, Task Execution, Map Reduce Types and Formats, Map Reduce Features.	14	6
7	**Understanding Big Data Technology Foundations, Storing Data in Databases and Data Warehouses** Understanding Big Data Technology Foundations: Exploring the Big Data Stack, Virtualization and Big Data, Virtualization Approaches, Summary, Quick Revise Storing Data in Databases and Data Warehouses: RDBMS and Big Data, Non-Relational Database, Polyglot Persistence, Integrating Big Data with Traditional Data Warehouses, Big Data Analysis and Data Warehouse, Changing Deployment Models in Big Data Era.	16	8
8	**Processing Your Data with MapReduce** Recollecting the Concept of MapReduce Framework,Developing Simple MapReduce Application, Points to Consider while Designing MapReduce.	16	8
	Total	100	48

Reference Books

1.	**Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses** By Michael Minelli, Michelle Chambers, and Ambiga Dhiraj | Wiley, Pub. Year 2013
2.	**Big-Data Black Book** By DT Editorial Services | Wiley India
3.	**Hadoop Operations, Eric Sammer, O'Reilley.** By O'Reilley

4.	**Hadoop: The Definitive Guide by Tom White, Third Edition, O'Reilley.** By Tom White

Course Outcome

After Learning the Course the students shall be able to:
1. Work with big data platform and explore the big data analytics techniques business applications. 2. Understand the fundamentals of various big data analytics techniques. 3. Analyse th 4. e HADOOP and Map Reduce technologies associated with big data analytics. 5. Have knowledge on accessing, storing and manipulating the huge data from different resources.

List of Practical

1.	**Practical-1** (i) Perform setting up and Installing Hadoop in its two operating modes: • Pseudo distributed, • Fully distributed. (ii) Use web based tools to monitor your Hadoop setup.
2.	**Practical-2** (i) Implement the following file management tasks in Hadoop: • Adding files and directories • Retrieving files • Deleting files ii) Benchmark and stress test an Apache Hadoop cluster
3.	**Practical-3** Run a basic Word Count Map Reduce program to understand Map Reduce Paradigm. • Find the number of occurrence of each word appearing in the input file(s) • Performing a Map Reduce Job for word search count (look for specific keywords in a file)
4.	**Practical-4** To understand the overall programming architecture using Map Reduce API
5.	**Practical-5** Stop word elimination problem: • Input: A large textual file containing one sentence per line

	A small file containing a set of stop words (One stop word per line) • Output: A textual file containing the same sentences of the large input file without the words appearing in the small file.
6.	**Practical-6** Write a Map Reduce program that mines weather data. Weather sensors collecting data every hour at many locations across the globe gather large volume of log data, which is a good candidate for analysis with MapReduce, since it is semi structured and record- oriented. Data available at: https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all. • Find average, max and min temperature for each year in NCDC data set? Filter the readings of a set based on value of the measurement, Output the line of input files associated with a temperature value greater than 30.0 and store it in a separate file.
7.	**Practical-7** Store the basic information about students such as roll no, name, date of birth , and address of student using various collection types such as List, Set and Map
8.	**Practical-8** Basic CRUD operations in MongoDB
9.	**Practical-9** Retrieve various types of documents from students collection
10.	**practical-10** To find documents from Students collection
11.	**Practical-11** Develop Map Reduce Work Application
12.	**Practical-12** Creating the HDFS tables and loading them in Hive and learn joining of tables in Hive

05201296 – Cyber Security and Forencics – I

Course: MCA (A.Y.-II) 2020

Prerequisite: Fundamental knowledge of computer network.

Rationale : The key objectives of this course are to develop an understanding of information assurance as practiced in computer operating systems, networks and representative applications and to gain familiarity with prevalent attacks, defenses against them.

Course Learning Objective:

CLOBJ 1	To introduce fundamental concepts of cyber security, security principles, cryptography, and secure communication models.
CLOBJ 2	To provide knowledge of various security threats, malware types, and common cyber-attacks affecting systems and networks.
CLOBJ 3	To understand operating system security mechanisms, memory and file protection methods, and system hardening techniques.
CLOBJ 4	To explain wireless network security protocols and vulnerabilities in modern wireless communication environments.
CLOBJ 5	To familiarize students with cyber forensics processes, digital evidence handling, and forensic investigation tools and practices.
CLOBJ 6	To develop practical skills in network security, virtualization technologies, mobile security issues, and Kali Linux for security testing and system administration.

Course Learning Outcomes:

CLO 1	Explain core cyber security principles, CIA triad, cryptographic techniques, encryption standards (DES, AES, RSA), and key management systems.
CLO 2	Identify, classify, and analyze different types of security threats, malware, and cryptographic attacks.
CLO 3	Apply operating system security concepts including access control, memory protection, file security, and patch management to secure computing environments.
CLO 4	Evaluate wireless network security protocols (WEP, WPA, WPA2) and assess common wireless attack techniques.
CLO 5	Demonstrate understanding of cyber forensics methodologies, digital evidence handling, chain of custody, and network forensic investigations.
CLO 6	Configure and use networking concepts, virtualization platforms, mobile device security controls, and Kali Linux tools to perform basic security analysis and testing.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	1	2	-	5	20	20	20	60	30	150

Course Content **W** - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Security Principles and Practices** Basic Cyber Security Concepts, layers of security, CIA Triad, Information security, Network security Model, Cryptography, Attacks on Cryptosystem, Traditional Cryptography, Modern cryptography methods (block cipher & stream cipher), Symmetric & Asymmetric Key Encryption, Feistel Cipher, DES- (Data Encryption Standards), 3DES, AES- (Advanced Encryption Standards), Block Cipher modes , Introduction to Public key encryption, Public key infrastructure, RSA algorithm, Model and Introduction to Hash, MAC and Digital Signature	15	9
2	**Security Threats** Types of security threats- worms, viruses, Trojan horse, malware, malicious spyware, adware, botnet, spam, phishing, stack and buffer overflow	8	3
3	**Operating System Security** Role of operating systems in information systems applications, Operating systems security, Patched operating systems, Protected objects and methods of protection, Memory address protection, File protection mechanism.	10	4
4	**Wireless Networks Security** Overview of wireless technology, Wireless security protocols -Wired Equivalent Privacy (WEP), Wi-Fi Protected Access (WPA), WPA2, Attacks on wireless networks.	10	4
5	**Understanding Cyber Forensics** Computer forensics, Cyber forensics and Digital evidence, Rules of evidence, Forensics analysis of e-mail- RFC282, Digital forensics life cycle, Chain of custody concept, Network forensics, Setting up a computer forensics laboratory, Computer forensics and steganography, Rootkits, Information hiding, Relevance of the OSI layer model to computer forensics, Forensics and social networking sites - security, privacy, threats.	15	9
6	**Introduction to Network Security** Networking Introduction ,Working IP address and its classes, Hidden Concept of ip using integer format, DNS and working , Ports and Services, Network topology, HTTP Method , NAT ,Bridge , Host to Host in virtual box , Network manager, 3 way handshake, tcp header , http header	10	5
7	**Cybercrime on Mobile and Wireless Devices** Introduction, Proliferation of Trends in Mobility Credit card Frauds in the mobile Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication service Security, Attacks on Mobile/Cell Phones, Organizational security Policies and Measures in the mobile Era.	12	6
8	**Concept of Virtualization** Software Virtualization, Hardware Virtualization, OS Partitioning, VM Ware Windows, Linux	10	4
9	**Introduction to kali linux**Linux Distributions, Introduction to command line, Command Line Arguments and Options, Commands for Linux Linux Filesystem Hierarchy Standard, Devices, Partitions, and Mounting , Absolute and Relative Paths, Working with Files and Directories, Users and Groups, Package managers, Shell and Access, Linux Default misconfiguration	10	4
	Total	100	48

Reference Books	
1.	**Information systems security (TextBook)** By Nina Godbole | Wiley Publications,2008
2.	**Cyber Security understanding Cyber Crimes, Computer forensics and Legal Perspectives (TextBook)** By Nina Godbole and Sunit Belapure
3.	**Cryptography and Network Security Principles and Practices** By W. Stallings | Prentice-Hall of India, 2006 | 4th Edition
4.	**Information Security: Principles and Practices** By M. Merkow and J. Breithaupt | Pearson Education, 2006

Course Outcome
After Learning the Course the students shall be able to:
1. Recognize the significance of information system security in terms of threats and attacks. 2. Describe current techniques and tools for cyberforensic examination. 3. Identify various approaches for improvement of security aspects in operating systems and wireless network protocols. 4. Explain the significance of cyber forensics and digital evidence.

List of Practical	
1.	**Configuration of Virtual Laboratory for Mobile Forensic and Pen Testing, Configuration of Genymotion in virtual machine, Configuration of Santoku OS in virtual machine, Configuration of Appie.**
2.	**Use various functionality of ADB, Starts two instances of Emulator in Gynemotion, Connect adb through Santoku and Appie and perform, and its commands.**
3.	**Configuration of target vulnerable mobile application, DIVA (Damn insecure and vulnerable App), OWASP GoatDroid**
4.	**Understanding of Android Application architecture using santoku / Unzipping Archive android application file**
5.	**Reversing Engineering of Android Application using APKtools and JaDX Decompiler which a part of Santoku APKTool**
6.	**Practical related to the analysis of Dex file using Dexdump**
7.	**Practical related to insecure logging**

05201294 – Artificial Intelligence – I

Course: MCA (A.Y.-II) 2020
Prerequisite: Knowledge of data structures and discrete mathematics**.**
Rationale : To make students aware of knowledge representation, problem solving, heuristic search techniques, learning and development of expert system using prolog, machine learning in AI.

Course Learning Objective:

CLOBJ 1	To introduce the fundamental concepts, history, and applications of Artificial Intelligence.
CLOBJ 2	To develop understanding of problem solving using heuristic search and state-space search techniques.
CLOBJ 3	To explain methods of knowledge representation using logic, rules, and frames.
CLOBJ 4	To familiarize students with symbolic reasoning under uncertainty using probabilistic and non-monotonic approaches.
CLOBJ 5	To provide practical knowledge of logic programming using Prolog for AI problem solving.
CLOBJ 6	To introduce basic concepts and importance of Machine Learning in intelligent systems.

Course Learning Outcomes:

CLO 1	Explain the fundamentals, evolution, applications, and techniques of Artificial Intelligence.
CLO 2	Analyze and solve problems using heuristic search methods, problem spaces, and production systems.
CLO 3	Represent knowledge effectively using predicate logic, rules, frames, and inference mechanisms.
CLO 4	Apply reasoning techniques under uncertainty using probability theory, Bayesian networks, certainty factors, and non-monotonic logic.
CLO 5	Develop simple AI programs using Prolog by implementing facts, rules, recursion, and backtracking.
CLO 6	Describe Machine Learning concepts, classifications, and applications in real-world intelligent systems.

Teaching and Examination Scheme

Teaching Scheme					**Examination Scheme**					**Total**
					Internal Marks			**External Marks**		
Lecture Hrs/Week	**Tutorial Hrs/Week**	**Lab Hrs/Week**	**Seminar Hrs/Week**	**Credit**	**T**	**CE**	**P**	**T**	**P**	
3	1	2	-	5	20	20	20	60	30	150

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Introduction** Definition, Basic elements of Artificial Intelligence (AI), Applications of AI,History of AI,AI techniques.	10	5
2	**Heuristic Search Techniques** Generate and test, Hill climbing, Best-First search, Problem reduction, Constraint satisfaction, Means-Ends analysis	13	6
3	**Problems, Problems Spaces and Search** Production systems, Problem characteristics, Production system characteristics, Design issues of search algorithms.	10	5
4	**Knowledge Representation** **Knowledge & Knowledge representation issues:** General concepts, Definition and importance of knowledge, Representations and mappings, Approaches to knowledge representation, Frame problem. **Using predicate knowledge:** Representing instance and Is-a relationship, Computable functions and predicates, Resolutions, Natural deduction.	13	6
5	**Representing Knowledge Using Rules** Procedural Versus Declarative Knowledge, Logic Programming, Forward Versus Backward Reasoning.	10	5
6	**Symbolic Reasoning Under Uncertainty** Introduction To Nonmonotonic Reasoning, Logics For Non-monotonic Reasoning	10	5
7	**Statistical Reasoning** Probability And Bays' Theorem, Certainty Factors And Rule-Base Systems, Bayesian Networks, DempsterShafer Theory	10	5
8	**Prolog** Introduction, Converting English to facts and rules, Prolog terminology, Variables, Control structures, Arithmetic operators, Matching, Backtracking, Cuts	17	8
9	**Machine Learning** Introduction Features of ML,Needs of ML,Classification of ML,Applications of ML	7	3
	Total	100	48

Reference Books

1.	**Artificial Intelligence (TextBook)** By Elaine Rich and Kevin Knight | TMH
2.	**Artificial Intelligence :A Modern Approach** By Stuart J. Russell and Peter Norvig | PEARSON EDUCATION LIMITED
3.	**Artificial Intelligence and Expert Systems** By D.W.Patterson | PHI
4.	**Expert Systems Principles and Programming** By Giarratano & Riley son | Vikas Publishing House | 3rd Edition

Course Outcome
After Learning the Course the students shall be able to:
1. Summarize elements, techniques and applications of Artificial Intelligence. 2. Recognize problem characteristics and design issues in search algorithms. 3. Identify problem using suitable approach for representing knowledge into logic. 4. Describe various techniques for game playing and planning. 5. Describe essentials of natural language processing and prolog.

List of Practical	
1.	**Write a program to implement Tic Tac Toe game.**
2.	**Write a program to implement BFS 8 Puzzle problem.**
3.	**Write a program to implement DFS Water Jug Problem.**
4.	**Write a program to implement N Queens Problem.**
5.	**Write a program to implement Tower of Hanoi Problem.**
6.	**Write a prolog program for the family tree.**
7.	**Write a program to solve N-Queens problem using Prolog.**
8.	**Write a program to solve 8 puzzle problem using Prolog.**
9.	**Case Study: Recent application of Artificial Intelligence in daily life. Ex. Voice recognition, face detection etc.**

05201293 – Full Stack Web Development – I

Course: MCA (A.Y.-II) 2020
Prerequisite: knowledge of Core JAVA
Rationale : To make the students aware of the data-driven web applications, Testing, Designing, and database**.**

Course Learning Objective:

CLOBJ 1	To introduce modern Java Date & Time API and data type conversion techniques for effective application development.
CLOBJ 2	To understand JSON structure, formats, and data exchange mechanisms across different programming environments.
CLOBJ 3	To develop responsive and user-friendly web interfaces using Bootstrap components and utilities.
CLOBJ 4	To learn automated software testing concepts and tools using Selenium IDE and Selenium WebDriver.
CLOBJ 5	To gain practical knowledge of Node.js architecture, core modules, and server-side programming.
CLOBJ 6	To implement database connectivity and CRUD operations using Node.js with MySQL.

Course Learning Outcomes:

CLO 1	Apply Java Date & Time classes and perform various data type conversions in real-world applications.
CLO 2	Create and manipulate JSON objects and arrays, and implement JSON in Java, PHP, and AJAX-based applications.
CLO 3	Design responsive web pages using Bootstrap components, layouts, forms, and utilities.
CLO 4	Develop and execute automated test cases using Selenium IDE and Selenium WebDriver for web applications.
CLO 5	Build server-side applications using Node.js core modules such as File System, Events, Streams, Buffers, and Networking.
CLO 6	Perform database operations including create, read, update, and delete (CRUD) using Node.js with MySQL integration.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	1	2	-	5	20	20	20	60	30	150

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Java Date & Time** Introduction, Local date, Local Time, Local dateTime, Monthday, OffsetTime, offsetDateTime, Clock, year, yearMonth, period, Duration,	5	2
2	**Java Conversion:** String to int, int to String, String to long, long to String, String to double, double to string, String to date , date to string, char to string, string to char, string to object, object to string , int to long, long to int, int to double , double to int, char to int, int to char, string to Boolean, Boolean to string, date to timrstamp, timestamp to date, binary to decimal, decimal to binary, hex to decimal, decimal to hex, octal to decimal , decimal to octal.	10	5
3	**Json:** Introduction, json vs XMl, JSON example, JSON Object, JSON Array, JSON Comments, JSON example using php, JSON example using JAVA, JSON example using AJAX	15	7
4	**Bootstrap** Introduction, example, container, jumbotron, button, grid, table, form, alert, wells, badge & Label, panels, pagination, pager, image, progress bar, list group, Dropdown, Collapse, tabs/pills, NavBar, Input Types, Models	15	7
5	**Bootstrap utilities** border, clearfix, color Icons, image replacement, invisible content, position, screen readers, sizing, spacing, star rating	10	5
6	**Selenium** Introduction, features, limitations, selenium vs QTP, tool suite, Selenium IDE: installation, features, test case, login test, commands, creating test cases,	10	5
7	**Selenium Web Driver** selenium web driver, webdriver vs RC, web driver installation, running test case on chrome, firefox, Web Driver- drag & drop, alerts, browser commands, navigation commands, radio buttons, checkbox	15	7
8	**Node JS** Installation on windows, example, console, REPL, NPM, CL options, globals, timer, errors, DNS, Net, Buffers, Streams, File system, path, String Decoder, Query String, event	10	5
9	**Node JS with MySQL** Create connection, create database, table, insert record, update, delete, select record, select unique, drop.	10	5
	Total	100	48

Reference Books

1.	**Java Server Programming Java EE6 Black Book (TextBook)** Dreamtech Press
2.	**RESTful Java with JAX-RS 2.0** By Bill Burke | O'Reilly | 2nd Edition
3.	**Core Servlets and Java Server Pages Volume-1**

	By Mary Hall and Larry Brown | Prentice Hall | 2nd Edition
4.	**Core Servlets and Java Server Pages Volume-2** By Marty Hall, Larry Brown and Yaakov Chaikin | Prentice Hall | 2nd Edition

Course Outcome
After Learning the Course the students shall be able to:
1. design and manage data driven applications using date and time. 2. develop the applications to implement logic of hardware concepts. 3. Manage various kind of bugs and errors using testing/selenioum. 4. develop effective web application using bootstrap. 5. develop the client side logic using JS.

List of Practical	
1.	**WAP to implement Date and Time**
2.	**WAP to implement various conversion**
3.	**WAP to implemt JSON object and Array**
4.	**Installation of selenium**
5.	**WAP to implement various test cases.**
6.	**WAP to implement various Web Driver.**
7.	**WAP to implement String Decoder**
8.	**WAP to implement Query String**
9.	**WAP to implement events.**
10.	**CRUD Operation in Node JS with Mongo DB**

05201289 – Cloud Computing – I

Course: MCA (A.Y.-II) 2020

Prerequisite: Fundamental knowledge of Networking and Computing, Basic Knowledge of Web Services

Rationale : The key objectives of this course are to provide an understanding of the basic concepts of parallel and distributed computing and their role in cloud computing, to study the concept of virtualization and relevant technologies available in the market, to understand the importance of cloud computing for higher throughput, to make aware about availability of various cloud platforms, to study different application of cloud and cloud management techniques.

Course Learning Objective:

CLOBJ 1	To introduce modern Java Date & Time API and data type conversion techniques for effective application development.
CLOBJ 2	To understand JSON structure, formats, and data exchange mechanisms across different programming environments.
CLOBJ 3	To develop responsive and user-friendly web interfaces using Bootstrap components and utilities.
CLOBJ 4	To learn automated software testing concepts and tools using Selenium IDE and Selenium WebDriver.
CLOBJ 5	To gain practical knowledge of Node.js architecture, core modules, and server-side programming.
CLOBJ 6	To implement database connectivity and CRUD operations using Node.js with MySQL.

Course Learning Outcomes:

CLO 1	Apply Java Date & Time classes and perform various data type conversions in real-world applications.
CLO 2	Create and manipulate JSON objects and arrays, and implement JSON in Java, PHP, and AJAX-based applications.
CLO 3	Design responsive web pages using Bootstrap components, layouts, forms, and utilities.
CLO 4	Develop and execute automated test cases using Selenium IDE and Selenium WebDriver for web applications.
CLO 5	Build server-side applications using Node.js core modules such as File System, Events, Streams, Buffers, and Networking.
CLO 6	Perform database operations including create, read, update, and delete (CRUD) using Node.js with MySQL integration.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	1	2	-	5	20	20	20	60	30	150

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Introduction** Cloud computing introduction, history, characteristics, Pros and cons of cloud computing, Nature of cloud, Technologies in cloud computing, Migrating into the cloud, Cloud applications Working of cloud computing, Trends in cloud computing.	15	8
2	**Cloud Computing Models** Types of Cloud, Cloud service models, Cloud deployment models, Cloud Service models - Software as a Service (SaaS), Platform as a Service (PaaS), Infrastructure as a Service (IaaS), Other cloud services, Cloud architecture, Cloud computing reference model.	15	8
3	**Virtualization Concepts** Virtualization - definition, architecture, software, applications, Virtual clustering, Anatomy of cloud infrastructure, Virtual infrastructures, CPU virtualization, Network and storage virtualization.	15	8
4	**Cloud Storage** Data Storage - Introduction to cloud data storage, Storage options in Cloud, Structured and Unstructured Storage in the Cloud, Exploring and Cloud Storage Services, data storage management, Cloud data stores, Provisioning cloud storage, Data-intensive technologies for cloud computing, Cloud storage from LANs to WANs - cloud characteristic, distributed data storage, applications utilizing cloud storage.	15	8
5	**Cloud Risks and Cloud Security** Risks in cloud computing - introduction, risk management, cloud impact, enterprise wide risk management, types of risks in cloud. Data security in cloud - Introduction to Cloud Security, digital persona and data security, content level security, Understanding the shared security model Cloud security services, Understand authentication and security authorization, challenges in the cloud, Secure cloud - software requirements, software testing.	20	8
6	**Google Cloud Platform (GCP)** The GCP console, Understanding GCP Projects, Install and Configure Cloud SDK, Use Cloud Shell, GCP APIs, Cloud Console Mobile App, Use GCP to build your Apps: Compute Options in the Cloud Exploring IaaS with Compute Engine [With Lab], Configuring Elastic Apps with Autoscaling.	20	8
	Total	100	48

Reference Books

1.	**Cloud Computing - A practical approach for learning and implementation** By A.Srinivasan and J.Suresh | Pearson Publications
2.	**Cloud Computing: A practical approach** By Anthony T. Vetle | TMH
3.	**Cloud Computing For Dummies** By Judith Hurwitz, Robin Bloor, Marcia Kaufman, Fern Halper | Wiley India Pvt Ltd
4.	**Cloud Computing: SaaS, PaaS, IaaS, Virtualization, Business Models, Mobile, Security and More** By Kris Jamsa | Jones & Bartlett Learning
5.	**Cloud Computing Bible** By Barrie Sosinsky | Wiley India Pvt Ltd

Course Outcome

After Learning the Course the students shall be able to:
• define cloud computing and its applications. • describe role of virtualization in establishing cloud. • understand principles and service models of cloud computing. • familiarize with various cloud computing platforms. • conceptualize cloud storage management and data security. • identify feasible cloud platforms for solutions to be provided. • deploy services and resources on cloud platforms.

List of Practical

1.	**Cloud Storage: Sign Up for a free account on two or more providers of your choice. Store any 10 files that are larger than 10 MB each in at least two providers of your choice.**
2.	**Cloud Models: Create or Sign Up for PaaS, Saas and IaaS services on platform of your choice and upload at least one of your resource or application on each platform.**
3.	**Para-Virtualization: Implement one VMs in LINUX/Windows using VM software on your local machine.**
4.	**Start a Windows Virtual Machine on Google Compute Engine.**
5.	**Start a LINUX Virtual Machine on Google Compute Engine.**
6.	**Build a Docker Application on GCP. You can use Git Repository to build.**
7.	**Develop, Build and Deploy a container Application On Google Compute Engine.**
8.	**Explore GCP. Perform at least one Case Study on GCP services and prepare detailed report on your Case Study.**

05201288 – Gaming Technology – I

Course: MCA (A.Y.-II) 2020
Prerequisite: Basic knowledge of Computer, IT
Rationale : The subject will provide awareness about the Gaming technology.

Course Learning Objective:

CLOBJ 1	Understand Artificial Intelligence fundamentals and apply problem-solving approaches such as heuristic search, simulated annealing, and game-based decision making.
CLOBJ 2	Develop strong foundations in game design, including design theory, MDA framework, player experience, gameplay mechanics, and visual style.
CLOBJ 3	Apply professional graphic design techniques using GIMP/Photoshop for asset creation, editing, masking, color correction, and optimization for games.
CLOBJ 4	Gain proficiency in Unity game engine and C# programming to build interactive 2D games with physics, UI, animations, and game logic.
CLOBJ 5	Design, document, and manage complete game projects using standard game design documentation and production workflows.
CLOBJ 6	Implement audio systems and multiplayer features including sound design, networking, data persistence, and real-time communication in Unity games.

Course Learning Outcomes:

CLO 1	Apply AI problem-solving techniques such as heuristic search, IDA*, simulated annealing, and action-state modeling in game and software scenarios.
CLO 2	Design engaging game concepts and mechanics using MDA framework, game design theory, and structured idea development processes.
CLO 3	Create and optimize digital game assets using GIMP/Photoshop with layers, masks, smart objects, filters, text, and batch processing.
CLO 4	Develop functional 2D games in Unity using C#, implementing sprites, animations, physics, UI systems, and scene management.
CLO 5	Prepare professional game design documentation, including GDD, LDD, TDD, production documents, and audio design plans.
CLO 6	Build multiplayer-enabled games with networking, data persistence, server-client communication, and integrated audio systems using Audacity and Unity.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	1	2	-	5	20	20	20	60	30	150

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Introduction to Artificial Intelligence** Definition, AI approaches : heuristic approach, Simulated annealing,IDA, measuring problem solving approach, choosing state and action approach , Generating action sequence approach.	10	5
2	**Game design** Principles of game design,Game Design Theory,MDA,8 type of Fun in Game,Visual style, Gameplay	5	2
3	**Generate ideas for a game concept** Idea Development Process, Stimulus, Genre Market Research, Target platform,Creating physical Games: Board Game, Card Game, Party Games and etc	5	3
4	**Gimp / Photoshop Basics** Mastering the effects of the clone and healing brush tools , working with Layers and the Adjustments Panel, basics of Masking,Transforming and maximizing Smart Objects, Employing Smart Filters to create interesting effects, Color correction, Working with text and vector shapes in Gimp, File formats, resizing, and saving, Actions and Batch Processing	10	5
5	**Introduction to Unity Game Engine** Intro to Tools & navigation, Terrain system in Unity, Camera control in Unity, Scene Navigation, Project setting / Player setting, Game publishing using Unity	10	5
6	**Intro to C# programming in Unity** Constants and variables, Integers, Floats and Strings, Arrays and Lists, Arithmetical operators, Using if statements, Writing while statements, Writing for statements, & all Other Basic C# Concept in Unity	15	7
7	**Unity Game Engine for Developing 2D Games** Intro to 2D Game system in unity, Sprite Editor in Unity, Sprite Animation in Unity, 2D Physics in Unity, 2D Components, UI system in Unity, 2D Game Project	15	6
8	**Prepare game design documentation** Purpose of design documents Document layout Level Desing Document Technical Desing Document ProductionDocument,ADR, sound effects, foley, Ambience recording, sampling, synthesis	10	5
9	**Audacity software** Recording & editing sounds, creating a flawless loop, laying sounds on time line, mixing sounds together, inventing new sounds using sampling, inventing new sounds using synthesis	10	5
10	**Multiplayer Programming** Carry forward data to one scene to another, Using Singleton method to save the data,saving Game data to local server while exiting , how to retrieve data from local server,Implement server creation & joining an existing host, How to find active host , how objects can be created on the network, Network communication, Remote Procedure Calls, Creating multiplayer Game using Unity	10	5
	Total	100	48

Reference Books	
1.	**The Art of Game Design: A Book of Lenses, Second Edition** By Jesse Schell | A K Peters/CRC Press | 2nd, Pub. Year 2014
2.	**Getting Started with Unity 5 From Zero to Proficiency**
3.	**Learning C# by Developing Games with Unity 3D Beginner's Guide** By Norton Terry | Packt Publishing Limited
4.	**Game Sound Games As A Service: How Free to Play Design Can Make Better Games**
5.	**Unity Multiplayer Games**

Course Outcome
After Learning the Course the students shall be able to:
1. With the bachelor's degree in Game Design you have many opportunities to enter the professional life of game designer, system designer, level designer, narrative designer, game writer, game artist or gaming programmer. 2. Become a creative director of your own studio or enter the professional fields of game journalism and game research.

List of Practical	
1.	**Design a visual style for game**
2.	**Create a Board Game, Card Game, Party Games**
3.	**Mastering the effects of the clone and healing brush tools**
4.	**Resizing, and saving, Actions and Batch Processing**
5.	**Camera control in Unity**
6.	**Project setting / Player setting for game**
7.	**Game publishing using Unity**
8.	**2D Game Project**

MSc in Data Science Syllabus

05203201 – Data Analysis and Statistical Modelling

Course: M.Sc. (Data Science)

Prerequisite: Basic knowledge of Statistics (descriptive statistics, probability, random variables), Fundamentals of Mathematics (linear algebra and calculus). Basic programming skills (Python/R)

Rationale : - This course provides students with a strong foundation in data analysis and statistical modelling, which are essential for extracting meaningful insights from real-world data. It focuses on developing the ability to explore, visualize, and summarize data, followed by building and evaluating statistical models for prediction and inference.

Course Learning Objective:

CLOBJ 1	Understand Artificial Intelligence fundamentals and apply problem-solving approaches such as heuristic search, simulated annealing, and game-based decision making.
CLOBJ 2	Develop strong foundations in game design, including design theory, MDA framework, player experience, gameplay mechanics, and visual style.
CLOBJ 3	Apply professional graphic design techniques using GIMP/Photoshop for asset creation, editing, masking, color correction, and optimization for games.
CLOBJ 4	Gain proficiency in Unity game engine and C# programming to build interactive 2D games with physics, UI, animations, and game logic.
CLOBJ 5	Design, document, and manage complete game projects using standard game design documentation and production workflows.
CLOBJ 6	Implement audio systems and multiplayer features including sound design, networking, data persistence, and real-time communication in Unity games.

Course Learning Outcomes:

CLO 1	Apply AI problem-solving techniques such as heuristic search, IDA*, simulated annealing, and action-state modeling in game and software scenarios.
CLO 2	Design engaging game concepts and mechanics using MDA framework, game design theory, and structured idea development processes.
CLO 3	Create and optimize digital game assets using GIMP/Photoshop with layers, masks, smart objects, filters, text, and batch processing.
CLO 4	Develop functional 2D games in Unity using C#, implementing sprites, animations, physics, UI systems, and scene management.
CLO 5	Prepare professional game design documentation, including GDD, LDD, TDD, production documents, and audio design plans.

CLO 6	Build multiplayer-enabled games with networking, data persistence, server-client communication, and integrated audio systems using Audacity and Unity.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	-	-	-	3	20	20	-	60	-	100

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Unit- I:Foundational Statistics:** **Descriptive Statistics:** Summarizing data using measures like mean, median, mode, and standard deviation. **Inferential Statistics:** Making inferences about populations based on sample data, including hypothesis testing and confidence intervals. **Probability and Distributions:** Understanding probability, different probability distributions (e.g., normal, binomial), and their applications. **Sampling Methods:** Learning about different sampling techniques and their implications for data analysis.	**23**	**10**
2	**Unit-II:Data Analysis Techniques:** Regression Analysis: Exploring the relationships between variables using linear and non-linear regression models. **Time Series Analysis:** Analyzing data that changes over time, including forecasting techniques. **Multivariate Analysis:** Analyzing multiple variables simultaneously, including techniques like Principal Component Analysis (PCA) and Factor Analysis. **Exploratory Data Analysis (EDA)**: Using various methods, including visualization, to understand and summarize data.	**26**	**12**
3	**Unit-III:Statistical Modelling:** **Model Building:** Learning about different types of statistical models, including linear models, logistic regression, and more advanced models. **Model Evaluation:** Assessing the performance of models using various metrics, including R-squared, AIC, and BIC. **Model Selection:** Choosing the best model for a given dataset and problem.	**26**	**12**
4	**Unit-IV:Software and Tools:** **R:** Basic of R Programming language and its usage in statistical computing. **Python:** Basic of Python, Libraries like Pandas and Scikit-learn for data analysis and machine learning. **SQL:** Basic of SQL, Use of SQL to manage and manipulate databases. **Data Visualization Tools:** Introduction to Tableau, Power BI, and SPSS	**25**	**11**
	Total	**100**	**45**

05203202 – Data Mining and Machine Learning

Course: M.Sc. (Data Science)

Prerequisite: Students should have basic knowledge of statistics and probability, fundamental concepts of linear algebra and calculus, and programming skills in Python or R. Prior exposure to data analysis or database systems is recommended.

Rationale : This course is designed to provide students with a comprehensive understanding of data mining and machine learning techniques used to extract meaningful patterns and knowledge from large and complex datasets. It covers essential methods such as classification, regression, clustering, association rule mining, and dimensionality reduction, along with feature engineering and model evaluation. The course emphasizes both theoretical foundations and practical implementation through real-world datasets and case studies. It also introduces ethical considerations including data privacy, bias, and model interpretability. By completing this course, students will be equipped to develop, evaluate, and deploy machine learning models for applications in domains such as business analytics, healthcare, finance, cybersecurity, and scientific research, thereby strengthening their analytical and problem-solving capabilities for advanced studies and professional careers in data science.

Course Learning Objective:

CLOBJ 1	Understand fundamentals of data mining including goals, processes, tasks, knowledge representation, and its relationship with data warehousing and related technologies such as machine learning, DBMS, OLAP, and statistics.
CLOBJ 2	Study and apply core data mining algorithms such as association rule mining, classification, prediction, decision trees, Bayesian methods, and instance-based learning using tools like WEKA.
CLOBJ 3	Develop foundational knowledge of machine learning concepts including supervised, unsupervised, and reinforcement learning, generalization, overfitting, and the mathematical principles behind learning models
CLOBJ 4	Gain practical understanding of artificial neural networks and deep learning including perceptrons, multilayer networks, backpropagation, CNNs, RNNs, transformers, and optimization techniques.
CLOBJ 5	Learn data handling and preprocessing techniques such as data cleaning, feature engineering, transformation, and dataset partitioning for training, validation, and testing.
CLOBJ 6	Acquire skills in model preparation, evaluation, and optimization including model selection, interpretability, performance metrics, cross-validation, and hyperparameter tuning for building robust predictive systems.

Course Learning Outcomes:

CLO 1	**Explain data mining concepts and tasks** and describe how knowledge is represented, evaluated, and visualized in real-world applications.
CLO 2	**Implement and analyze data mining algorithms** for association rule discovery, classification, and prediction using suitable datasets and tools such as WEKA.
CLO 3	**Apply machine learning techniques** to solve practical problems while addressing issues of overfitting, generalization, and model complexity.
CLO 4	**Design and train neural network models** and demonstrate understanding of deep learning architectures, NLP basics, and

	reinforcement learning concepts.
CLO 5	**Prepare datasets and evaluate models effectively** using appropriate preprocessing methods, validation strategies, performance measures, and optimization techniques to improve model accuracy and reliability.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	-	-	-	3	20	20	-	60	-	100

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Unit- I: Introduction to Data Mining :** What is data mining? Related technologies - Machine Learning, DBMS, OLAP, Statistics, Data Mining Goals, Stages of the Data Mining Process, Data Mining Techniques, Knowledge Representation Methods, Applications, Example: weather data Data mining relationship with data warehousing, why data mining is a discipline? Overview of data mining tasks: Clustering, Classifications, Rules learning etc. **Data mining knowledge representation:** Task relevant data, Background knowledge, Interestingness measures, Representing input data and output knowledge, Visualization techniques, Experiments with Weka – visualization	**20**	**9**
2	**Unit-II:Data mining algorithms:** Association rules, Motivation and terminology, Example: mining weather data, **Basic idea:** item sets, Generating item sets and rules efficiently Correlation analysis, Experiments with Weka - mining association rules **Data mining algorithms:** Classification, Basic learning/mining tasks, **Inferring rudimentary rules:** 1R algorithm, Decision trees Covering rules, Experiments with Weka - decision trees, rules **Data mining algorithms:** Prediction, The prediction task, Statistical (Bayesian) classification, Bayesian networks, Instance- based methods (nearest neighbor), Linear models, Experiments with Weka - Prediction	**20**	**9**
3	**Unit-III:Foundational Concepts:** Introduction to machine learning, different types of learning (supervised, unsupervised, reinforcement), and the role of mathematics (linear algebra, probability, statistics), Role of machine learning in computer science and problem solving **Supervised Learning:** Different algorithms- Linear regression, logistic regression, decision trees, and ensemble methods (bagging, boosting). **Unsupervised Learning:** Different techniques- clustering (K-means, hierarchical), and dimensionality reduction (PCA).Notion of Generalization and concern of Overfitting	**20**	**9**
4	**Unit-IV: Artificial Neural Network:** Introduces neural networks, Single Layer Neural Networks, Multi-layer perceptron, biological neuron, Artificial neuron, Activation functions, Architectures of neural network, Perceptron, Learning process in ANN, Role of Loss Functions and	**20**	**9**

	Optimization, Gradient Descent and Perceptron/Delta Learning, Back propagation. **Advance topics in Machine Learning (Deep Learning):** Convolutional neural networks (CNNs), Recurrent neural networks (RNNs), Advances in Backpropagation and Optimization for Neural Networks Adversarial Learning, Recurrent Neural Networks, Transformer **Natural Language Processing (NLP):** Focuses on text analytics, sentiment analysis, and word embeddings. Reinforcement Learning: Explores concepts like Markov Decision Processes and Q-learning. **Data Handling and Preprocessing:** Covers data cleaning, feature engineering, and data transformation techniques. Notion of Training, Validation and Testing; Connect to generalization and overfitting.		
5	**Unit-V:**Model Preparation, Evaluation and feature engineering: Types of data in machine learning, Exploring structure of data, Data pre-processing, Model selection and training (for supervised learning), Model representation and interpretability, **Model Evaluation and Optimization:** Includes methods for evaluating model performance, cross-validation, and hyperparameter tuning. Evaluating machine learning algorithms and performance enhancement of models	**20**	**9**
	Total	**100**	**45**

05203205 – Python and R Programming

Course: M.Sc. (Data Science)

Prerequisite: Students should have basic knowledge of computer programming concepts such as variables, control structures, functions, and data types. Familiarity with mathematics and statistics is beneficial but not mandatory.

Rationale : This course aims to develop strong programming proficiency in Python and R, the two most widely used languages in data science and statistical computing. It introduces students to core programming constructs, data structures, and libraries required for data manipulation, visualization, statistical analysis, and machine learning. The course emphasizes hands-on practice using real-world datasets to build skills in writing efficient code, automating data analysis workflows, and implementing analytical models. By learning both Python and R, students gain flexibility in selecting appropriate tools for different data science tasks, including data preprocessing, exploratory analysis, modeling, and result interpretation. This course forms a foundational pillar for advanced subjects such as data mining, machine learning, big data analytics, and research-oriented data analysis.

Course Learning Objective:

CLOBJ 1	To introduce students to the fundamentals of Python and R programming languages for data analysis tasks.
CLOBJ 2	To develop proficiency in using core programming constructs such as data types, control structures, functions, and file handling.
CLOBJ 3	To enable students to work effectively with built-in and advanced data structures in Python and R.
CLOBJ 4	To provide practical knowledge of data manipulation using Pandas, NumPy, and Tidyverse/dplyr libraries.
CLOBJ 5	To train students in data visualization techniques using Matplotlib, Seaborn, and ggplot2.
CLOBJ 6	To build foundational understanding of statistical analysis and machine learning methods using Python (scikit-learn) and R (caret).

Course Learning Outcomes:

CLO 1	Apply Python and R syntax, variables, operators, and control structures to solve basic programming problems.
CLO 2	Implement functions, modules, packages, and file handling mechanisms for real-world data processing tasks.
CLO 3	Use appropriate data structures (lists, tuples, dictionaries, sets, vectors, matrices, data frames, etc.) to store and manage data efficiently.
CLO 4	Perform data cleaning, transformation, and analysis using Pandas, NumPy, and Tidyverse/dplyr libraries.
CLO 5	Create meaningful data visualizations and graphical reports using Matplotlib, Seaborn, and ggplot2.
CLO 6	Develop and evaluate basic machine learning models for regression, classification, and clustering using scikit-learn and caret.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	-	4	-	5	40		60	60	-	100

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Python-I Fundamentals**Installation, syntax, variables, data types, operators, control structures (if, while, for), functions, modules, packages, file handling, error handling. Data Structures: Lists, tuples, dictionaries, sets. Data Manipulation: Libraries like Pandas for data manipulation and analysis, NumPy for numerical computing	26	12
2	**Python-II Data Visualization:**Libraries like Matplotlib, Seaborn for creating charts and graphs. Data Analysis and Machine Learning: Basic statistical concepts, linear regression, classification, clustering, and advanced techniques using libraries like scikit-learn. Data Import and Export: Reading and writing data from various formats (CSV, Excel, databases).	22	10
3	**R-I Fundamentals**Installation, syntax, data types, operators, control structures, functions, packages. Data Structures: Vectors, lists, matrices, arrays, data frames, factors. Data Manipulation: Tidy verse packages for data wrangling and cleaning, dplyr for data manipulation.	26	11
4	**R- II Data Visualization**ggplot2 for creating publication-quality graphics. Data Analysis and Statistics: Descriptive statistics, hypothesis testing, regression, ANOVA, and other statistical methods. Machine Learning: Using packages like caret for building machine learning models.	26	12
	Total	100	45

List of Practical

1.	**Installation of Python environment (e.g., Anaconda), basic syntax, variables, data types, and operators.**
2.	**Implementing conditional statements (if-else, elif) and loops (for, while) to solve problems like finding GCD, Fibonacci series, or prime numbers.**
3.	**Creating and manipulating lists, tuples, sets, and dictionaries (add, delete, slice, index elements).**
4.	**Defining and calling user-defined functions, using arguments and return values, and importing standard modules.**
5.	**Performing string manipulations (slicing, concatenation) and reading from/writing to text and CSV files.**
6.	**Implementing classes and objects for real-world scenarios (e.g., student information system, employee payroll).**
7.	**Data Analysis Libraries- Using NumPy for array manipulation and Pandas for data loading, cleaning, preparation, and manipulation.**
8.	**Data Visualization- Creating basic plots and graphs using libraries like Matplotlib.**
9.	**Setting up R and RStudio, understanding R sessions, basic syntax, variables, and data types**
10.	**Implementing control statements (if, else, switch) and user-defined functions (mean, sum, min, max, etc.).**
11.	**Working with vectors, lists, matrices, arrays, factors, and data frames (accessing elements, adding/deleting elements, merging).**

12.	**Loading, handling, and exploring data from various file formats (CSV, Excel).**
13.	**Using packages like dplyr for data transformation and tidyr for tidying data.**
14.	**Calculating descriptive statistics such as mean, median, mode, standard deviation, and correlation.**
15.	**Calculating descriptive statistics such as mean, median, mode, standard deviation, and correlation.**
16.	**Write a program to read a csv file and analyze the data in the file in R**
17.	**Create pie charts and bar charts using R**

05203187 – Data Analytics and Visualization

Course: M.Sc. (Data Science)

Prerequisite: Students should have basic knowledge of statistics, including descriptive measures and probability concepts, and fundamental programming skills in Python or R. Familiarity with spreadsheets or databases is an added advantage.

Rationale : This course focuses on developing the ability to analyze, interpret, and communicate data effectively through analytical techniques and visual representations. It introduces students to data preprocessing, exploratory data analysis, and analytical methods for identifying trends, patterns, and relationships within datasets. The course emphasizes the design of meaningful and interactive visualizations using modern tools and libraries to support clear storytelling and informed decision-making. By combining analytical reasoning with visualization principles, students learn to transform raw data into actionable insights that can be understood by both technical and non-technical audiences. This course prepares students for advanced work in data science, business intelligence, and research, where effective data communication is as critical as accurate analysis.

Course Learning Objective:

CLOBJ 1	To introduce students to the complete **Data Analytics Lifecycle** and the roles involved in successful analytics projects.
CLOBJ 2	To provide a strong foundation in **data mining algorithms**, regression techniques, time series analysis, and text analytics.
CLOBJ 3	To develop the ability to perform **data preparation, exploration, modeling, and evaluation** using structured methodologies.
CLOBJ 4	To familiarize students with **statistical analysis and visualization using R** for exploratory data analysis.
CLOBJ 5	To train students in using **Python libraries (Pandas, NumPy, SciPy, Matplotlib, Seaborn)** for data analytics and visualization.
CLOBJ 6	To enable students to **communicate analytical results effectively** and deploy models in real-world operational environments.

Course Learning Outcomes:

CLO 1	Explain and apply the phases of the **Data Analytics Lifecycle** to define business problems and identify suitable data sources.
CLO 2	Implement **data preparation and conditioning techniques**, including ETLT, data cleaning, visualization, and feature selection.
CLO 3	Apply **data mining and predictive modeling algorithms** such as association rules, linear regression, logistic regression, and ARIMA models.
CLO 4	Perform **text analytics tasks** including text representation, TF-IDF computation, topic classification, and sentiment analysis.
CLO 5	Conduct **exploratory data analysis and visualization using R**, and interpret statistical summaries and graphical outputs.
CLO 6	Develop end-to-end analytics solutions using **Python data libraries and visualization tools**, and present insights for decision-making.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	-	4	-	5	40		60	60	-	100

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Unit- I: Data Analytics Lifecycle overview**Key Roles for a Successful Analytics, Background and Overview of Data Analytics Lifecycle Project Phase 1: Discovery: Learning the Business Domain, Resources Framing the Problem, Identifying Key Stakeholders. Interviewing the Analytics Sponsor, Developing Initial Hypotheses Identifying Potential Data Sources Phase 2: Data Preparation: Preparing the Analytic Sandbox, Performing ETLT, Learning About the Data, Data Conditioning, Survey and visualize, Common Tools for the Data Preparation Phase Phase 3: Model Planning: Data Exploration and Variable Selection, Model Selection, Common Tools for the Model Planning Phase Phase 4: Model Building: Common Tools for the Model Building Phase Phase 5: Communicate Results Phase 6: Operationalize	**22**	**10**
2	**Unit-II: Data mining algorithms** Association rules, Motivation and terminology, Example: mining weather data, Regression Models **Introduction to simple Linear Regression:** The Regression Equation, Fitted value and Residuals, Least Square, Introduction to **Multiple Linear Regression:** Assessing the Model, Cross-Validation, Model Selection and Stepwise Regression, Prediction Using Regression **Logistic Regression:** Logistic Response function and logit, Logistic Regression and GLM, Generalized Linear model, Predicted values from Logistic Regression, Interpreting the coefficients and odds ratios, Linear and Logistic Regression: similarities and Differences, Assessing the models. **Time Series:** Overview of Time Series Analysis Box-Jenkins Methodology, ARIMA Model Autocorrelation Function (ACF) , Autoregressive Models, Moving Average Models, ARMA and ARIMA Models, Building and Evaluating an ARIMA Model, Reasons to Choose and Cautions. **Text Analytics:** History of text mining, Roots of text mining overview of seven practices of text analytic, Application and use cases for Text mining: extracting meaning from unstructured text, Summarizing Text. Text Analysis Steps, A Text Analysis Example, Collecting Raw Text, Representing Text, Term Frequency—Inverse Document Frequency (TFIDF), Categorizing Documents by Topics, Determining Sentiments , Gaining Insights.	**32**	**15**
3	**Unit-III: Data analytics and visualization with R** Introduction to R: Data Import and Export, Attribute and Data type, Descriptive statistics. Exploratory Data Analysis:	**23**	**10**

	Visualization before analysis, Dirty Data, visualizing single variable, examining Multiple variables, Data Exploration versus presentation.		
4	**Unit-IV: Data analytics and Visualization with Python** Essential Data Libraries for data analytics: Pandas, NumPy, SciPy. Plotting and visualization with python: Introduction to Matplotlib, Basic Plotting with Matplotlib, Create Histogram, BarChart, Pie chart, Box Plot, violin plot using Matplotlib. Introduction to seaborn Library, Multiple Plots, Regression plot, regplot.	**23**	**10**
	Total	**100**	**45**

05203188– Predictive Analytics/ Spatial Modelling

Course: M.Sc. (Data Science)

Prerequisite: Basic knowledge of statistics and probability, linear algebra, and programming in Python or R.

Rationale : This course introduces students to predictive analytics and spatial modelling techniques for analyzing historical and geospatial data to forecast outcomes and identify location-based patterns. It focuses on building and interpreting predictive models using statistical and machine learning methods, enabling data-driven decision-making in areas such as business forecasting, urban planning, environmental analysis, and public policy.

Course Learning Objective:

CLOBJ 1	To introduce students to the fundamentals of predictive analytics, business modeling, and regression techniques for data-driven decision making.
CLOBJ 2	To develop practical understanding of data preparation, exploration, and feature engineering methods for building reliable analytical models.
CLOBJ 3	To familiarize learners with classification, ensemble, time series, clustering, and recommendation models used in real-world business applications.
CLOBJ 4	To provide conceptual and practical knowledge of segmentation techniques using supervised and unsupervised learning approaches.
CLOBJ 5	To introduce foundational concepts of Geographic Information Systems (GIS), spatial data models, cartography, and geospatial database management.
CLOBJ 6	To train students in advanced spatial analysis, modeling, and simulation techniques for environmental, urban, and business problem solving.

Course Learning Outcomes:

CLO 1	Explain core concepts of analytics, business modeling, data types, regression theory, and model assumptions used in predictive analytics.
CLO 2	Prepare and explore datasets using cleaning, imputation, outlier handling, visualization, and summary statistics to support model development.
CLO 3	Apply data mining and predictive modeling algorithms such as association rules, linear regression, logistic regression, and ARIMA models.
CLO 4	Build and evaluate predictive models using linear regression, logistic regression, decision trees, SVM, Naive Bayes, neural networks, and ensemble methods.
CLO 5	Apply segmentation techniques using clustering and decision tree models while addressing overfitting, pruning, and model complexity.

CLO 6	Build and evaluate predictive models using linear regression, logistic regression, decision trees, SVM, Naive Bayes, neural networks, and ensemble methods.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	-	4	-	5	40		60	60	-	100

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Unit- I: Introduction to Predictive Analytics & Linear Regression:** What and Why Analytics, Introduction to Tools and Environment, Application of Modeling in Business, Databases & Types of data and variables, Data Modeling Techniques, Missing imputations etc. Need for Business Modeling, Regression — Concepts, Blue property-assumptions-Least Square Estimation, Variable Rationalization, and Model Building etc. Regression: Model Theory, Model fit Statistics, Model Conclusion, Analytics applications to various Business Domains etc. Data preparation and exploration: Cleaning, wrangling, and visualizing data, dealing with missing values and outliers, and using summary statistics. Simple and multiple linear regression, assessing model fit, and using cross-validation. Classification: Logistic regression, support vector machines (SVM), decision trees, and Naive Bayes classifiers. Advanced and ensemble methods: Neural networks, boosting, bagging, and model stacking. Other models: Time series analysis, clustering, and recommendation systems. Model evaluation: Confusion matrices, performance metrics, and comparing models.	24	11
2	**Unit-II: Objective Segmentation:** Regression Vs Segmentation — Supervised and Unsupervised Learning, Tree Building — Regression, Classification, over fitting, Pruning and complexity, Multiple Decision Trees etc. Develop Knowledge, Skill and Competences, Introduction to Knowledge skills & competences, Training & Development, Learning & Development, Policies and Record keeping. etc.	20	8
3	**Unit-III: Foundational concepts:** : History and theory of spatial analysis, archaeological data, and spatial legacy data. Introduction to GIS and spatial data: Data types, data models (vector/raster), coordinate systems, and projections. **Geospatial data input and editing:** Data creation, digitization, attribute linking, and database management. **Cartography and visualization**: Basic cartographic principles and map design. **GIS software:** Familiarity with software such as ArcGIS, including its tools and functions. **Spatial data and visualization**: Spatial legacy data, data visualization, and time and space in analysis. **Geographical modeling:**Settlement geography, settlement and environment, and historical landscape analysis. **Modeling techniques:** Spatial agent-based modeling, spatial networks, and spatial interaction modeling.	28	13

	Spatial statistics: Analyzing spatial point patterns and processes, and using space syntax. **Data-driven spatial analysis:** Spatial statistics and data analysis in general.		
4	**Unit-IV: Core spatial analysis techniques--Queries and reasoning:** Spatial and attribute queries to answer simple questions. **Measurements:** Calculating distance, density, and neighborhood statistics. **Transformations:** Data transformation and manipulation, including vector overlays and raster calculations. **Spatial autocorrelation:** Understanding and measuring spatial relationships using statistics like Moran's I. **Point pattern analysis:** Analyzing the distribution of point data. Advanced modeling and analysis **Surface analysis:** Analyzing elevation data using tools like TIN, slope, aspect, and watershed modeling. **Network analysis:** Performing routing and allocation analyses in networks. **Suitability and multicriteria analysis:** Combining multiple criteria to evaluate sites for a specific purpose. **Simulation and modeling**: Building models to simulate spatial processes and project future changes. **Hydrological and environmental modeling:** Specific applications of modeling in environmental and ecological contexts.	**28**	**13**
	Total	**100**	**45**

05203189 – Cloud Native Development

Course: M.Sc. (Data Science)
Prerequisite: Basic knowledge of programming, operating systems, and computer networks. Familiarity with web technologies is helpful.
Rationale : This course introduces cloud-native concepts and tools for building scalable and reliable applications using microservices, containers, and cloud platforms. It enables students to develop and deploy data-driven applications suitable for modern data science and big-data environments.

Course Learning Objective:

CLOBJ 1	To introduce students to the fundamentals of predictive analytics, business modeling, and regression techniques for data-driven decision making.
CLOBJ 2	To develop practical understanding of data preparation, exploration, and feature engineering methods for building reliable analytical models.
CLOBJ 3	To familiarize learners with classification, ensemble, time series, clustering, and recommendation models used in real-world business applications.
CLOBJ 4	To provide conceptual and practical knowledge of segmentation techniques using supervised and unsupervised learning approaches.
CLOBJ 5	To introduce foundational concepts of Geographic Information Systems (GIS), spatial data models, cartography, and geospatial database management.
CLOBJ 6	To train students in advanced spatial analysis, modeling, and simulation techniques for environmental, urban, and business problem solving.

Course Learning Outcomes:

CLO 1	Explain core concepts of analytics, business modeling, data types, regression theory, and model assumptions used in predictive analytics.
CLO 2	Prepare and explore datasets using cleaning, imputation, outlier handling, visualization, and summary statistics to support model development.
CLO 3	Apply data mining and predictive modeling algorithms such as association rules, linear regression, logistic regression, and ARIMA models.
CLO 4	Build and evaluate predictive models using linear regression, logistic regression, decision trees, SVM, Naive Bayes, neural networks, and ensemble methods.
CLO 5	Apply segmentation techniques using clustering and decision tree models while addressing overfitting, pruning, and model complexity.
CLO 6	Build and evaluate predictive models using linear regression, logistic regression, decision trees, SVM, Naive Bayes, neural networks, and ensemble methods.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
3	-	4	-	5	40		60	60	-	100

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Unit- I: Cloud Deployment Models:** -Public Cloud, Private Cloud, Community Cloud, Hybrid Cloud, Multi-Cloud, Cloud Platforms AWS, Google Cloud, Azure, OpenStack Cloud Service Models: - IaaS, PaaS, SaaS, Managed/Serverless Services, IT Ops, SysOps, DevOps, Low Ops, No Ops Activity: Planning the Case Study Architecture What Is Cloud-Native? Introduction to Cloud Native: Principles, goals, and key technologies. Traditional Development vs. Cloud-Native, Agile Practices, Feature-Driven Development Value-Driven Development, Agile Teams, Writing User Stories Activity: Analyzing the Case Study Requirements, **Automation and DevOps:** Infrastructure as Code (IaC), Continuous Integration and Continuous Delivery (CI/CD) **Application Architecture:** Microservices versus monoliths, API integration, and designing for resilience. **Containerization:** Using Docker to build, run, and distribute container images. **Container Orchestration:** Deploying and managing containers at scale using Kubernetes. **DevOps and CI/CD:** Implementing continuous integration, continuous delivery, and automation to streamline the development lifecycle.	26	12
2	**Unit-II: Building and Deploying Applications Programming:** Using languages like JavaScript, Node.js, and NoSQL databases for backend development. **Cloud Platforms:** Choosing and using services from public, private, and hybrid cloud environments. **Data and Storage:** Leveraging cloud-based data storage services. **Serverless Computing:** Simplifying deployments with PaaS and serverless environments. **Application Life Cycle Management:** Package Management, Managing Application Dependencies, Maven and Gradle Pip, NPM, NuGet, Source Control, Git- asic Git Commands, GitHub-Version Control with Git on the Cloud **Activity:** Using Git with the Case Study Version Control and Collaboration Branching and Merging Strategies **Activity:** Branching and Collaboration	24	11
3	**Unit-III:Docker:** Understanding Docker, Containers, Advantages of Containers, Images, Using Docker Basic Docker Commands, Building Docker Images, Dockerfile, Starting Containers, Stopping Containers, Deleting Containers and Images **Activity:** Containerizing the Case Study **Deploying Docker Containers:** Container Registries, Push and Pull **Activity:** Managing Docker Containers in a Container Registry	25	11

	Management and Operations **Infrastructure as Code (IaC):** Automating infrastructure provisioning. **Observability:** Implementing monitoring, logging, and tracing. **Security:** Addressing security basics, compliance, and access management. **Reliability and Disaster Recovery:** Planning for fault tolerance and business continuity.		
4	**Unit-IV: Some Advanced Topics:** **Resilience and Fault Tolerance:** Building applications that can withstand failures. **AI and Machine Learning:** Integrating AI to make applications "smarter". **Observability:** Ensuring applications are easy to monitor and troubleshoot.	**25**	**11**
	Total	**100**	**45**

05203184 – Power BI for Data Analytics

Course: M.Sc. (Data Science)
Prerequisite: Some foundational knowledge in data analytics, Data Visualization Concepts and Mathematical Formulas and Functions
Rationale : Upon completion of this course, students will be able to:students will be able to:Import, transform and cleanse data using Power Query Editor, build a data model for self-service reporting and manipulate the model with DAX, publish and share visualizations

Course Learning Objective:

CLOBJ 1	To introduce students to data analysis using MS Excel tools such as Pivot Tables, Charts, Slicers, Goal Seek, Scenarios, and VLOOKUP.
CLOBJ 2	To develop skills in designing interactive dashboards in Excel and Power BI for analyzing trends and performance over time.
CLOBJ 3	To enable learners to apply Power BI visualization techniques including KPI cards, line charts, donut charts, maps, gauges, and scatter plots.
CLOBJ 4	To enhance the ability to use filters, slicers, QnA, and conditional formatting for effective user-driven analysis.
CLOBJ 5	To familiarize students with data modeling and DAX functions for computing cumulative sales, moving averages, and business metrics.
CLOBJ 6	To cultivate data storytelling and decision-support skills through real-world dataset selection and comprehensive dashboard development.

Course Learning Outcomes:

CLO 1	Create Excel-based analytical dashboards using Pivot Tables, Charts, and Slicers to evaluate sales trends over time.
CLO 2	Apply Excel what-if analysis tools (Goal Seek, Scenario Manager) and lookup functions (VLOOKUP) to support conditional business decisions.
CLO 3	Design interactive Power BI dashboards using cards, multi-row cards, KPI visuals, line charts, donut charts, and advanced formatting techniques.
CLO 4	Implement DAX functions to calculate cumulative totals, moving averages, and key performance indicators.
CLO 5	Utilize filters, slicers, maps, conditional formatting, and Top-N analysis to extract actionable insights from large datasets.
CLO 6	Develop a complete data visualization project demonstrating data cleaning, visualization design, analytical reasoning, and storytelling using Excel and Power BI.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
-	-	4	-	2	-	20	20	-	60	100

List of Practical	
1.	**Using MSEXCEL, create a dashboard to display the implementation of the Pivot Table, Slicers and Charts to show the change in trends of Sales Data Analysis with the change over time.**
2.	**Using MSEXCEL, create a Sheet to display the implementation of the Goal Seek, Scenario, and VLOOKUP to show the data as per the condition and User's Request.**
3.	**Using POWER BI, create a dashboard to display the implementation of the followings: Cards and Multi Row Cards, Building and Formatting Charts, Line Charts, KPI Cards, Basic Filtering Options**
4.	**Using POWER BI create a dashboard to display the implementation of the followings: Use of Donus and Filters in Power BI, Table & Matrix Visuals, Conditional Formatting, Top N Filters, Map Visuals, Slicers, Gauge Charts, Use of QnA Feature, Use of Gauge Charts, Waterfall Chart, Scatter Chart.**
5.	**Using POWER BI, Implement the Functions in DAX and show the Cumulative Sales and Moving Average.**
6.	**Students will choose a dataset and create a comprehensive data visualization project. The project should demonstrate their understanding of data visualization techniques and data storytelling**
7.	**Using Power BI for Creating and Formatting Visualizations**
8.	**Using Slicers and Filters to interact with your reports.**
9.	**Using Power BI to implement Conditional Formatting to highlight specific data points.**

Course Outcome
After Learning the Course the students shall be able to:
1: Do the Data Handling and Preparation. 2: Prepare the Report and Dashboard Creation. 3: Do the Data Analysis and Interpretation. 4: Do the Sharing and Collaboration of Reports and Dashboard

05203186 – Time Series Analysis

Course: M.Sc. (Data Science)
Prerequisite: Solid foundation in front-end technologies like HTML, CSS, and JavaScript, including frameworks like React or Angular
Rationale : Upon completion of this course, students will be able to:become knowledgeable about the most recent web development technologies, build Idea for creating two tier and three tier architectural web applications, Design and analyze real time web applications, Constructing suitable client and server-side applications and to learn core concept of both front end and back-end programming**.**

Course Learning Objective:

CLOBJ 1	To introduce the fundamental concepts of time series data, its components (trend, seasonal, cyclical, and random), and visualization techniques.
CLOBJ 2	To develop understanding of trend estimation methods including graphical, semi-average, moving average, and mathematical curve fitting techniques .
CLOBJ 3	To provide a strong foundation in probability, statistics, stationarity, ergodicity, and correlation structures required for time series modeling.
CLOBJ 4	To explain the identification and measurement of seasonal components using classical statistical methods.
CLOBJ 5	To familiarize students with stochastic models such as AR, MA, ARMA, ARIMA, and SARIMA for modeling time-dependent data.
CLOBJ 6	To equip students with forecasting techniques including exponential smoothing, short-term forecasting methods, and advanced volatility models such as ARCH and GARCH.

Course Learning Outcomes:

CLO 1	Explain the structure, components, and applications of time series data across various domains such as economics, finance, and engineering.
CLO 2	Estimate and analyze trends using free-hand curve fitting, semi-averages, moving averages, and growth curve models (exponential, Gompertz, logistic).
CLO 3	Assess stationarity and dependence in time series data using ACF and PACF, and apply appropriate statistical concepts.
CLO 4	Compute seasonal indices using ratio-to-trend, ratio-to-moving average, simple averages, and link relative methods.
CLO 5	Build and evaluate models including AR, MA, ARMA, ARIMA, and SARIMA for stationary and non-stationary time series.
CLO 6	Apply forecasting techniques such as exponential smoothing, short-term forecasting, and ARCH/GARCH models to generate and interpret reliable predictions.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
-	-	4	-	2	-	40	60	-	60	100

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Unit 1 . Introduction To Trend** Definition, components (trend, seasonality, cyclicity, and noise), and visualization, Introduction to times series data, application of time series from various fields, Components of a time series, Decomposition of time series. Trend: Estimation of trend by free hand curve method, method of semi averages, fitting a various mathematical curve, and growth curves. **Probability and statistics review:** Foundational concepts required for understanding statistical models. **Stationarity and ergodicity**: Key properties of time series data. **Autocorrelation functions:** Auto-correlation function (ACF) and partial auto-correlation function (PACF).		
2	**Unit 2 - Trend And Seasonal Component** Method of moving averages. Detrending. Effect of elimination of trend on other components of the time series. Seasonal Component: Estimation of seasonal component by Method of simple averages, Ratio to Trend, Ratio to moving average and Link relatives. Modelling and forecasting: **Random processes:** Linear processes, autoregressive (AR) models, moving average (MA) models, and ARMA models. **Non-stationary models:** Trend and seasonality modelling, ARIMA, and SARIMA models. **Forecasting techniques:** Exponential smoothing, short-term forecasting methods, and forecasting for trend and seasonal processes. **Advanced models:** Introduction to ARCH and GARCH models, particularly for finance applications.		
3	**Fitting and plotting of modified exponential curve**		
4	**Fitting and plotting of Gompertz curve**		
5	**Fitting and plotting of logistic curve**		
6	**Fitting of trend by Moving Average Method**		
7	**Measurement of Seasonal indices Ratio-to-Trend method**		
8	**Measurement of Seasonal indices Ratio-to-Moving Average method.**		
9	**Measurement of seasonal indices Link Relative method**		
10	**Calculation of variance of random component by variate difference method**		
11	**Forecasting by exponential smoothing**		
12	**Forecasting by short term forecasting methods**		
	Total		

List of Practical

1.	**Introduction To Trend** Definition, components (trend, seasonality, cyclicity, and noise), and visualization, Introduction to times series data, application of time series from various fields, Components of a time series, Decomposition of time series. Trend: Estimation of trend by free hand curve method, method of semi averages, fitting a various mathematical curve, and growth curves. Probability and statistics review: Foundational concepts required for understanding statistical models. Stationarity and ergodicity: Key properties of time series data.

	Autocorrelation functions: Auto-correlation function (ACF) and partial auto-correlation function (PACF).
2.	**Trend And Seasonal Component** Method of moving averages. Detrending. Effect of elimination of trend on other components of the time series. Seasonal Component: Estimation of seasonal component by Method of simple averages, Ratio to Trend, Ratio to moving average and Link relatives. Modelling and forecasting: Random processes: Linear processes, autoregressive (AR) models, moving average (MA) models, and ARMA models. Non-stationary models: Trend and seasonality modelling, ARIMA, and SARIMA models. Forecasting techniques: Exponential smoothing, short-term forecasting methods, and forecasting for trend and seasonal processes. Advanced models: Introduction to ARCH and GARCH models, particularly for finance applications.
3.	**Fitting and plotting of modified exponential curve**
4.	**Fitting and plotting of Gompertz curve**
5.	**Fitting and plotting of logistic curve**
6.	**Fitting of trend by Moving Average Method**
7.	**Measurement of Seasonal indices Ratio-to-Trend method**
8.	**Measurement of Seasonal indices Ratio-to-Moving Average method.**
9.	**Measurement of seasonal indices Link Relative method**
10.	**Calculation of variance of random component by variate difference method**
11.	**Forecasting by exponential smoothing**
12.	**Forecasting by short term forecasting methods.**

NPTEL / Swayam / MOOCs Courses List and Details

Sr. No.	Program	Semester	Course Name	NPTEL Course Title
1	MCA & MScIT All	02	MCA: Data Communication and Networking MScIT: Computer Networking and Network Securty	Computer Networks and Network Protocol By Prof. Soumya Kanti Ghosh, Prof. Sandip Chakraborty | IIT Kharagpur (12 Weeks)

All MCA & MScIT Sem-2 Students must join following MOOC Course:
Course Name: Programming, Data Structures and Algorithms using Python
Course Link: Computer Networks and Internet Protocol - Course
Offered by: NPTEL
Start Date: 20 Jan 2025
End Date: 11 Apr 2025
Duration: 12 Weeks
Exam Date: 04 May 2025 IST

Recommended Books and Course Mapping – Semester 2

05201251 (MCA) / 05202156 (MSc IT) – Advanced Java Programming

Relevant Course / Area	Book Title	Author(s)	Publisher
Advanced Java Programming	Java Server Programming Java EE 6 Black Book	Dreamtech Press	Dreamtech
Advanced Java Programming	Core Servlets and JavaServer Pages (Vol. 1)	Marty Hall, Larry Brown	Prentice Hall
Advanced Java Programming	Core Servlets and JavaServer Pages (Vol. 2)	Marty Hall et al.	Prentice Hall
Advanced Java Programming	RESTful Java with JAX-RS 2.0	Bill Burke	O'Reilly
Advanced Java Programming	Spring in Action	Craig Walls	Manning
Advanced Java Programming	Hibernate in Action	Christian Bauer, Gavin King	Manning

05202158 – Computer Networks and Network Security

Relevant Course / Area	Book Title	Author(s)	Publisher
Computer Networks	Computer Networking: A Top-Down Approach	James F. Kurose, Keith W. Ross	Pearson
Computer Networks	Computer Networks	Andrew S. Tanenbaum	Prentice Hall
Computer Networks	Computer Networks: A Systems Approach	Larry Peterson, Bruce Davie	Morgan Kaufmann
Network Security	Cryptography and Network Security	William Stallings	Pearson
Network Security	Network Security Essentials	William Stallings	Pearson

05201254 / 05202154 – Comprehensive Assessment

Relevant Course / Area	Book Title	Author(s)	Publisher
Computer Organization	Structured Computer Organization	Andrew S. Tanenbaum	PHI
Programming in C	Programming in ANSI C	E. Balagurusamy	McGraw Hill
Data Structures	Fundamentals of Data Structures	Ellis Horowitz	McGraw Hill
Algorithms	Introduction to Algorithms	Cormen et al.	PHI
Operating Systems	Operating System Concepts	Silberschatz et al.	Wiley
Software Engineering	Software Engineering	Pressman	TMH
Databases	Database System Concepts	Silberschatz et al.	McGraw Hill

05201255 / 05202155 – Seminar

Relevant Course / Area	Book Title	Author(s)	Publisher
Seminar / ICT Topics	Research Methodology	C.R. Kothari	New Age
Seminar / ICT Topics	How to Write and Publish a Scientific Paper	Robert A. Day	Cambridge Univ. Press
Seminar / ICT Topics	Technical Communication	Mike Markel	Bedford/St. Martin’s

05201256 / 05202186 – Graphics Programming

Relevant Course / Area	Book Title	Author(s)	Publisher
Computer Graphics	Computer Graphics	Donald Hearn, M. Pauline Baker	PHI
Computer Graphics	Computer Graphics: A Programming Approach	Steven Harrington	McGraw Hill
Computer Graphics	Computer Graphics with OpenGL	Hearn, Baker, Carithers	Pearson
OpenGL Programming	Learn OpenGL	Joey de Vries	Open Source
Computer Graphics	Procedural Elements for Computer Graphics	David F. Rogers	McGraw Hill

05202108 – Software Engineering with Agile

Relevant Course / Area	Book Title	Author(s)	Publisher
Software Engineering	Software Engineering: A Practitioner’s Approach	R.S. Pressman	TMH
Software Engineering	Software Engineering	Ian Sommerville	Pearson
Agile Methodology	Agile Software Development	Robert C. Martin	Pearson
Agile / Scrum	Scrum: The Art of Doing Twice the Work	Jeff Sutherland	Crown

05293252 – English and Mathematical Aptitude

Relevant Course / Area	Book Title	Author(s)	Publisher
Verbal Ability	Word Power Made Easy	Norman Lewis	Goyal
English Grammar	Understanding and Using English Grammar	Betty Azar	Pearson
Quantitative Aptitude	Quantitative Aptitude	R.S. Aggarwal	S. Chand
Programming Basics	Let Us C++	Yashavant Kanetkar	BPB

05201295 – Big Data Analytics – I

Relevant Course / Area	Book Title	Author(s)	Publisher
Big Data Analytics	Big Data, Big Analytics	Minelli et al.	Wiley
Big Data	Big Data Black Book	DT Editorial Services	Wiley
Hadoop	Hadoop: The Definitive Guide	Tom White	O'Reilly
Hadoop Administration	Hadoop Operations	Eric Sammer	O'Reilly

05201296 – Cyber Security and Forensics – I

Relevant Course / Area	Book Title	Author(s)	Publisher
Cyber Security	Cryptography and Network Security	William Stallings	Pearson
Cyber Security	Network Security Essentials	William Stallings	Pearson
Cyber Forensics	Digital Forensics and Incident Response	Jason Luttgens	Addison-Wesley
Ethical Hacking	The Web Application Hacker's Handbook	Dafydd Stuttard	Wiley
Linux Security	Kali Linux Revealed	Raphaël Hertzog	Offensive Security

05201294 – Artificial Intelligence – I

Relevant Course / Area	Book Title	Author(s)	Publisher
Artificial Intelligence	Artificial Intelligence	Elaine Rich, Kevin Knight	Tata McGraw-Hill
Artificial Intelligence	Artificial Intelligence: A Modern Approach	Stuart J. Russell, Peter Norvig	Pearson Education
Artificial Intelligence & Expert Systems	Artificial Intelligence and Expert Systems	D. W. Patterson	PHI Learning
Expert Systems / Prolog	Expert Systems: Principles and Programming	Giarratano, Riley	Vikas Publishing House

05201289 – Cloud Computing – I

Relevant Course / Area	Book Title	Author(s)	Publisher
Cloud Computing	Cloud Computing: A Practical Approach	A. Srinivasan, J. Suresh	Pearson
Cloud Computing	Cloud Computing: A Practical Approach	Anthony T. Velte	TMH
Cloud Fundamentals	Cloud Computing For Dummies	Judith Hurwitz et al.	Wiley India
Cloud Services & Security	Cloud Computing: SaaS, PaaS, IaaS	Kris Jamsa	Jones & Bartlett
Cloud Architecture	Cloud Computing Bible	Barrie Sosinsky	Wiley India

05201288 – Gaming Technology – I

Relevant Course / Area	Book Title	Author(s)	Publisher
Game Design	The Art of Game Design: A Book of Lenses	Jesse Schell	CRC Press
Unity Game Engine	Getting Started with Unity 5	Various Authors	Packt
Game Programming	Learning C# by Developing Games with Unity	Norton Terry	Packt
Game Monetization	Games as a Service	Various Authors	CRC Press
Multiplayer Games	Unity Multiplayer Games	Various Authors	Packt

05203201 – Data Analysis and Statistical Modelling (M.Sc. Data Science)

Relevant Course / Area	Book Title	Author(s)	Publisher
Statistics	Applied Statistics and Probability	Douglas Montgomery	Wiley
Statistical Modelling	An Introduction to Statistical Learning	Gareth James et al.	Springer
Data Analysis	Practical Statistics for Data Scientists	Peter Bruce, Andrew Bruce	O’Reilly
Data Visualization	Storytelling with Data	Cole Nussbaumer Knaflic	Wiley

05203202 – Data Mining and Machine Learning (M.Sc. Data Science)

Relevant Course / Area	Book Title	Author(s)	Publisher
Data Mining	Data Mining: Concepts and Techniques	Jiawei Han, Micheline Kamber	Morgan Kaufmann
Machine Learning	Machine Learning	Tom Mitchell	McGraw-Hill
Neural Networks	Neural Networks and Deep Learning	Michael Nielsen	Determination Press
ML Algorithms	Pattern Recognition and Machine Learning	Christopher Bishop	Springer

05203205 – Python and R Programming (M.Sc. Data Science)

Relevant Course / Area	Book Title	Author(s)	Publisher
Python Programming	Python for Data Analysis	Wes McKinney	O’Reilly
Python & ML	Hands-On Machine Learning with Scikit-Learn	Aurélien Géron	O’Reilly
R Programming	R for Data Science	Hadley Wickham, Garrett Grolemund	O’Reilly
Data Visualization	ggplot2: Elegant Graphics for Data Analysis	Hadley Wickham	Springer

05203187 – Data Analytics and Visualization (M.Sc. Data Science)

Relevant Course / Area	Book Title	Author(s)	Publisher
Data Analytics	Data Analytics Made Accessible	Anil Maheshwari	Pearson
Predictive Analytics	Predictive Analytics	Eric Siegel	Wiley
R Visualization	Applied Data Visualization	Morrison	O’Reilly
Python Visualization	Python Data Science Handbook	Jake VanderPlas	O’Reilly

05203188 – Predictive Analytics / Spatial Modelling (M.Sc. Data Science)

Relevant Course / Area	Book Title	Author(s)	Publisher
Predictive Analytics	Applied Predictive Modeling	Max Kuhn, Kjell Johnson	Springer
Business Analytics	Business Analytics	James Evans	Pearson
GIS & Spatial Analysis	Geographic Information Systems and Science	Longley et al.	Wiley
Spatial Statistics	Spatial Data Analysis	Robert Haining	Cambridge Press

05203189 – Cloud Native Development (M.Sc. Data Science)

Relevant Course / Area	Book Title	Author(s)	Publisher
Cloud-Native Systems	Cloud Native Patterns	Cornelia Davis	Manning
Kubernetes	Kubernetes in Action	Marko Lukša	Manning
DevOps	The DevOps Handbook	Gene Kim et al.	IT Revolution
Microservices	Building Microservices	Sam Newman	O'Reilly
Containers	Docker Deep Dive	Nigel Poulton	Packt

Note: Preferable to purchase the first 1–2 books

Course Lecture/Laboratory/Tutorial Planning
Lesson Plan – 05201251 (MCA) / 05202156 (MScIT) - Advanced Java Programming

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to JDBC, need for JDBC, overview of JDBC architecture	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	JDBC components – DriverManager, Driver, Connection	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Statement, ResultSet, SQLException handling	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Types of JDBC drivers, database connectivity setup	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Establishing connection, submitting queries, retrieving results	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	ResultSet and ResultSetMetaData interface	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Precompiled statements (PreparedStatement) and CallableStatement	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	JDBC transactions and batch processing	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Web basics, client–server model, configuring Java & Apache Tomcat	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Introduction to servlets, servlet architecture and life cycle	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Servlet API and basic servlet structure	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Handling client requests (GET/POST) and generating responses	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Cookies and session tracking techniques	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	URL rewriting and hidden form fields	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Introduction to servlet filter	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Creating and mapping custom filters	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Filters with URL patterns and real-time use cases	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Introduction to JSP, need and advantages over servlets	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	JSP life cycle and basic structure	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	JSP scripting elements – declaration, scriptlet, expression tags	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	JSP implicit objects – request, response, out, session, application, etc.	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	JavaBeans – concept and usage in JSP	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Reading and writing JavaBeans properties in JSP	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	JSP Expression Language (EL) – basics and usage	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Scoped variables and preventing EL evaluation	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Custom tag libraries – tag handlers and TLD	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	JSTL – installation and core tags (c:out, c:if, c:forEach, etc.)	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Introduction to web services and architecture	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Types – SOAP vs REST, comparison	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Introduction to RESTful services and principles	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Configuring REST using Jersey framework	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	REST architecture, URI design, HTTP methods	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	JAX-RS injection – PathParam, QueryParam, FormParam	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Server responses and exception handling	23/04/26	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	JAX-RS Client API	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Introduction to Spring framework and Spring MVC	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Spring MVC architecture and components	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Advantages and modules of Spring framework	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Spring application configuration (XML & annotation-based)	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Introduction to Hibernate and ORM concept	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Hibernate architecture	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Hibernate configuration and environment setup	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Implementing Hibernate O/R mapping	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	CRUD operations using Hibernate	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Hibernate Query Language (HQL) and integration with Spring	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lesson Plan – 05202158 – Computer Networks and Network Security (MCA/MSc IT))

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to computer networks, data vs information, importance of networking	22/01/26	21/01/26	20/01/26	19/01/26	22/01/26	21/01/26	20/01/26	19/01/26	22/01/26
2	Data communication – characteristics and components	23/01/26	24/01/26	21/01/26	22/01/26	23/01/26	24/01/26	21/01/26	22/01/26	23/01/26
3	Data representation and data flow (simplex, half duplex, full duplex)	24/01/26	28/01/26	23/01/26	23/01/26	24/01/26	28/01/26	23/01/26	23/01/26	24/01/26
4	Categories of networks (LAN, MAN, WAN), network topologies	29/01/26	31/01/26	27/01/26	29/01/26	29/01/26	31/01/26	27/01/26	29/01/26	29/01/26
5	Protocol concepts and elements of a protocol	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Networking standards and standard organizations	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Reference models – OSI model and TCP/IP model, comparison	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Data Link Layer – functions and design issues overview	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26
9	Framing techniques	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Error control mechanisms	12/02/26	14/02/26	10/02/26	12/02/26	12/02/26	14/02/26	10/02/26	12/02/26	12/02/26
11	Flow control techniques	13/02/26	16/02/26	11/02/26	13/02/26	13/02/26	16/02/26	11/02/26	13/02/26	13/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
12	DLL protocols – HDLC and PPP	14/02/26	18/02/26	13/02/26	16/02/26	14/02/26	18/02/26	13/02/26	16/02/26	14/02/26
13	High-speed DLL technologies – FDDI, CDDI	19/02/26	21/02/26	17/02/26	19/02/26	19/02/26	21/02/26	17/02/26	19/02/26	19/02/26
14	Frame Relay and ATM	20/02/26	23/02/26	18/02/26	20/02/26	20/02/26	23/02/26	18/02/26	20/02/26	20/02/26
15	Multiple access protocols – ALOHA (pure & slotted)	21/02/26	25/02/26	20/02/26	23/02/26	21/02/26	25/02/26	20/02/26	23/02/26	21/02/26
16	CSMA and CSMA/CD	26/02/26	28/02/26	24/02/26	26/02/26	26/02/26	28/02/26	24/02/26	26/02/26	26/02/26
17	Ethernet basics and frame structure	27/02/26	03/03/26	25/02/26	27/02/26	27/02/26	03/03/26	25/02/26	27/02/26	27/02/26
18	Switched Ethernet, Fast Ethernet, Gigabit Ethernet	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Internetworking devices – Repeaters, hubs, bridges	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Switches, routers, gateways	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Virtual LANs (VLANs) and DLL switching	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Network layer functions and design issues	12/03/26	14/03/26	10/03/26	12/03/26	12/03/26	14/03/26	10/03/26	12/03/26	12/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
23	Routing algorithms – Distance vector routing	13/03/26	16/03/26	11/03/26	13/03/26	13/03/26	16/03/26	11/03/26	13/03/26	13/03/26
24	Routing algorithms – Link state routing	14/03/26	18/03/26	13/03/26	16/03/26	14/03/26	18/03/26	13/03/26	16/03/26	14/03/26
25	Flooding and congestion overview	19/03/26	21/03/26	17/03/26	19/03/26	19/03/26	21/03/26	17/03/26	19/03/26	19/03/26
26	Routing protocols – RIP and IGRP	20/03/26	23/03/26	18/03/26	20/03/26	20/03/26	23/03/26	18/03/26	20/03/26	20/03/26
27	Routing protocols – EIGRP and OSPF	21/03/26	25/03/26	20/03/26	23/03/26	21/03/26	25/03/26	20/03/26	23/03/26	21/03/26
28	Internetworking concepts – Tunneling and fragmentation	26/03/26	28/03/26	24/03/26	26/03/26	26/03/26	28/03/26	24/03/26	26/03/26	26/03/26
29	IPv4 addressing and basics	27/03/26	30/03/26	25/03/26	27/03/26	27/03/26	30/03/26	25/03/26	27/03/26	27/03/26
30	IPv6 basics and comparison with IPv4	28/03/26	01/04/26	27/03/26	30/03/26	28/03/26	01/04/26	27/03/26	30/03/26	28/03/26
31	Border Gateway Protocol (BGP)	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26
32	Transport layer overview and services	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26
33	UDP – features and header format	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26
34	TCP – features and header format	23/04/26	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26	23/04/26
35	DNS – namespace and name servers	24/04/26	13/04/26	08/04/26	10/04/26	24/04/26	13/04/26	08/04/26	10/04/26	24/04/26
36	Email protocols – SMTP and POP3	25/04/26	15/04/26	10/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26	25/04/26
37	Web and file protocols – HTTP, FTP, Telnet	30/04/26	18/04/26	14/04/26	16/04/26	30/04/26	18/04/26	14/04/26	16/04/26	30/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
38	Security architecture and authentication overview	01/05/26	20/04/26	15/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26	01/05/26
39	Influence Competen- cies	02/05/26	22/04/26	17/04/26	20/04/26	02/05/26	22/04/26	17/04/26	20/04/26	02/05/26
40	Authentication applications – Kerberos and X.509	02/05/26	25/04/26	21/04/26	23/04/26	02/05/26	25/04/26	21/04/26	23/04/26	02/05/26
41	Email security – PGP and S/MIM	08/05/26	27/04/26	22/04/26	24/04/26	08/05/26	27/04/26	22/04/26	24/04/26	08/05/26
42	IP security – Authentication Header, ESP, Security Associations, Key management	07/05/26	29/04/26	24/04/26	27/04/26	07/05/26	29/04/26	24/04/26	27/04/26	07/05/26
43	Project Management Tools and Techniques	07/05/26	01/05/26	28/04/26	30/04/26	07/05/26	01/05/26	28/04/26	30/04/26	07/05/26
44	Project Evaluation and Review Tech- nique (PERT Chart)	08/05/26	04/05/26	29/04/26	04/05/26	08/05/26	04/05/26	29/04/26	04/05/26	08/05/26
45	System security – intruders and viruses	08/05/26	06/05/26	05/05/26	07/05/26	08/05/26	06/05/26	05/05/26	07/05/26	08/05/26

Lesson Plan – 05201256[MCA]/05202186[MSc IT] – Graphics Programming

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Computer Graphics and its applications	22/01/26	21/01/26	20/01/26	19/01/26	22/01/26	21/01/26	20/01/26	19/01/26	22/01/26
2	Raster-scan system and Random-scan system	23/01/26	24/01/26	21/01/26	22/01/26	23/01/26	24/01/26	21/01/26	22/01/26	23/01/26
3	Graphics primitives and scan conversion concepts	24/01/26	28/01/26	23/01/26	23/01/26	24/01/26	28/01/26	23/01/26	23/01/26	24/01/26
4	Line drawing algorithms – DDA algorithm	29/01/26	31/01/26	27/01/26	29/01/26	29/01/26	31/01/26	27/01/26	29/01/26	29/01/26
5	Bresenham's line drawing algorithm	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Circle generating algorithms – Midpoint circle algorithm	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Bresenham's circle algorithm	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Ellipse generating algorithm – Midpoint ellipse algorithm	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26
9	Attributes of output primitives: line style, color, intensity	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Area filling algorithms – Scan line algorithm	12/02/26	14/02/26	10/02/26	12/02/26	12/02/26	14/02/26	10/02/26	12/02/26	12/02/26
11	Boundary fill and Flood fill algorithms	13/02/26	16/02/26	11/02/26	13/02/26	13/02/26	16/02/26	11/02/26	13/02/26	13/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
12	Antialiasing techniques	14/02/26	18/02/26	13/02/26	16/02/26	14/02/26	18/02/26	13/02/26	16/02/26	14/02/26
13	2D transformations – Translation and Scaling	19/02/26	21/02/26	17/02/26	19/02/26	19/02/26	21/02/26	17/02/26	19/02/26	19/02/26
14	Rotation and Reflection	20/02/26	23/02/26	18/02/26	20/02/26	20/02/26	23/02/26	18/02/26	20/02/26	20/02/26
15	Shearing and Composite transformations	21/02/26	25/02/26	20/02/26	23/02/26	21/02/26	25/02/26	20/02/26	23/02/26	21/02/26
16	Transformation commands and character generation	26/02/26	28/02/26	24/02/26	26/02/26	26/02/26	28/02/26	24/02/26	26/02/26	26/02/26
17	Viewing concepts and viewing coordinates	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26
18	Window–Viewport concept	28/02/26	03/04/26	27/02/26	02/03/26	28/02/26	03/04/26	27/02/26	02/03/26	28/02/26
19	Window to Viewport transformation	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Line clipping – Cohen Sutherland algorithm	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Polygon clipping – Sutherland Hodgman algorithm	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	3D clipping – Normalized view volume	12/03/26	14/03/26	10/03/26	12/03/26	12/03/26	14/03/26	10/03/26	12/03/26	12/03/26
23	Viewport clipping and clipping in homogeneous coordinates	13/03/26	16/03/26	11/03/26	13/03/26	13/03/26	16/03/26	11/03/26	13/03/26	13/03/26
24	Introduction to illumination models	14/03/26	18/03/26	13/03/26	16/03/26	14/03/26	18/03/26	13/03/26	16/03/26	14/03/26
25	Light sources and diffuse reflection	19/03/26	21/03/26	17/03/26	19/03/26	19/03/26	21/03/26	17/03/26	19/03/26	19/03/26
26	Color models – RGB, YIQ	20/03/26	23/03/26	18/03/26	20/03/26	20/03/26	23/03/26	18/03/26	20/03/26	20/03/26
27	Color models – CMY and HSV	21/03/26	25/03/26	20/03/26	23/03/26	21/03/26	25/03/26	20/03/26	23/03/26	21/03/26
28	Introduction to 3D concepts and 3D display techniques	26/03/26	28/03/26	24/03/26	26/03/26	26/03/26	28/03/26	24/03/26	26/03/26	26/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
29	3D representation and transformations	27/03/26	30/03/26	25/03/26	27/03/26	27/03/26	30/03/26	25/03/26	27/03/26	27/03/26
30	3D rotation about arbitrary axis	28/03/26	01/04/26	27/03/26	30/03/26	28/03/26	01/04/26	27/03/26	30/03/26	28/03/26
31	Composite 3D transformations	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26
32	Parallel projection	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26
33	Perspective projection	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26
34	Hidden surface and hidden line removal concepts	23/04/26	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26	23/04/26
35	Back face removal and Depth buffer method	24/04/26	13/04/26	08/04/26	10/04/26	24/04/26	13/04/26	08/04/26	10/04/26	24/04/26
36	Scan line method for hidden surface removal	25/04/26	15/04/26	10/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26	25/04/26
37	Introduction to OpenGL, OpenGL pipeline	30/04/26	18/04/26	14/04/26	16/04/26	30/04/26	18/04/26	14/04/26	16/04/26	30/04/26
38	OpenGL coordinate system and setup	01/05/26	20/04/26	15/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26	01/05/26
39	OpenGL programming basics – drawing shapes	02/05/26	22/04/26	17/04/26	20/04/26	02/05/26	22/04/26	17/04/26	20/04/26	02/05/26
40	Event handling and rendering pipeline	07/05/26	25/04/26	21/04/26	23/04/26	07/05/26	25/04/26	21/04/26	23/04/26	07/05/26
41	Lighting and material properties in OpenGL	08/05/26	27/04/26	22/04/26	24/04/26	08/05/26	27/04/26	22/04/26	24/04/26	08/05/26
42	Texture mapping techniques and OpenGL texture	07/05/26	29/04/26	24/04/26	27/04/26	07/05/26	29/04/26	24/04/26	27/04/26	07/05/26
43	Shading techniques – Flat, Gouraud, Phong	07/05/26	01/05/26	28/04/26	30/04/26	07/05/26	01/05/26	28/04/26	30/04/26	07/05/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
44	OpenGL animations, double buffering, frame rate control	08/05/26	04/05/26	29/04/26	04/05/26	08/05/26	04/05/26	29/04/26	04/05/26	08/05/26
45	Modern OpenGL (VBOs, VAOs, shaders), optimization techniques & mini project overview	08/05/26	06/05/26	05/05/26	07/05/26	08/05/26	06/05/26	05/05/26	07/05/26	08/05/26

Lesson Plan – 05201296 – Cyber Security and Forencics – I

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Cyber Security and Security Layers	19/01/26	19/01/26	19/01/26	19/01/26	19/01/26	19/01/26	19/01/26	19/01/26	19/01/26
2	CIA Triad, Information Security Concepts	21/01/26	21/01/26	21/01/26	21/01/26	21/01/26	21/01/26	21/01/26	21/01/26	21/01/26
3	Network Security Model and Security Architecture	23/01/26	23/01/26	23/01/26	23/01/26	23/01/26	23/01/26	23/01/26	23/01/26	23/01/26
4	Basics of Cryptography and Types of Attacks on Cryptosystems	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26
5	Traditional vs Modern Cryptography	28/01/26	28/01/26	28/01/26	28/01/26	28/01/26	28/01/26	28/01/26	28/01/26	28/01/26
6	Symmetric and Asymmetric Key Encryption	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26
7	Block Cipher and Stream Cipher Concepts	02/02/26	02/02/26	02/02/26	02/02/26	02/02/26	02/02/26	02/02/26	02/02/26	02/02/26
8	Feistel Cipher Structure	04/02/26	04/02/26	04/02/26	04/02/26	04/02/26	04/02/26	04/02/26	04/02/26	04/02/26
9	DES Algorithm	06/02/26	06/02/26	06/02/26	06/02/26	06/02/26	06/02/26	06/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
10	3DES and AES Algorithms	09/02/26	09/02/26	09/02/26	09/02/26	09/02/26	09/02/26	09/02/26	09/02/26	09/02/26
11	Block Cipher Modes of Operation	11/02/26	11/02/26	11/02/26	11/02/26	11/02/26	11/02/26	11/02/26	11/02/26	11/02/26
12	Public Key Encryption, PKI, RSA Algorithm	13/02/26	13/02/26	13/02/26	13/02/26	13/02/26	13/02/26	13/02/26	13/02/26	13/02/26
13	Hash Functions and Message Authentication Codes (MAC)	16/02/26	16/02/26	16/02/26	16/02/26	16/02/26	16/02/26	16/02/26	16/02/26	16/02/26
14	Digital Signatures and Applications	18/02/26	18/02/26	18/02/26	18/02/26	18/02/26	18/02/26	18/02/26	18/02/26	18/02/26
15	Security Services Using Hash and Digital Signature	20/02/26	20/02/26	20/02/26	20/02/26	20/02/26	20/02/26	20/02/26	20/02/26	20/02/26
16	Overview of Security Threats and Malware	23/02/26	23/02/26	23/02/26	23/02/26	23/02/26	23/02/26	23/02/26	23/02/26	23/02/26
17	Viruses, Worms, Trojan Horses	25/02/26	25/02/26	25/02/26	25/02/26	25/02/26	25/02/26	25/02/26	25/02/26	25/02/26
18	Spyware, Adware, Botnets, Spam	27/02/26	27/02/26	27/02/26	27/02/26	27/02/26	27/02/26	27/02/26	27/02/26	27/02/26
19	Phishing and Social Engineering Attacks	03/03/26	03/03/26	03/03/26	03/03/26	03/03/26	03/03/26	03/03/26	03/03/26	03/03/26
20	Stack and Buffer Overflow Attacks	05/03/26	05/03/26	05/03/26	05/03/26	05/03/26	05/03/26	05/03/26	05/03/26	05/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
21	Role of OS in Information Security	09/03/26	09/03/26	09/03/26	09/03/26	09/03/26	09/03/26	09/03/26	09/03/26	09/03/26
22	OS Security Mechanisms and Patch Management	11/03/26	11/03/26	11/03/26	11/03/26	11/03/26	11/03/26	11/03/26	11/03/26	11/03/26
23	Memory Address Protection and File Protection	13/03/26	13/03/26	13/03/26	13/03/26	13/03/26	13/03/26	13/03/26	13/03/26	13/03/26
24	Protected Objects and Access Control Methods	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26
25	Wireless Technologies and Security Challenges	17/03/26	17/03/26	17/03/26	17/03/26	17/03/26	17/03/26	17/03/26	17/03/26	17/03/26
26	Wireless Security Protocols – WEP, WPA, WPA2	18/03/26	18/03/26	18/03/26	18/03/26	18/03/26	18/03/26	18/03/26	18/03/26	18/03/26
27	Attacks on Wireless Networks	19/03/26	19/03/26	19/03/26	19/03/26	19/03/26	19/03/26	19/03/26	19/03/26	19/03/26
28	Introduction to Computer and Cyber Forensics	27/04/26	28/04/26	27/04/26	27/04/26	28/04/26	27/04/26	27/04/26	28/04/26	27/04/26
29	Digital Evidence and Rules of Evidence	29/04/26	29/04/26	30/04/26	29/04/26	29/04/26	30/04/26	29/04/26	29/04/26	30/04/26
30	Email Forensics, RFC 2822	01/05/26	02/05/26	01/05/26	01/05/26	02/05/26	01/05/26	01/05/26	02/05/26	01/05/26

Lesson-Plan 05201254 (MCA)/05202154s (MScIT) - Comprehensive Assessment

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Computer Organization, Block diagram of computer, Data representation	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Number Systems – Binary, Octal, Decimal, Hexadecimal, Conversions	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Boolean Algebra – Laws, Truth tables, Logic gates	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Flip-Flops – SR, JK, D, T Flip-Flops, Registers, Counters	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Memory & Buses – RAM, ROM, Cache memory, System buses	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Programming basics, Data types, Decision & control structures	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Looping constructs – for, while, do-while	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Arrays (1D & 2D), Functions, Pass by value	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Pointers, Pass by pointer, Structures & Unions	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	File handling – File operations, Modes, Examples	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Internet basics, HTML structure, Common HTML tags	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Linking documents, Frames, Media elements	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Images, Image maps, Forms	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	CSS styles, Introduction to JavaScript	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	OOP concepts – Class, Object, Encapsulation, Abstraction	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Functions, Constructors & Destructors	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Operator overloading, Inheritance	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Runtime polymorphism, Virtual functions	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Exception handling, IO operations	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Operating System concepts & types	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Process management, Threads, CPU scheduling	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Memory management – Paging, Segmentation	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Inter-process communication & Synchronization	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	I/O system, File system, Disk scheduling	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Java basics, Inheritance & Sub classing	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Packages, Interfaces	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Exception handling in Java	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Collection framework – ArrayList, Vector, Set, Map	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	I/O, Multithreading, Synchronization	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Software process models, SDLC	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Requirements engineering, Data dictionary	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	UML – Class diagram, Use case diagram	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Activity diagram, Sequence diagram	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Object relational & Extended relational database	23/04/26	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	PL/SQL basics, Cursors	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Triggers & Stored procedures	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Database tuning, Indexing & Security	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Backup & Recovery, Grant & Revoke privileges	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Open source software, PHP configuration & syntax	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	PHP variables, Control structures	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Sessions & Cookies, MySQL with PHP	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Working with user data, Error handling	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Algorithm analysis, Time & Space complexity, Asymptotic notations	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Divide & Conquer, Dynamic Programming	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Graph algorithms, Backtracking, Branch & Bound	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lesson-Plan 05202108 - Software Engineering with Agile

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to system and basic components of a system,	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Information systems categories and need of information system development,	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Study of software characteristics, components and applications,	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Layered technologies, software processes, methods and tools,	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Generic view of software engineering and Classical SDLC overview,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Requirement analysis using Classical SDLC method,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Requirement determination and System Requirement Specification (SRS),	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Fact finding techniques and requirements documentation,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Waterfall and Incremental process models,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Evolutionary process models – Prototyping and Spiral model,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Concurrent development model and comparison of process models,	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Requirement engineering – problem recognition and RE tasks,	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Use cases, functional specification and requirement modeling,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Requirements validation and verification techniques,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Introduction to Structured System Analysis Development Methodology (SSADM),	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Decision trees, decision tables and structured English,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Data Flow Diagrams (DFD) – concepts and levels,	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Entity Relationship (ER) diagrams and data dictionary,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	System design concepts and design modeling	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Software architecture and architectural styles and patterns,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Data design and procedural design,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Object oriented design concepts and modeling,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Agile methodology and agile process principles,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Extreme Programming (XP) and Scrum framework,	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Adaptive software development and other agile models,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	User Interface design concepts and interface design model,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Internal and external design, evaluation and interaction design,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Software project planning and management spectrum,	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	People, product, process and project management,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	W5HH principle, scope and feasibility analysis,	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Effort estimation, scheduling, staffing and quality planning,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Risk management – identification, assessment and control,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Introduction to CASE tools and integrated CASE environment,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Object Oriented Programming concepts, functions, constructors and destructors,	23/04/26	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Inheritance, polymorphism, operator overloading and exception handling,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Operating system concepts, process management and threads,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Memory management, IPC, synchronization and disk scheduling,	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Java programming – inheritance, packages and exception handling,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Java collections, I/O and multithreading,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Software engineering diagrams – class and use case diagrams,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Activity and sequence diagrams with applications,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Advanced DBMS – object relational and extended relational databases,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	PL/SQL, cursors, triggers, stored procedures and database tuning,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Open source technology using PHP, PHP syntax, sessions and MySQL integration,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Analysis and design of algorithms – complexity, dynamic programming, backtracking and graph algorithms	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lesson-Plan 05293252 – English and Mathematical Aptitude

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Aptitude and Placement Examination Pattern	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Number System – basics, types of numbers and divisibility rules,	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Number System – LCM, HCF and remainder theorem,	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Number System – problems on digits and shortcuts,	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Reading Comprehension – introduction and question types,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Reading Comprehension – speed reading techniques,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Reading Comprehension – inference and vocabulary-based questions,	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Time and Work – basic concepts and formulas,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Time and Work – problems on efficiency and work distribution,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Time and Work – shortcut methods and mixed problems,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Sentence Correction – grammar fundamentals and error identification,	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Sentence Correction – common placement exam patterns,	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Coding and Decoding – basic letter and number coding,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Series – number series and alphabet series,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Analogy and Odd Man Out problems,	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Visual Reasoning – patterns, figures and mirror images,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Time, Speed and Distance – basic concepts and formulas,	03/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Time, Speed and Distance – trains and relative speed,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Time, Speed and Distance – boats, upstream and downstream problems,	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Vocabulary – techniques to learn and remember new words,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Vocabulary – synonyms, antonyms and contextual usage,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Vocabulary – anagrams, word puzzles and jumbled words,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Ratio and Proportion – fundamentals and applications,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Ratio and Proportion – mixture and comparison problems,	16/03/2026	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Logical Connectives – statements and conclusions,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Syllogism – rules and Venn diagram method,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Venn Diagrams – multi-set problem solving,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Worksheet on Adjectives and Adverbs,	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Worksheet on Collocations, Gerunds and Infinitives,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Worksheet on Determiners and Articles,	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Worksheet on Nouns and Pronouns,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Worksheet on Prepositions,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Worksheet on Pronoun–Antecedent Agreement,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Worksheet on Subject–Verb Agreement,	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Worksheet on Tenses,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Recap on C++ – syntax, operators and control structures,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Recap on C++ – functions, OOP concepts and examples,	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	C++ Case Studies and MCQs from technical exams,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Recap on Data Structures – arrays, stacks and queues,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Data Structures – linked lists and trees,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Data Structures – searching and sorting techniques,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Coding problems in C and C++ for technical tests,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Logical and Technical Mixed MCQs for Placement Exams,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Mock Aptitude Test and Solution Discussion,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Full Placement Test Simulation and Final Revision	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lesson-Planing 05201295 – Big Data Analytics – I

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Big Data and its significance,	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Evolution of Big Data and growth of data sources,	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Structuring of Big Data – structured, semi-structured and unstructured data,	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Fundamentals of Big Data – characteristics and challenges,	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Big Data Analytics and analytical approaches,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Career opportunities and future scope in Big Data,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Discovering Big Data usage in business context,	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Big Data in social networking platforms,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Big Data in preventing fraudulent activities,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Big Data in detecting fraud in insurance sector,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Use of Big Data in retail industry and e-commerce,	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Distributed and parallel computing for Big Data,	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Introduction to Hadoop and its advantages,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Cloud computing and its role in Big Data,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	In-memory computing technology for Big Data,	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	History and evolution of Hadoop,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Hadoop ecosystem and its core components,	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Analysing data using Unix tools,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Analysing data using Hadoop framework,	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Hadoop Streaming and its applications,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Hadoop ecosystem tools overview,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	IBM Big Data strategy and platforms,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Introduction to Infosphere BigInsights and Big Sheets,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Design and architecture of HDFS,	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	HDFS concepts – blocks, replication and fault tolerance,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	HDFS command line interface and file operations,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Hadoop file system interfaces and APIs,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	HDFS data flow and metadata management,	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Data ingest using Flume and Sqoop,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Hadoop archives and data management,	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Hadoop I/O – compression and serialization,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Avro and file-based data structures,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Anatomy of a MapReduce job run,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	MapReduce failures, job scheduling and fault handling,	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Shuffle and sort phase in MapReduce,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Task execution lifecycle in MapReduce,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	MapReduce types, input/output formats and features,	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Understanding Big Data technology foundations and Big Data stack,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Virtualization concepts and Big Data virtualization approaches,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Storing data in databases and data warehouses – overview,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	RDBMS and Big Data integration,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Non-relational databases and polyglot persistence,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Integrating Big Data with traditional data warehouses,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Big Data analysis, data warehouse and changing deployment models,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Processing data with MapReduce – developing simple MapReduce applications and design considerations	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lab Plan – 05201256[MCA]/05202186[MSc IT] – Graphics Programming

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Line Drawing using Digital Differential Analyzer (DDA) Algorithm	12/01/26	13/01/26	14/01/26	15/01/26	19/01/26	20/01/26	21/01/26	22/01/26	23/01/26
2	Line Drawing using Bresenham's Algorithm	27/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26
3	Circle Drawing using Midpoint Circle Algorithm	02/02/26	04/02/26	06/02/26	02/02/26	04/02/26	06/02/26	02/02/26	04/02/26	06/02/26
4	Circle Drawing using Bresenham's Circle Algorithm	06/03/26	09/03/26	10/03/26	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26
5	Ellipse Drawing using Midpoint Ellipse Algorithm	06/03/26	09/03/26	10/03/26	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26
6	Polygon Filling using Scan-line and Boundary Fill Algorithms	19/03/26	20/03/26	23/03/26	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26
7	Flood Fill Algorithm for Closed Shapes	02/04/26	03/04/26	06/04/26	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26
8	2D Geometric Transformations	16/04/26	17/04/26	20/04/26	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26
9	Line Clipping using Cohen–Sutherland Algorithm	29/04/26	30/04/26	01/05/26	04/05/26	05/05/26	06/05/26	07/05/26	08/05/26	08/05/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
10	Polygon Clipping using Sutherland–Hodgman Algorithm	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26	16/03/26
11		27/04/26	28/04/26	27/04/26	27/04/26	28/04/26	27/04/26	27/04/26	28/04/26	27/04/26
12	3D Geometric Transformations	29/04/26	29/04/26	30/04/26	29/04/26	29/04/26	30/04/26	29/04/26	29/04/26	30/04/26
13	Hidden Surface Removal Techniques	04/05/26	05/05/26	06/05/26	07/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
14	Color Models and Anti-Aliasing	27/04/26	28/04/26	27/04/26	27/04/26	28/04/26	27/04/26	27/04/26	28/04/26	27/04/26
15	Animated Scene Creation	01/05/26	02/05/26	01/05/26	01/05/26	02/05/26	01/05/26	01/05/26	02/05/26	01/05/26

Lesson-Plan 05201294 – Artificial Intelligence – I

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Big Data and its significance,	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Evolution of Big Data and growth of data sources,	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Structuring of Big Data – structured, semi-structured and unstructured data,	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Fundamentals of Big Data – characteristics and challenges,	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Big Data Analytics and analytical approaches,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Career opportunities and future scope in Big Data,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Discovering Big Data usage in business context,	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Big Data in social networking platforms,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Big Data in preventing fraudulent activities,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Big Data in detecting fraud in insurance sector,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Use of Big Data in retail industry and e-commerce,	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Distributed and parallel computing for Big Data,	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Introduction to Hadoop and its advantages,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Cloud computing and its role in Big Data,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	In-memory computing technology for Big Data,	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	History and evolution of Hadoop,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Hadoop ecosystem and its core components,	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Analysing data using Unix tools,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Analysing data using Hadoop framework,	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Hadoop Streaming and its applications,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Hadoop ecosystem tools overview,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	IBM Big Data strategy and platforms,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Introduction to Infosphere BigInsights and Big Sheets,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Design and architecture of HDFS,	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	HDFS concepts – blocks, replication and fault tolerance,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	HDFS command line interface and file operations,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Hadoop file system interfaces and APIs,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	HDFS data flow and metadata management,	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Data ingest using Flume and Sqoop,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Hadoop archives and data management,	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Hadoop I/O – compression and serialization,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Avro and file-based data structures,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Anatomy of a MapReduce job run,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	MapReduce failures, job scheduling and fault handling,	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Shuffle and sort phase in MapReduce,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Task execution lifecycle in MapReduce,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	MapReduce types, input/output formats and features,	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Understanding Big Data technology foundations and Big Data stack,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Virtualization concepts and Big Data virtualization approaches,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Storing data in databases and data warehouses – overview,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	RDBMS and Big Data integration,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Non-relational databases and polyglot persistence,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Integrating Big Data with traditional data warehouses,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Big Data analysis, data warehouse and changing deployment models,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Processing data with MapReduce – developing simple MapReduce applications and design considerations	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

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Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Java Date & Time API and overview,	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	LocalDate, LocalTime and LocalDateTime classes,	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	MonthDay, Year, YearMonth and Clock,	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	OffsetTime, OffsetDateTime and time zones,	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Period and Duration with examples,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Java type conversion overview and importance,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	String to double, double to String, int to long and long to int,	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	int to double, double to int, char to int and int to char,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	String to Boolean, Boolean to String and object conversions,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	String to date, date to String and date to timestamp,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Timestamp to date and numeric base conversions,	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Binary to decimal, decimal to binary,	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Problems and problem spaces in AI,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Hexadecimal to decimal and decimal to hexadecimal,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Octal to decimal and decimal to octal,	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Introduction to JSON and need of JSON,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	JSON vs XML comparison,	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	JSON syntax and simple JSON examples,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	JSON Object and JSON Array,	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	JSON comments and nested JSON structures,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	JSON example using PHP,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	JSON example using Java,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	JSON example using AJAX,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Introduction to Bootstrap and setup,	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Bootstrap containers, grid system and layout,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Bootstrap jumbotron, buttons and tables,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Bootstrap forms, input types and validation,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Alerts, wells, badges and labels,	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Panels, pagination and pager,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Images, progress bar and list groups,	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Dropdowns, collapse and accordion,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Tabs, pills and navigation bar,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Bootstrap modal and responsive design,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Bootstrap utilities – borders, clearfix and colors,	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Icons, image replacement and invisible content,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Positioning, sizing, spacing and screen readers,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Star rating and utility use cases,	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Introduction to Selenium, features and limitations,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Selenium vs QTP and Selenium tool suite overview,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Selenium IDE installation and features,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Creating test cases, login test and Selenium IDE commands,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Introduction to Selenium WebDriver and WebDriver vs RC,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	WebDriver installation and running tests on Chrome and Firefox,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	WebDriver commands – navigation, alerts, drag & drop, radio button and checkbox handling,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Introduction to Node.js, installation on Windows, basic example, NPM, core modules overview and Node.js with MySQL CRUD operations	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

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Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	To write a Java program to implement Date and Time API.	19/01/26	20/01/26	25/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	To implement various data type conversions in Java.	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	To implement JSON Object and JSON Array.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	To install and configure Selenium environment.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	To write and execute Selenium test cases.	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26	19/03/26	20/03/26	23/03/26
6	To implement various Selenium WebDriver functionalities.	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26	02/04/26	03/04/26	06/04/26
7	To implement String Decoder in Node.js.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
8	To implement Query String module in Node.js.	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26
9	To implement events in Node.js.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	To perform CRUD operations using Node.js and MongoDB	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	To develop a mini application using Node.js and MongoDB.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

Lesson-Plan 05201289 – Cloud Computing – I

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Cloud Computing, Definition, Basic Concepts	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	History and Evolution of Cloud Computing	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Characteristics of Cloud Computing	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Pros and Cons of Cloud Computing	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Nature of Cloud and Enabling Technologies	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Technologies Used in Cloud Computing	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Migrating Applications and Data into the Cloud	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Cloud Applications and Working of Cloud Computing	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Trends and Future Scope of Cloud Computing	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Introduction to Cloud Computing Models	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Types of Cloud – Public, Private, Hybrid, Community	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Cloud Service Models – Overview	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Software as a Service (SaaS) Model	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Platform as a Service (PaaS) Model	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Infrastructure as a Service (IaaS) Model	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Other Cloud Services and Emerging Models	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Cloud Architecture and Components	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Cloud Computing Reference Model	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Introduction to Virtualization Concepts	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Virtualization – Definition, Architecture and Software	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Virtualization Applications and Benefits	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Virtual Clustering Concepts	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Anatomy of Cloud Infrastructure	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Virtual Infrastructures in Cloud Environment	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	CPU Virtualization Techniques	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Network and Storage Virtualization	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Introduction to Cloud Data Storage	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Storage Options in Cloud Computing	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Structured and Unstructured Storage in Cloud	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Exploring Cloud Storage Services	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Cloud Data Stores and Storage Management	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Provisioning and Managing Cloud Storage	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Data-Intensive Technologies for Cloud Computing	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Cloud Storage from LANs to WANs, Distributed Data Storage	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Applications Utilizing Cloud Storage	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Introduction to Cloud Risks and Risk Management	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Enterprise-Wide Risk Management, Types of Cloud Risks	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Cloud Security and Data Security Concepts	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Digital Persona, Content-Level Security, Shared Security Model	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Cloud Security Services, Authentication and Authorization	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Security Challenges and Secure Cloud Software Requirements	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Introduction to Google Cloud Platform (GCP) and GCP Console	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	GCP Projects, Cloud SDK Installation, Cloud Shell	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	GCP APIs and Cloud Console Mobile App	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Compute Engine, Autoscaling, Building Applications on GCP	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lab-Plan 05201289 – Cloud Computing – I

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	To create cloud storage accounts on multiple cloud providers.	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	To store and manage large files on cloud platforms.	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	To understand and use Software as a Service (SaaS).	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	To work with Platform as a Service (PaaS).	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	To explore Infrastructure as a Service (IaaS).	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	To implement para-virtualization on a local machine.	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	To start and manage a Windows VM on GCP.	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	To start and manage a Linux VM on GCP.	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	To build a Docker application using GCP.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	To deploy a containerized application on GCE.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	To explore Google Cloud Platform services and prepare a case study.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

Lesson-Plan 05201288 – Gaming Technology – I

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Artificial Intelligence and definition of AI,	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	AI approaches – heuristic approach and problem-solving overview,	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Simulated annealing and Iterative Deepening A* (IDA*),	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Measuring problem-solving approaches in AI,	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Choosing state and action approach, generating action sequence approach,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Introduction to game design and principles of game design,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Game design theory and MDA framework,	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Eight types of fun in games and player motivation,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Visual se and gameplay mechanics,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Introduction to game idea generation,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Idea development process and stimulus techniques,	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Genre analysis and market research,	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Target platform selection and audience analysis,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Physical games – board games, card games, party games and hybrids,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Introduction to GIMP / Photoshop interface and tools,	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Clone tool and healing brush techniques,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Working with layers and adjustment panel,	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Masking basics and smart object transformations,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Smart filters, color correction and visual effects,	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Working with text and vector shapes, file formats, resizing and saving,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Actions and batch processing for game assets,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Introduction to Unity Game Engine and interface navigation,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Unity tools, scene navigation and camera control,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Terrain system and environment setup in Unity,	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Project settings, player settings and build configuration,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Game publishing using Unity,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Introduction to C# scripting in Unity,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Variables, constants, integers, floats and strings,	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Arrays, lists and arithmetic operators,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Conditional statements – if, else and switch,	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Looping constructs – while, for and basic C# concepts in Unity,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Introduction to 2D game development in Unity,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Sprite editor and sprite animation,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	2D physics system and 2D components,	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	UI system in Unity and 2D game project development,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Game design documentation – purpose and importance,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Design document layout, level design document and technical design document	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Production document, ADR, sound effects, foley and ambience recording,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Introduction to Audacity software,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Recording and editing sounds, looping and timeline arrangement,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Mixing sounds, sampling and synthesis techniques,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Introduction to multiplayer programming concepts,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Scene data transfer, singleton pattern and local data storage,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Server creation, joining hosts and discovering active hosts,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Network object creation, RPCs and developing a multiplayer game in Unity	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lab-Plan 05201288 – Gaming Technology – I

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	To design a visual style for a game.	19/01/26	20/01/26	25/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	To design and create a board game.	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	To design and create a card game.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	To design a party game.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	To master clone and healing brush tools in GIMP/Photoshop.	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26	19/03/26	20/03/26	23/03/26
6	To manage game assets efficiently.	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26	02/04/26	03/04/26	06/04/26
7	To implement camera control in Unity.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
8	To configure project and player settings.	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26
9	To publish a game using Unity.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	To develop a complete 2D game project.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	To finalize and present the 2D game project.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

MSc in Data Science Syllabus
Lesson-Plan05203201 – Data Analysis and Statistical Modelling

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to statistics, types of data and role of statistics in data analysis,	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Descriptive statistics – measures of central tendency (mean, median, mode),	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Descriptive statistics – measures of dispersion (range, variance, standard deviation),	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Data summarization techniques and graphical representation,	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Introduction to inferential statistics and population vs sample,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Sampling distributions and central limit theorem,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Hypothesis testing – null and alternative hypotheses,	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Hypothesis testing – Z-test and t-test,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Confidence intervals and interpretation,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Introduction to probability concepts and rules of probability,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Probability distributions – discrete distributions (binomial, Poisson),	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Continuous distributions – normal distribution and sampling methods (simple random, stratified, systematic, cluster),	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Introduction to data analysis and data preprocessing,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Exploratory Data Analysis (EDA) – concepts and importance,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	EDA techniques – summary statistics and visual exploration,	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Data visualization techniques for EDA,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Introduction to regression analysis and correlation,	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Simple linear regression – model building and interpretation,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Multiple linear regression and non-linear regression models,	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Regression diagnostics and assumptions,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Introduction to time series data and components,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Time series analysis – trend, seasonality and decomposition,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Forecasting techniques – moving averages and exponential smoothing,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Introduction to multivariate analysis,	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Principal Component Analysis (PCA) and Factor Analysis,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Introduction to statistical modelling and modeling process,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Linear models and assumptions,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Logistic regression and classification models,	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Advanced statistical models overview,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Model evaluation concepts and performance metrics	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Evaluation metrics – R-squared, adjusted R-squared,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Model evaluation using AIC and BIC,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Model selection techniques and overfitting vs underfitting,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Model validation and interpretation of results,	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Introduction to statistical software tools,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Basics of R programming – syntax, data types and operators,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Data manipulation and visualization using R,	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Introduction to Python programming for data analysis,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Python libraries – NumPy and Pandas,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Data analysis and preprocessing using Pandas,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Introduction to Scikit-learn and basic machine learning workflow,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Basics of SQL – database concepts and SQL queries,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Using SQL for data retrieval and manipulation,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Introduction to data visualization tools – Tableau and Power BI,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Overview of SPSS and comparative study of analytics tools,	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lesson-Plan 05203202 – Data Mining and Machine Learning

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to data mining, definition and need	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Related technologies – Machine Learning, DBMS, OLAP and Statistics,	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Data mining goals and stages of the data mining process,	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Data mining techniques and knowledge representation methods,	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Applications of data mining with example of weather data,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Relationship between data mining and data warehousing,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Why data mining is a discipline and overview of data mining tasks,	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Knowledge representation – task relevant data, background knowledge and interestingness measures,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Representing input data, output knowledge, visualization techniques and Weka visualization experiments,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Association rules – motivation, terminology and applications,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Mining weather data using association rule examples,	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Item sets – basic concepts and generation of frequent item sets,	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Efficient generation of item sets and association rules,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Correlation analysis and interestingness measures,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Experiments with Weka – mining association rules,	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Classification – basic learning and mining tasks,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Rule learning – 1R algorithm and covering rules,	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Decision tree learning and rule extraction,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Experiments with Weka – decision trees and rule-based classifiers,	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Prediction – prediction task, Bayesian classification and Bayesian networks,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Instance-based learning – nearest neighbor methods,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Linear models and prediction techniques, Weka experiments for prediction,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Introduction to machine learning and types of learning,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Supervised, unsupervised and reinforcement learning concepts,	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Role of mathematics – linear algebra, probability and statistics in ML,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Supervised learning algorithms – linear and logistic regression,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Decision trees and ensemble methods – bagging and boosting,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Unsupervised learning – clustering (K-means and hierarchical clustering),	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Dimensionality reduction (PCA), generalization and overfitting,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Introduction to artificial neural networks and biological neuron,	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Artificial neuron model, activation functions and network architectures,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Single-layer neural networks and perceptron model,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Multi-layer perceptron and learning process in ANN,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Loss functions, optimization techniques and gradient descent,	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Backpropagation algorithm and delta learning rule,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Deep learning – CNNs and advances in backpropagation,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	RNNs, transformers and adversarial learning,	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	NLP concepts – text analytics, sentiment analysis, word embeddings, reinforcement learning – MDPs and Q-learning,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Types of data in machine learning and data exploration,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Data cleaning, preprocessing and feature engineering,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Data transformation techniques and feature selection,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Model selection, training and interpretability for supervised learning,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Model evaluation techniques and performance metrics,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Cross-validation and hyperparameter tuning,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Performance enhancement, training–validation–testing concepts and handling overfitting,	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lesson-Plan 05203205 – Python and R Programming

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Python, installation and environment setup,	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Python syntax, keywords and indentation rules,	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Variables and basic data types (int, float, string, boolean),	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Operators in Python (arithmetic, relational, logical),	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Control structures – if, if-else and nested conditions,	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Looping constructs – while loop and for loop,	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Functions – definition, parameters, return values,	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Modules and packages in Python,	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	File handling – reading and writing text files,	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Error and exception handling,	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Python data structures – lists and list operations,	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Tuples and sets with use cases,	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Dictionaries and dictionary operations,	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Introduction to NumPy – arrays and numerical computing,	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Introduction to Pandas – Series, DataFrames and basic operations,	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Data manipulation using Pandas – filtering, sorting, grouping,	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Data import and export – CSV and Excel files,	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Introduction to data visualization concepts,	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Matplotlib – line plots, bar charts and histograms,	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Seaborn – statistical plots and styling,	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Basic statistical concepts for data analysis,	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Linear regression using scikit-learn,	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Classification techniques – basic classifiers,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Clustering techniques – K-means and overview,	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Advanced machine learning workflow using scikit-learn,	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Introduction to R, installation and R environment,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	R syntax, data types and operators,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Control structures in R – if, loops and apply family,	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Functions and packages in R,	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	R data structures – vectors and lists,	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Matrices, arrays and data frames, Factors and categorical data handling,	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Introduction to tidyverse and data wrangling concepts,	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Data manipulation using dplyr – select, filter, mutate, summarize,	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Data cleaning and transformation using tidyverse,	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Introduction to data visualization in R,	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	ggplot2 basics – grammar of graphics,	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Creating bar charts, line plots and scatter plots using ggplot2,	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Advanced ggplot2 – faceting and customization,	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Descriptive statistics using R,	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Advanced ggplot2 – faceting and customization,	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Hypothesis testing – t-test and chi-square test,	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Regression analysis and ANOVA in R,	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Advanced statistical methods overview,	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Introduction to machine learning concepts in R,	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Machine learning using caret – model building and evaluation,	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lab-Plan 05203205 – Python and R Programming

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	To install Python environment and understand basic syntax.	19/01/26	20/01/26	25/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	To implement decision making and looping constructs.	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	To create and manipulate Python data structures.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	To define and use functions and modules.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	To perform string manipulation and file operations.	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26	19/03/26	20/03/26	23/03/26
6	To implement classes and objects.	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26	02/04/26	03/04/26	06/04/26
7	To perform data analysis using Python libraries.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
8	To visualize data using Matplotlib.	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26
9	To set up R and understand basic programming concepts.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	To work with R data structures and data manipulation.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	To analyze and visualize data using R.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

Lesson-Plan 05203187 – Data Analytics and Visualization

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Data Analytics and analytics lifecycle overview	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Key roles for a successful analytics project	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Background and overview of the Data Analytics Lifecycle	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Phase 1 – Discovery: understanding business domain	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Framing the problem and identifying key stakeholders	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Interviewing the analytics sponsor and defining objectives	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Developing initial hypotheses	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Identifying potential data sources	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Phase 2 – Data Preparation: analytic sandbox and ETLT process	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Learning about the data and data conditioning	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Surveying, visualizing data and handling dirty data	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Common tools used in data preparation phase	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Phase 3 – Model Planning: data exploration and variable selection	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Model selection techniques and tools for model planning	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Phase 4 – Model Building, Phase 5 – Communicating results and Phase 6 – Operationalizing analytics	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Introduction to data mining algorithms	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Association rules – motivation, terminology and concepts	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Mining weather data using association rules	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Introduction to regression models	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Simple linear regression – regression equation, fitted values and residuals	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Least squares method and interpretation	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Multiple linear regression – assessing the model	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Classification techniques – basic classifiers,	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Prediction using regression models	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Logistic regression – logistic response function and logit	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Logistic regression, GLM and interpretation of coefficients	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Linear vs logistic regression – similarities and differences, model assessment	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Introduction to time series analysis	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Box-Jenkins methodology and ARIMA models	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	ACF, AR, MA, ARMA and ARIMA modeling	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Building, evaluating ARIMA models and cautions	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Introduction to text analytics and history of text mining	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Text mining practices and applications	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Text analytics steps – collecting, representing text and TF-IDF	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Document categorization, sentiment analysis and gaining insights	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Introduction to R – data import, export and data types	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Descriptive statistics using R	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Exploratory data analysis – visualization before analysis	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Handling dirty data and visualizing single variables	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Examining multiple variables and data exploration vs presentation	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Introduction to Python data analytics libraries – NumPy, Pandas, SciPy	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Introduction to Matplotlib and basic plotting	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Creating histogram, bar chart, pie chart and box plot	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Violin plots and advanced plotting using Matplotlib	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Introduction to Seaborn – multiple plots and regression plots (regplot)	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lesson-Plan 05203188 – Predictive Analytics/ Spatial Modelling

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	What is Analytics and why analytics is important in business decision-making	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Introduction to analytics tools and working environment	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Applications of predictive modeling in business domains	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Databases, types of data and variables used in analytics	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Data modeling techniques and business modeling need	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Data preparation and exploration – data cleaning and wrangling	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Handling missing values, outliers and summary statistics	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Data visualization techniques for analytics	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Introduction to regression – concepts and business relevance	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Assumptions of regression, BLUE property and least squares estimation	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Variable rationalization and regression model building	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Simple linear regression – theory and interpretation	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Multiple linear regression – model fit statistics and cross-validation	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Regression diagnostics, model conclusions and business insights	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Classification models – logistic regression, decision trees, Naive Bayes, SVM	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Advanced & ensemble methods – neural networks, bagging, boosting, stacking; other models (time series, clustering, recommendation systems); model evaluation using confusion matrix and performance	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26

	metrics									
17	Regression vs segmentation – supervised and unsupervised learning	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Tree-based models – regression trees and classification trees	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Overfitting, pruning techniques and model complexity	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Multiple decision trees and ensemble tree approaches	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Knowledge development through analytics – skill and competence framework	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Training & development strategies and learning methodologies	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Policies, documentation and record keeping for analytics projects	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	History and theory of spatial analysis and archaeological data	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Spatial legacy data and its importance	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Introduction to GIS and spatial data concepts	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Spatial data types and models – vector and raster	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Coordinate systems and map projections	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Geospatial data input and editing – digitization and attribute linking	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Database management for spatial data	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Cartography fundamentals and map design principles	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	GIS software overview – ArcGIS tools and functionalities	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Spatial data visualization, time and space in analysis	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Geographical modeling – settlement geography, environment and historical landscape analysis	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Spatial and attribute queries for geographic reasoning	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Spatial measurements – distance, density and neighborhood analysis	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Spatial transformations – vector overlays and raster calculations	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Spatial autocorrelation and Moran’s I	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Point pattern analysis and spatial processes	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Surface analysis – TIN, slope, aspect and watershed modeling	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Network analysis – routing, allocation and accessibility modeling	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Suitability and multi-criteria decision analysis	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Simulation and spatial process modeling	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Hydrological and environmental modeling applications	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Data-driven spatial analysis, spatial statistics and future trends in spatial analytics	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lesson-Plan 05203189 – Cloud Native Development

Lecture No.	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to cloud computing and cloud deployment models overview	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Public cloud – characteristics, benefits and use cases	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Private cloud – architecture, benefits and challenges	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Community cloud and hybrid cloud models	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Multi-cloud strategy and cloud platforms overview	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Cloud platforms – AWS, Google Cloud, Azure and OpenStack	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Cloud service models overview – IaaS, PaaS and SaaS	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Managed services, serverless computing, IT Ops, SysOps, DevOps	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Low-Ops and No-Ops concepts	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Activity: Planning the case study cloud architecture	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	What is cloud-native computing – principles and goals	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Key cloud-native technologies and ecosystem	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Traditional application development vs cloud-native development	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Agile practices and feature-driven development	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Value-driven development and agile team structures	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Writing effective user stories	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Activity: Analyzing case study requirements using user stories	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Automation and DevOps overview – Infrastructure as Code (IaC), CI/CD	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Application architecture – monolith vs microservices	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	API integration and designing resilient applications	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Containerization concepts – Docker overview and use cases	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Container orchestration – Kubernetes fundamentals	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	DevOps and CI/CD pipelines for cloud-native applications	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Backend programming for cloud – JavaScript, Node.js fundamentals	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	NoSQL databases and backend data modeling	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	Choosing cloud services for public, private and hybrid environments	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Cloud data and storage services	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Serverless computing – PaaS and function-as-a-service concepts	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	624/03/26	26/03/26
29	Application lifecycle management – package and dependency management	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	625/03/26	27/03/26
30	Build tools – Maven, Gradle, Pip, NPM and NuGet	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Source control basics – Git fundamentals and basic commands	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	GitHub and version control in the cloud; activity: using Git with the case study	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Version control workflows and collaboration strategies	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Branching and merging strategies	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Activity: Branching, merging and collaboration using Git	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	608/04/26	10/04/26
36	Introduction to Docker – containers, images and advantages	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Basic Docker commands and container lifecycle	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	614/04/26	16/04/26
38	Dockerfile creation and building Docker images	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Running, stopping and deleting containers and images	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	617/04/26	20/04/26
40	Activity: Containerizing the case study application	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	621/04/26	23/04/26
41	Container registries – push and pull images; activity: managing Docker containers in a registry	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	622/04/26	24/04/26
42	Infrastructure as Code (IaC) – automating infrastructure provisioning	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	624/04/26	27/04/26
43	Observability – monitoring, logging and tracing; security basics, compliance and access management	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Reliability, resilience, fault tolerance and disaster recovery planning	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Advanced trends – AI & ML integration in cloud applications and enhanced observability	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Lab-Plan 05203184 – Power BI for Data Analytics

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	To analyze sales data using Pivot Tables in MS Excel.	19/01/26	20/01/26	25/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	To create an interactive Excel dashboard	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	To implement Goal Seek in MS Excel.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	To implement Scenario Manager and VLOOKUP.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	To build a Power BI dashboard using basic visuals.	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26	19/03/26	20/03/26	23/03/26
6	To explore advanced Power BI visuals.	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26	02/04/26	03/04/26	06/04/26
7	To implement interactive features in Power BI.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
8	To visualize performance and change analysis.	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26
9	To implement DAX calculations in Power BI.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	To create fully interactive Power BI reports.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	To develop a complete data visualization project.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

Lab-Plan 05203186 – Time Series Analysis

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	To understand time series data and its components.	19/01/26	20/01/26	25/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	To decompose a time series into its components.	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	To estimate trend using simple techniques.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	To fit mathematical curves to time series data.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	To study statistical properties of time series.	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26	19/03/26	20/03/26	23/03/26
6	To estimate trend and seasonal components.	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26	02/04/26	03/04/26	06/04/26
7	To compute seasonal indices using different methods.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
8	To compute seasonal indices using link relatives.	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26
9	To analyze the random component of a time series.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	To perform forecasting using smoothing techniques.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	To apply advanced forecasting and financial models.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

Lab Plan – Computer Networks and Network Security (MCA/MSc IT))

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Study different types of network cables (UTP, STP, coaxial, fiber optic) and identify their use cases.	19/01/26	13/01/26	19/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	Prepare a straight-through Ethernet cable using RJ-45 connectors and a clamping (crimping) tool.	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	Prepare a cross-wired Ethernet cable using RJ-45 connectors and test cable connectivity.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	Test straight-through and cross-wired cables using a LAN cable tester and troubleshoot faults.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	Study and demonstrate the working of network devices such as hub, switch, router, bridge, modem, repeater, and access point.	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26	26/01/26
6	Understand IPv4 addressing, classes, subnet mask, private and public IP addresses.	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26	30/01/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
7	Configure static and dynamic IP addresses on a computer using network settings.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26
8	Configuring and Troubleshooting a Switched Network	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26
9	Connecting a Switch	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	Physically connect a switch to end devices, verify port status, and test network connectivity.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

LAB- Plan 05202108 - Software Engineering with Agile

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Analyze a Library Management System and identify its system components.	19/01/26	20/01/26	25/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	Examine a College Examination System and identify applicable Information System types.	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	Apply the Classical SDLC to an Online Admission System.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	Elicit requirements for an Online Voting System using interview technique.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	Elicit requirements for an Online Voting System using questionnaire method.	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26	19/03/26	20/03/26	23/03/26
6	Classify requirements for an Online Examination System.	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26	02/04/26	03/04/26	06/04/26
7	Prepare SRS for a Library Management System. Algorithm to fill a closed shape with color.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
8	Design Context Diagram and Level-0 DFD for Student Information System.	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26
9	Design Level-1 DFD with proper balancing.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	Develop ER Diagram for a Banking System.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	Prepare Agile user stories and a mini Software Project Plan.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

Lab Plan – 05201294 – Artificial Intelligence – I

	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	To design and implement a Tic Tac Toe game using appropriate data structures and game logic.	12/01/26	13/01/26	19/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	To solve the 8-puzzle problem using the BFS algorithm and understand state space search.	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	To apply DFS technique to solve the classic water jug problem and analyze solution paths.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	To place N queens on a chessboard such that no two queens threaten each other using backtracking.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	To solve the Tower of Hanoi problem using recursion and analyze time complexity.	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26	19/03/26	20/03/26	23/03/26
6	To represent family relationships using facts and rules in Prolog and perform logical queries.	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26	02/04/26	03/04/26	06/04/26
7	To implement the N-Queens problem using logical programming and constraint satisfaction in Prolog.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26
8	Support Vector MachAine (SVM) and Expectation Maximization (EM)	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	To study recent advancements and real-world applications of AI in voice recognition systems.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	Unit-wise Review and Practical Record Verification	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	To analyze AI techniques used in face detection and recognition systems in daily life	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

Lab-Plan 05201296 – Cyber Security and Forencics – I

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	To configure a virtual laboratory environment for mobile forensics and penetration testing.	19/01/26	20/01/26	25/01/26	20/01/26	21/01/26	22/01/26	23/01/26	27/01/26	28/01/26
2	To install and configure Genymotion Android Emulator.	29/01/26	30/01/26	02/02/26	03/02/26	04/02/26	05/02/26	06/02/26	09/02/26	10/02/26
3	To install and configure Santoku OS for Android forensics.	11/02/26	12/02/26	13/02/26	16/02/26	17/02/26	18/02/26	19/02/26	20/02/26	23/02/26
4	To install and configure Appie for Android penetration testing.	24/02/26	25/02/26	27/02/26	02/03/26	03/03/26	05/03/26	06/03/26	09/03/26	10/03/26
5	To explore and use various ADB functionalities.	11/03/26	12/03/26	13/03/26	16/03/26	17/03/26	18/03/26	19/03/26	20/03/26	23/03/26
6	To start multiple emulator instances and connect through ADB.	24/03/26	25/03/26	27/03/26	30/03/26	31/03/26	01/04/26	02/04/26	03/04/26	06/04/26
7	To configure intentionally vulnerable Android applications.	07/04/26	08/04/26	09/04/26	10/04/26	13/04/26	15/04/26	16/04/26	17/04/26	20/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
8	To understand Android application architecture.	21/04/26	22/04/26	23/04/26	24/04/26	27/04/26	28/04/26	29/04/26	30/04/26	01/05/26
9	To reverse engineer Android applications.	04/05/26	04/05/26	05/05/26	05/05/26	06/05/26	06/05/26	07/05/26	07/05/26	08/05/26
10	To analyze Dalvik Executable (DEX) files.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26
11	o identify insecure logging vulnerabilities in Android applications.	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26	08/05/26

Lesson-Plan 05203186 – Time Series Analysis

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
1	Introduction to Time Series Analysis and its Importance	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26	21/01/202	620/01/26	19/01/26	22/01/26
2	Definition of Time Series and Real-World Applications	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26	24/01/202	621/01/26	22/01/26	23/01/26
3	Components of Time Series: Trend, Seasonality, Cyclicity, Noise	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26	28/01/202	623/01/26	23/01/26	24/01/26
4	Visualization of Time Series Data	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26	31/01/202	627/01/26	29/01/26	29/01/26
5	Applications of Time Series in Economics, Finance, Weather, Healthcare	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26	02/02/26	28/01/26	30/01/26	30/01/26
6	Time Series Decomposition – Additive and Multiplicative Models	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26	02/04/26	30/01/26	02/02/26	31/01/26
7	Trend Estimation: Free-Hand Curve Method	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26	07/02/26	03/02/26	05/02/26	05/02/26
8	Trend Estimation: Method of Semi-Averages	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26	09/02/26	04/02/26	06/02/26	06/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
9	Trend Estimation: Fitting Linear Trend	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26	11/02/26	06/02/26	09/02/26	07/02/26
10	Fitting Non-Linear Trend Curves (Quadratic & Polynomial)	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26	14/02/202	610/02/26	12/02/26	12/02/26
11	Growth Curves: Concept and Applications	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26	16/02/202	611/02/26	13/02/26	13/02/26
12	Fitting Modified Exponential Curve	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26	18/02/202	613/02/26	16/02/26	14/02/26
13	Fitting Gompertz Curve	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26	21/02/202	617/02/26	19/02/26	19/02/26
14	Fitting Logistic Curve	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26	23/02/202	618/02/26	20/02/26	20/02/26
15	Review of Probability Concepts for Time Series	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26	25/02/202	620/02/26	23/02/26	21/02/26
16	Review of Statistical Measures (Mean, Variance, Correlation)	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26	28/02/202	624/02/26	26/02/26	26/02/26
17	Random Variables and Probability Distributions	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26	03/02/26	25/02/26	27/02/26	27/02/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H	Div I
18	Stationarity: Concept and Types	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26	04/03/26	27/02/26	02/03/26	28/02/26
19	Ergodicity and its Importance in Time Series	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26	07/03/26	03/03/26	05/03/26	05/03/26
20	Practical Examples of Stationary and Non-Stationary Series	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26	09/03/26	04/03/26	06/03/26	06/03/26
21	Moving Average Method for Trend Estimation	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26	11/03/26	06/03/26	09/03/26	07/03/26
22	Fitting Trend Using Moving Average Method	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26	14/03/202	610/03/26	12/03/26	12/03/26
23	Detrending of Time Series	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26	16/03/202	611/03/26	13/03/26	13/03/26
24	Effect of Trend Elimination on Seasonal and Random Components	15/03/26	18/03/202	613/03/26	16/03/26	14/03/26	18/03/202	613/03/26	16/03/26	14/03/26
25	Seasonal Component and Seasonal Indices	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26	21/03/202	617/03/26	19/03/26	19/03/26
26	HDFS command line interface and file operations,	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26	23/03/202	618/03/26	20/03/26	20/03/26
27	Hadoop file system interfaces and APIs,	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26	25/03/202	620/03/26	23/03/26	21/03/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
28	Measurement of Seasonal Indices – Simple Average Method	26/03/26	28/03/202	624/03/26	26/03/26	26/03/26	28/03/202	24/03/26	26/03/26
29	Seasonal Indices – Ratio-to-Trend Method	27/03/26	30/03/202	625/03/26	27/03/26	27/03/26	30/03/202	25/03/26	27/03/26
30	Seasonal Indices – Ratio-to-Moving Average Method and Seasonal Indices – Link Relative Method	28/03/26	04/01/26	27/03/26	30/03/26	28/03/26	04/01/26	27/03/26	30/03/26
31	Random Component and its Characteristics and Variate Difference Method	16/04/26	04/04/26	31/03/26	02/04/26	16/04/26	04/04/26	31/03/26	02/04/26
32	Calculation of Variance of Random Component	17/04/26	06/04/26	01/04/26	03/04/26	17/04/26	06/04/26	01/04/26	03/04/26
33	Random Processes and Linear Processes	18/04/26	08/04/26	03/04/26	06/04/26	18/04/26	08/04/26	03/04/26	06/04/26
34	Autoregressive (AR) Models	Worksheet on Tenses,	11/04/26	07/04/26	09/04/26	23/04/26	11/04/26	07/04/26	09/04/26
35	Moving Average (MA) Models	24/04/26	13/04/202	608/04/26	10/04/26	24/04/26	13/04/202	08/04/26	10/04/26
36	ARMA Models	25/04/26	15/04/202	610/04/26	13/04/26	25/04/26	15/04/26	10/04/26	13/04/26
37	Non-Stationary Models: Trend and Seasonality Modeling	30/04/26	18/04/202	614/04/26	16/04/26	30/04/26	18/04/202	14/04/26	16/04/26
38	ARIMA Model: Identification and Estimation	01/05/26	20/04/202	615/04/26	17/04/26	01/05/26	20/04/26	15/04/26	17/04/26

Lec	Topic	Div A	Div B	Div C	Div D	Div E	Div F	Div G	Div H
39	Seasonal ARIMA (SARIMA) Model	01/05/26	22/04/202	617/04/26	20/04/26	01/05/26	22/04/202	17/04/26	20/04/26
40	Forecasting Concepts and Accuracy Measures	02/05/26	25/04/202	621/04/26	23/04/26	02/05/26	25/04/202	21/04/26	23/04/26
41	Exponential Smoothing Techniques	07/05/26	27/04/202	622/04/26	24/04/26	07/05/26	27/04/202	22/04/26	24/04/26
42	Forecasting Using Exponential Smoothing	07/05/26	29/04/202	624/04/26	27/04/26	07/05/26	29/04/202	24/04/26	27/04/26
43	Short-Term Forecasting Methods	8/05/26	01/05/26	28/04/26	30/04/26	08/05/26	01/05/26	28/04/26	30/04/26
44	Forecasting for Trend and Seasonal Processes	08/05/26	04/04/26	29/04/26	04/05/26	08/05/26	04/04/26	29/04/26	04/05/26
45	Introduction to ARCH and GARCH Models with Financial Applications	08/05/26	06/04/26	05/05/26	07/05/26	08/05/26	06/04/26	05/05/26	07/05/26

Online Certificate Courses

Subject Code	Subject	Link of Online Certificate Program
05010202DS01	Fundamentals of Database Management Systems	https://cursa.app/en/free-course/database-management-systems-ebic
05010202DS03	Structured System Analysis and Design	https://www.careers360.com/university/technical-university-of-munich-munich/software-engineering-essentials-certification-course
05010102SE01	Data Visualization for Business Intelligence	https://www.simplilearn.com/free-data-visualization-course-online-skillup
05010102AM01	Artificial Intelligence and Machine Learning	https://www.simplilearn.com/free-ai-program-skillup
05010102CF01	Cyber Security and Forensics	https://cursa.app/en/free-course/cyber-security-jjg

Council Details and Planned Activity

CLUB Coordinator-ship

Sr. No.	Club Name	Coordinator Details
1	Cloud Computing Club	Prof. U. Arun Mob: 9626985016 Email: arun.udayasuriyan29426@paruluniversity.ac.in
2	Helping Hands Club	Prof. Sohil Parmar Mob: 6352559859 Email: sohil.parmar24381@paruluniversity.ac.in
3	Technical Skills Club	Prof. Megha Trivedi Mob: 9106805356 Email: megha.trivedi24314@paruluniversity.ac.in
		Prof. Pratik Parmar Mob: 9510615601 Email: pratik.parmar36283@paruluniversity.ac.in
4	AI and ML Club	Prof. Shikha Bansal Mob: 7026732615 Email: shikha.bansal27932@paruluniversity.ac.in
		Prof. Praveen Tomar Mob: 79873 31323 Email: praveen.tomar36403@paruluniversity.ac.in
5	Research Club	Ramchandran Mob No: 9384345662 Email Id: ramchandran.p34251@paruluniversity.ac.in
		Dr. Raj Sinha Mob No: 70615 49761 Email Id: raj.sinha40299@paruluniversity.ac.in
6	Big Data Analytics	Prof. Vipul Gamit Mob: 8200012226 Email: vipulbhai.gamit26053@paruluniversity.ac.in
		Prof. Sonali Kharade Mob: 8141613990 Email: sonali.kharade31678@paruluniversity.ac.in
		Jatin Morwal 87643 98766 jatin.morwal40859@paruluniversity.ac.in

Sr. No.	Club Name	Coordinator Details
7	IoT Club	Prof. Chaitali Nayka Mob: 9712009578 Email: chaitalikumari.nayka36591@paruluniversity.ac.in
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8	Full Stack Web Developer	Prof. Rinkal Sarvaiya Mob: 8799025175 Email: rinkal.sarvaiya31672@paruluniversity.ac.in
		Prof. Sathwik Chebrolu Mob: 8008469584 Email: sathwik.chebrolu40366@paruluniversity.ac.in
		Prof. Hardik Parmar Mob: 9586607479 Email: hardik.parmar26611@paruluniversity.ac.in
9	Image Processing and Co	Prof. Mehulkumar Dalwadi Mob: 8000131149 Email: mehulkumar.dalwadi33536@paruluniversity.ac.in
		Prof. Payal Parekh Mob: 8866573415 Email: payal.parekh40362@paruluniversity.ac.in
10	Placement Club	Prof. Vipul Gamit Mob: 8200012226 Email: vipulbhai.gamit26053@paruluniversity.ac.in
		Prof. Mehulkumar Dalwadi Mob: 8000131149 Email: mehulkumar.dalwadi33536@paruluniversity.ac.in
		Prof. Aniket Paul 63538 71530 aniket.paul36074@paruluniversity.ac.in
11	Software Development Club	Prof. Hardik Parmar Mob: 9586607479 Email: hardik.parmar26611@paruluniversity.ac.in
		Rama Deepak Yadagiri Mob No: 9494075286 Email Id: rama.yadagiri35463@paruluniversity.ac.in

Co-curricular and Extra-curricular Events During the Semester

CSR Activity (Monthly Basis):
Each Department must plan **minimum one CSR activity** on a monthly basis.

No.	Name of Activity	Defined Month/Day
1	Social Activity / NSS Camp	27/01/2026
2	National Youth Day	12/01/2026
3	Army Day	15/01/2026
4	National Voters Day	25/01/2026
5	Republic Day	26/01/2026
6	4th International Conference	Jan 2026 (Tentative)
7	Spoken Tutorial Workshop	Feb 1st Week, 2026
8	Birth Anniversary of Subhash Chandra Bose	Feb 2nd Week, 2026
9	International Mother Language Day	21/02/2026
10	National Science Day	28/02/2026
11	Seminar on Career Guidance	03/03/2026
12	Mental Health Awareness	02/03/2026
13	Specialization / Minor - Quiz Competition and Online Free Course Certification	03/03/2026 – 08/03/2026
14	International Women's Day	Mar 1st Week, 2026
15	Hackathon to Improve Technical Skills	Apr 2026
16	Anti-Tobacco Day	31/05/2026
17	Research Seminar	Mar 3rd Week, 2026
18	IPR Session	1st Week of March
19	Smart Tricks to Write Research Paper and Book Chapter	3rd Week of March
20	Library – Self Study Activities	A. Pick and Write Event B. Self-Study
21	Workshop on Research Paper Writing	March, 1st Week 2026

Details of Workshop During the Semester
Prominent academic competition (Outside PU)

S.No	Event Name	Month(s) & Year	Location / Mode	Link (Registration / Official)
1	National Workshop on Artificial Intelligence	15–16 March 2026	IIT Bombay, Maharashtra	https://example.com/ai-workshop
2	Smart India Hackathon Orientation Program	22 March 2026	NIT Surat, Gujarat	https://example.com/sih
3	Web Development Bootcamp using MERN Stack	05 April 2026	Online Mode	https://example.com/mern
4	Cyber Security Awareness and Ethical Hacking Seminar	12 April 2026	Ahmedabad, Gujarat	https://example.com/cyber
5	National Conference on Data Science and Analytics	20–21 April 2026	Bengaluru, Karnataka	https://example.com/datascience
6	Python Programming for Beginners Workshop	25 April 2026	Vadodara, Gujarat	https://example.com/python
7	Cloud Computing and AWS Fundamentals Webinar	02 May 2026	Online Mode	https://example.com/aws
8	National Level Coding Contest	10 May 2026	Pune, Maharashtra	https://example.com/coding
9	Seminar on Career Opportunities in IT Industry	18 May 2026	Indore, Madhya Pradesh	https://example.com/career

Continued on next page

S.No	Event Name	Month(s) & Year	Location / Mode	Link (Registration / Official)
10	Machine Learning Hands-on Training Program	25–26 May 2026	Hyderabad, Telangana	https://example.com/ml

Coordinators of Various Committees

Faculty of IT & Computer Science

Department of MCA (2025–26)

Sr. No.	Portfolio	Coordinator Details
1	CEP (Continuing Education Programme)	Nirmit Shah Mob No: 9898315486 Email Id: nirmitkumar.shah27546@paruluniversity.ac.in
2	Department Newsletter / College Magazine	Uma Bharti Mob No: 7440697996 Email Id: uma.bharti33724@paruluniversity.ac.in Shivani Bhadoria Mob No: 9499726372 Email Id: shivaniba.bhadoriya36466@paruluniversity.ac.in Khyati Kariya Mob No: 9898094186 Email Id: khyati.kariya33400@paruluniversity.ac.in Preeti Sharma Mob No: 8218864370 Email Id: preeti.sharma32496@paruluniversity.ac.in
3	NASSCOM / NSDC / Internshala / Intel Unnati	Vijya Tulsani Mob No: 8866665173 Email Id: vijya.tulsani42087@paruluniversity.ac.in Adarsh Ashok Mob No: 90336441202 Email Id: adarsh.ashok35533@paruluniversity.ac.in Akanksha Singh Mob No: 8273544152 Email Id: akanksha.singh40730@paruluniversity.ac.in Prof. Himanshu Kumar Yadav [1T+2P] Mob: 9534369065 Email: himanshu.yadav40349@paruluniversity.ac.in
4	Internship and Industrial Visit	Mehulkumar Dalwadi Mob No: 8000131149 Email Id: mehulkumar.dalwadi33536@paruluniversity.ac.in

Sr. No.	Portfolio	Coordinator Details
5	Internal / Mid-Term Exam	Manish Kumar Joshi Mob No: 9898530753 Email Id: manishkumar.joshi42083@paruluniversity.ac.in Prachi Mehta Mob No: 7698650313 Email Id: prachibahen.mehta35932@paruluniversity.ac.in Prof. Mini Bhole Mob No: 72030 83488 Email Id: miniben.bhola40198@paruluniversity.ac.in Prof. Anmol Singh Mob No: 63063 85299 Email Id: anmol.singh40298@paruluniversity.ac.in
6	Class Test	Manish Kumar Joshi Mob No: 9898530753 Email Id: manishkumar.joshi42083@paruluniversity.ac.in Prof. Anmol Singh Mob No: 63063 85299 Email Id: anmol.singh40298@paruluniversity.ac.in
7	Unnat Bharat Abhiyan / NSS / SRC	Sohil Parmar Mob No: 6352559859 Email Id: sohil.parmar24381@paruluniversity.ac.in Rama Deepak Yadagiri Mob No: 9494075286 Email Id: rama.yadagiri35463@paruluniversity.ac.in Prof. Priyanka Mod Mob No: 95300 96894 Email Id: priyanka.mod33321@paruluniversity.ac.in
8	Sports	Mayank Parashar Mob No: 8005596759 Email Id: mayank.parashar32541@paruluniversity.ac.in
9	Time Table & Resource Allocation	Jigar Bhawsar Mob No: 9229891004 Email Id: jigar.bhawsar24744@paruluniversity.ac.in Hardik Parmar Mob No: 9586607479 Email Id: hardik.parmar26611@paruluniversity.ac.in

Sr. No.	Portfolio	Coordinator Details
10	E-Content	Sathwik Chebrolu Mob No: 8008469584 Email Id: sathwik.chebrolu40366@paruluniversity.ac.in Shivani Bhadoria Mob No: 9499726372 Email Id: shivaniba.bhadoriya36466@paruluniversity.ac.in
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12	T&P / Impact Training / Placement Club	Vipul Gamit Mob No: 8200012226 Email Id: vipulbhai.gamit26053@paruluniversity.ac.in Mehulkumar Dalwadi Mob No: 8000131149 Email Id: mehulkumar.dalwadi33536@paruluniversity.ac.in Sonali Gupta 76520 38657 sonali.gupta36849@paruluniversity.ac.in Prof. Pratik Parmar Mob: 9510615601 Email: pratik.parmar36283@paruluniversity.ac.in Adarsh Ashok Mob No: 90336441202 Email Id: adarsh.ashok35533@paruluniversity.ac.in Prof. Aniket Paul 63538 71530 aniket.paul36074@paruluniversity.ac.in

Sr. No.	Portfolio	Coordinator Details
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15	Mentoring	Hardik Parmar Mob No: 9586607479 Email Id: hardik.parmar26611@paruluniversity.ac.in
16	NEP	Tanmay Shah Mob No: 9923238304 Email Id: tanmay.shah35142@paruluniversity.ac.in Rinku Patil Mob No: 9423284752 Email Id: rinku.patil31670@paruluniversity.ac.in

Sr. No.	Portfolio	Coordinator Details
17	Institute Ranking, Awards & Achievements	Abhishek Mehta Mob No: 9898735028 Email Id: abhishek.mehta3094@gmail.com Vijya Tulsani Mob No: 8866665173 Email Id: vijya.tulsani42087@paruluniversity.ac.in Dr. Raj Sinha Mob No: 70615 49761 Email Id: raj.sinha40299@paruluniversity.ac.in Khyati Kariya Mob No: 9898094186 Email Id: khyati.kariya33400@paruluniversity.ac.in Prof. Sahil Kumar Mob No: 75577 39975 Email Id: sahil.sah41068@paruluniversity.ac.in
18	NPTEL	Nirmit Shah Mob No: 9898315486 Email Id: nirmitkumar.shah27546@paruluniversity.ac.in Komal Bharti Mob No: 8340259116 Email Id: komal.pandey35785@paruluniversity.ac.in
19	Competitive Exam Activity	Honey Parmar Mob No: 9157176781 Email Id: honey.parmar36498@paruluniversity.ac.in Prof. Sahil Kumar Mob No: 75577 39975 Email Id: sahil.sah41068@paruluniversity.ac.in
20	PU Internship	U. Arun Mob No: 9626985016 Email Id: arun.udayasuriyan29426@paruluniversity.ac.in

Sr. No.	Portfolio	Coordinator Details
21	EDC / Entrepreneur-ship Club	U. Arun Mob No: 9626985016 Email Id: arun.udayasuriyan29426@paruluniversity.ac.in Sathwik Chebrolu Mob No: 8008469584 Email Id: sathwik.chebrolu40366@paruluniversity.ac.in
22	Industry Embedded Program	Faruk Abdulla Mob No: 9810470397 Email Id: faruk.abdulla30274@paruluniversity.ac.in
23	ICT and PULMS	Sathwik Chebrolu Mob No: 8008469584 Email Id: sathwik.chebrolu40366@paruluniversity.ac.in Shivani Bhadoria Mob No: 9499726372 Email Id: shivaniba.bhadoriya36466@paruluniversity.ac.in
24	Orientation Booklet	Payal Parekh Mob No: 8866573415 Email Id: payal.parekh40362@paruluniversity.ac.in Abhishek Mehta Mob No: 9898735028 Email Id: abhishek.mehta3094@gmail.com Hina Chokshi Mob No: 9227873387 Email Id: hina.chokshi42058@paruluniversity.ac.in Sathwik Chebrolu Mob No: 8008469584 Email Id: sathwik.chebrolu40366@paruluniversity.ac.in Aesh Gada Mob No: 9974954579 Email Id: aesh.gada36179@paruluniversity.ac.in Honey Parmar Mob No: 9157176781 Email Id: honey.parmar36498@paruluniversity.ac.in Prof. Himanshu Kumar Yadav Mob: 9534369065 Email: himanshu.yadav40349@paruluniversity.ac.in

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36	IRC / Student Ex-change	Vijya Tulsani Mob No: 8866665173 Email Id: vijya.tulsani42087@paruluniversity.ac.in Prof. Adarsh Ashok Mob No: 90336441202 Email: adarsh.ashok35533@paruluniversity.ac.in
38	Anti–Ragging	Rinkal Sarvaiya Mob No: 8799025175 Email Id: rinkal.sarvaiya31672@paruluniversity.ac.in Rinku Patil Mob No: 9423284752 Email Id: rinku.patil31670@paruluniversity.ac.in
40	PUMIS–BCA	Rinkal Sarvaiya Mob No: 8799025175 Email Id: rinkal.sarvaiya31672@paruluniversity.ac.in Prof. Sonali Kharade Mob: 8141613990 Email: sonali.kharade31678@paruluniversity.ac.in Prof. Chaitali Nayka Mob: 9712009578 Email: chaitalikumari.nayka36591@paruluniversity.ac.in
43	Spoken Tutorial	Shikha Bansal Mob No: 7026732615 Email Id: shikha.bansal27932@paruluniversity.ac.in Prof. Dipti Sharma Mob No: 76101 77528 Email Id: dipti.sharma40862@paruluniversity.ac.in

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46	Low Attendance	Preeti Sharma Mob No: 8218864370 Email Id: preeti.sharma32496@paruluniversity.ac.in Madhav Kapadia Mob No: 93132 30772 Email Id: madhavkumar.kapadia36168@paruluniversity.ac.in Vishakha Bathwar Mob No: 6352410619 Email Id: vishakha.bathwar32493@paruluniversity.ac.in
47	Scholarship	Prachi Mehta Mob No: 7698650313 Email Id: prachibahen.mehta35932@paruluniversity.ac.in
48	External Events	Saikat Paramanik Mob No: 94745 83427 Email Id: saikat.paramanik40297@paruluniversity.ac.in
49	Social Media Platform	Vijya Tulsani Mob No: 8866665173 Email Id: vijya.tulsani42087@paruluniversity.ac.in Prof. Ahikpa Djidji Etienne Mob: 91576 37567 Email: etienneahikpa@gmail.com
51	Event Coordinator (All Semesters)	Khyati Kariya Mob No: 9898094186 Email Id: khyati.kariya33400@paruluniversity.ac.in

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53	Online Certification Course	Mehulkumar Dalwadi Mob No: 8000131149 Email Id: mehulkumar.dalwadi33536@paruluniversity.ac.in
54	Free Value Added Course	Ramchandran Mob No: 9384345662 Email Id: ramchandran.p34251@paruluniversity.ac.in
55	Armed Force Motivation / NCC	Chandan Singh Mob No: 9508724215 Email Id: chandan.singh33017@paruluniversity.ac.in
56	Women Development Cell	Saswati Chatterjee Mob No: 9874410748 Email Id: saswati.chatterjee30729@paruluniversity.ac.in
60	Parent Teachers Meeting	Smarika Rai Mob No: 9023483944 Email Id: smarika.rai30842@paruluniversity.ac.in

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65	Global Certificate	Mehulkumar Dalwadi Mob No: 8000131149 Email Id: mehulkumar.dalwadi33536@paruluniversity.ac.in Madhav Kapadia Mob No: 93132 30772 Email Id: madhavkumar.kapadia36168@paruluniversity.ac.in
67	IPR	Hina Chokshi Mob No: 9227873387 Email Id: hina.chokshi42058@paruluniversity.ac.in
70	FITCS Conference	Ramchandran Mob No: 9384345662 Email Id: ramchandran.p34251@paruluniversity.ac.in
71	Students Health Checkup	Vishakha Bathwar Mob No: 6352410619 Email Id: vishakha.bathwar32493@paruluniversity.ac.in
72	UHV / Ekagrata	Aesh Gada Mob No: 9974954579 Email Id: aesh.gada36179@paruluniversity.ac.in

Anti-Ragging Committee Report

Academic Year 2025 - 2026

Faculty of IT and Computer Applications
Parul Institute of Computer Applications - BCA / BSCIT / IMCA

Note: *Anti-ragging is strictly prohibited in Parul University Campus.*

Sr. No	Name of the Committee Members with Designation	Role in the Committee	Mobile	Email ID
1	Dr. Priya R. Swaminarayan (Head of the Institution)	Dean (Chairperson)	9427857621	priya.swaminarayan@paruluniversity.ac.in
1	Dr. M.K.Nallakaruppan	Principal (Chairperson)	9940741714	nallakaruppan.k42707@paruluniversity.ac.in
2	Dr. Hina Chokshi (Head of Department)	Member	9227873387	hina.chokshi42058@paruluniversity.ac.in
3	Rinkal Sarvaiya (Asst. Professor)	Member	8799025175	rinkal.sarvaiya31672@paruluniversity.ac.in
4	Rinku Patil (Asst. Professor)	Member	9423284752	rinku.patil31670@paruluniversity.ac.in
5	Payal Joshi (Office Assistant)	Member	9737634949	payal.joshi@paruluniversity.ac.in
6	1. Deep Gujarati (2405101270026 - 3rd) 2. Anvi Pravin Bhai Patel (2205101110016 - 3rd)	Member(s) (Students)	1. 8320362337 2. 9328013040	1. 2405101270026@paruluniversity.ac.in 2. 2405101270010@paruluniversity.ac.in

7	1. Krishna Shankar (2305103140018 - 5th) 2. Anjali Savariya (2305101270016 - 5th)	Member(s) (Students)	1. 90264106225 2. 9328356573	1. 2305103140018@paruluniversity.ac.in 2. 2305101270016@paruluniversity.ac.in
8	1. Mavajibhai Gujarati (Deep's Father) 2. Pravinbhai Patel (Anvi's Father)	Member(s) (Parents)	1. 9427263552 2. 9712753199	1. gujaratdeep886@gmail.com 2. anvipatel1219@gmail.com
9	Shri Siddharth Maniyar, Sr. Journalist, Sandesh Newspaper	Member	8000050255	siddharth.maniyar@gmail.com
10	Smt. Swati S. Bedekar, Rep. of NGO, Vatsalya Foundation, 15/Vrindavan Estate, Pashabhai Patel Park, Race Course, Vadodara - 390007	Member	9824058675	swatibedekar@gmail.com
11	Police Inspector, Waghodia Police Station	Member	02668-262233	–
12	Any other member as the concerned HOI may deem fit	Member	–	–

Women Development Cell Report

Academic Year 2025 - 2026

Faculty of IT and Computer Applications
Parul Institute of Computer Applications - BCA / BSCIT / IMCA

Sr. No.	Name of the Committee Members with Designation	Role in the Committee	Mobile	Email ID
1	Dr. Priya Swaminarayan	Head of the Institute	94278 57621	priya.swaminarayan@paruluniversity.ac.in
2	Dr. Hina Chokshi	Head of the Department	92278 73387	hina.chokshi42058@paruluniversity.ac.in
3	Aesh Gada	Coordinator	99749 54579	aesh.gada36179@paruluniversity.ac.in
4	Dr. Saswati Chatterjee	Sub Coordinator	98744 10748	saswati.chatterjee30729@paruluniversity.ac.in

Grievance Redressal Cells (GRC)

Sr. No.	Name & Designation	Contact Details	Role
1.	Dr. Hemant Toshikhane, Dean, Faculty of Ayurved and Principal, Parul Institute of Ayurved	hemant.toshikhane@paruluniversity.ac.in (M) 8469496525	Chairperson
2.	Dr. M. N. Parmar, Dean, Faculty of Social Work and Principal, Parul Institute of Social Work	mnp.msu@gmail.com (M) 9824064291	Member
3.	Dr. Guno Chakraborty, Principal, Parul Institute of Pharmacy and Research	g.chakraborthy19159@paruluniversity.ac.in (M) 8800978930	Member
4.	Dr. Mehul Jadav, Principal, Parul Institute of Physiotherapy and Research	piptr@paruluniversity.ac.in (M) 9825314847	Member
5.	Dr. Rekha M. Parmar, Professor, Parul Institute of Ayurved	rekha.parmar@paruluniversity.ac.in (M) 9724306523	Member
6.	Susri. Khushi Singh, Second Year Student, B.Sc in Microbiology, Parul Institute of Applied Sciences	khushisinghrajput1210@gmail.com (M) 6359826234	Member
7.	Shri. Umrao Singh Rathore, Second Year Student, Parul Institute of Design	heavenartworkk@gmail.com (M) 8290583206	Member
8.	Shri. Dhruvil Shah, Dean, Students' Welfare	dsw@paruluniversity.ac.in (M) 7574809590	Member Convener

Alumni Association Cell

Parul University Alumni Association (PUAA) proudly represents a global family of 80,000+ alumni across 38 constituent institutions, creating a lifelong network that connects graduates with their peers, mentors, and the University. PUAA serves as a bridge between past and present, fostering collaboration, mentorship, and professional growth while celebrating the achievements of our alumni community across the world.

Sr. No.	Name & Designation	Contact Details	Role
1	Prof. Jigar Bhawsar	Mob No: 9229891004 Email: jigar.bhawsar24744@paruluniversity.ac.in	Alumni Co-ordinator
2	Prof. Sohil Parmar	Mob: 6352559859 Email: sohil.parmar24381@paruluniversity.ac.in	Alumni Co-ordinator

Center of Distance of Online Education (CDOE)

Sr. No.	Name & Designation	Contact Details	Role
1	Prof. Nirmit Shah	Mob: 9898315486 Email: nirmitkumar.shah27546@paruluniversity.ac.in	CDOE-Co-ordinator
2	Prof. Tanmay Shah	Mob: 99232 38304 Email ID: tanmay.shah35142@paruluniversity.ac.in	CDOE-Co-ordinator

Office of International Affairs

Sr. No.	Name & Designation	Contact Details	Role
1	Prof. Saswati Chatterjee	Mob No: 9874410748 Email: saswati.chatterjee30729@paruluniversity.ac.in	OIA- Co-ordinator

2	Prof. Shikha Bansal	Mob: 7026732615 Email: shikha.bansal27932@paruluniversity.ac.in	OIA- Co-ordinator
3	Prof. Ahikpa Djidji Etienne	Mob: 915 Email: etienneahikpa@gmail.com	OIA- Co-ordinator

Unfair Means(UFM)

Sr. No.	**Name & Desig-nation**	**Contact Details**	**Role**
1.	Prof. Sweta Jethava	Mob No: 9925045656 Email: sweta.jethava24460@paruluniversity.ac.in	UFM co-ordinator
2.	Prof. Saumil Trivedi	Mob No: 7984149533 Email Id: saumil.trivedi31228@paruluniversity.ac.in	UFM co-ordinator

Discipline Committee Coordinators

Sr. No.	**Coordinator**
1	Prof. Arun U.
2	Prof. RamDeepak Yadagiri
3	Prof. Pravin Singh
4	Prof. Sohil Parmar
5	Prof. Vishakha Bathwar

Departmental Achievement



Figure 1: Departmental Achievement

Departmental Achievement



Figure 2: Departmental Achievement

Internships/PPOs – AY 2025–26 (UG)

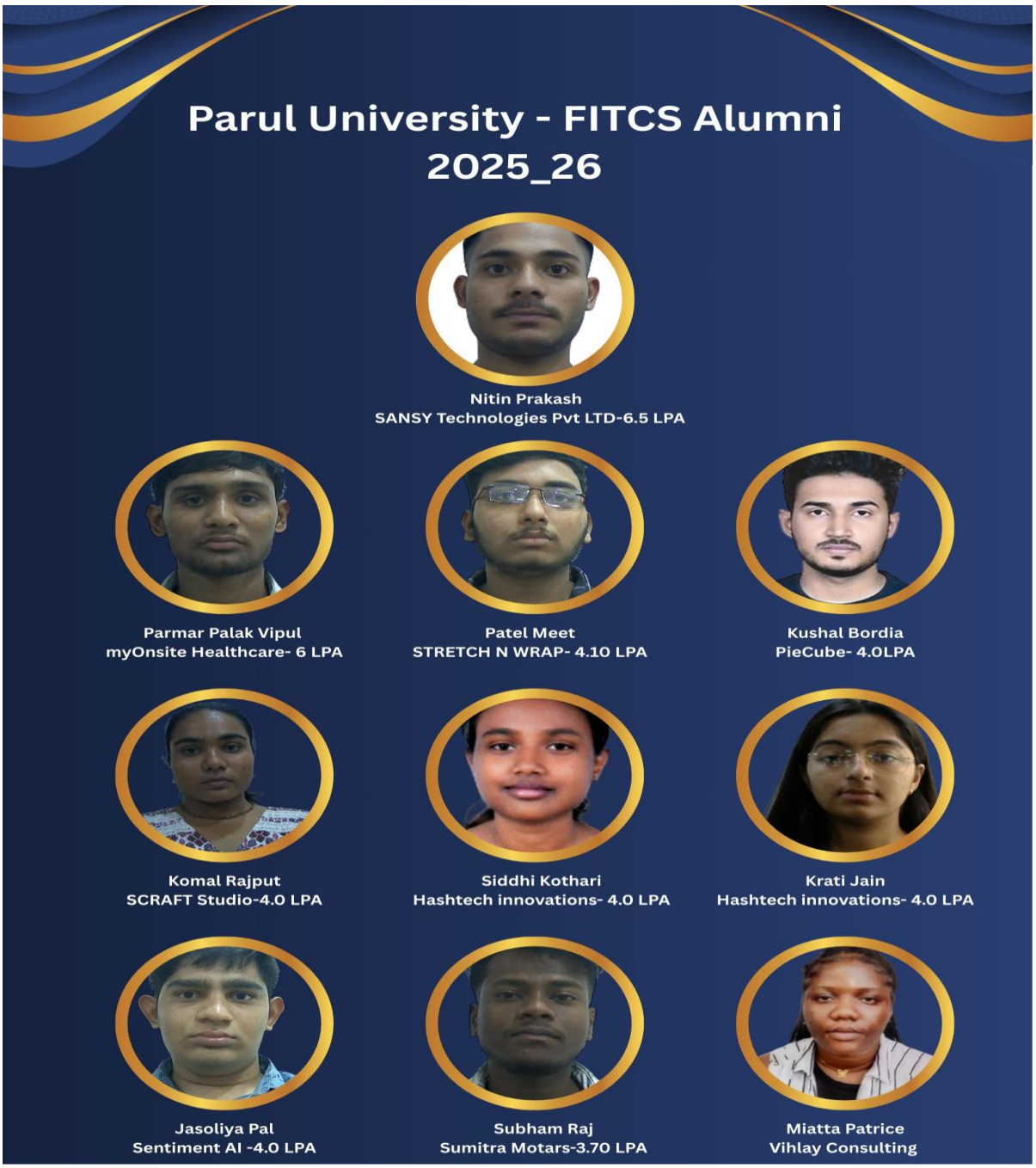


Figure 3: Departmental Achievement

Ranker List of Last Year Result

Top 10 Students by CGPA

Rank	Enrollment No.	Student Name	Institute	Specialization	CGPA
1	2405101010033	KAPADWALA AAFIYA MUSTAK AHEMAD	PICA	BCA	8.98
2	2405101010069	SHWETA THAKUR	PICA	BCA	8.73
3	2405101010041	KSTUTI SHARMA	PICA	BCA	8.36
4	2405101010163	NARAYN PRABHU	PICA	BCA	8.30
5	2405101010146	MAHMOUD JUMA AMEIR	PICA	BCA	8.23
6	2405101010021	DHARM PATEL	PICA	BCA	8.09
7	2405101010009	ANANYA YADAV	PICA	BCA	8.02
8	2405101010092	P S THARUN	PICA	BCA	7.93
9	2405101010077	YUGRAJ MEWARA	PICA	BCA	7.91
10	2405101010122	GEMBALI SAI BALAJI	PICA	BCA	7.86
10	2405101010162	METO SELDON	PICA	BCA	7.86

Interaction of Various Media Platforms



FITCS Instagram Page Link

https://www.instagram.com/fitcs_pu?igsh=bmQ2NHR1aGk5eTho

FITCS Facebook Page Link

https://www.facebook.com/pufoitcs?mibextid=LQQJ4d

FITCS WhatsApp Page Link

https://chat.whatsapp.com/GGG3sjgY0vy46oQjUFMort?mode=ems*copy*t

PU PICA Official Communication Channel

https://whatsapp.com/channel/0029Vb6UI1j7YScuKTJvcz1n

PU Instagram Page Link

https://www.instagram.com/paruluniversity/?hl=en

PU Facebook Page Link

https://www.facebook.com/ParulUniversity/

ANNEXURE

PARUL UNIVERSITY

R/Notification-1386/2024-25

Office of the Registrar
December 9, 2024

NOTIFICATION

Sub: Fees to be charged for different documents in the University
Ref: (i) R/Notification-155/2017-18 dated September 5, 2017
(ii) Proceedings of the Eighteenth Meeting of the Governing Body held on December 6, 2024
(ii) Orders of the President

In partial modification of the notification cited at ref.(i), one time fees at the rates shown below shall be paid by the students to obtain different types of credentials from the University. The revision shall come into effect from January 1, 2025.

Certificate/Document	*Fees*	*Mode of Payment*
Transcript	(i)Rs.2,000/- (First Set-4 copy) (ii)Additional Rs.2,000/- (multiple thereof)	Online Payment through Parul University Website
Transcript for MBBS	2000/- per copy	Online Payment through Parul University Website
Migration Certificate	Rs.1000/-	Online Payment through Parul University Website
N.O.C. Admission Cancellation	Rs.1000/-	Online Payment through Parul University Website
Statement of Mark	Rs.2000/-	Online Payment through Parul University Website
Document Verification of • Grade Card/Marks Card • Degree Certificate • Provisional Degree Certificate • Etc…	Rs.1000/- per document	Online Payment through Parul University Website Necessary documents will have to be produced by the student.
Correction of Student's Name/ ABC ID in Grade/Marks Card (in case the wrong entry of name is made by the student in MIS)	Rs.1000/-	Online Payment through Parul University Website. Necessary documents will have to be produced by the student .
Rank Certificate (University Exams)	Rs.500/-	Online Payment through Parul University Website
Language Certificate	Rs.500/-	Online Payment through Parul University Website
Backlog Certificate	Rs.500/-	Online Payment through Parul University Website

Attempt Certificate	Rs.500/-	Online Payment through Parul University Website
CGPA to Percentage Certificate	Rs.500/-	Online Payment through Parul University Website
Duplicate Grade Card/Mark Sheet	2000/-	Online Payment through Parul University Website Submit an original copy of challan for having registered a complaint on loss of original document/s with the police.
Duplicate Degree Certificate	5000/-	Online Payment through Parul University Website Submit an original copy of challan for having registered a complaint on loss of original document/s with the police.
Other Certificate (Please specify details of certificate required)	Rs.500/-	Online Payment through Parul University Website Necessary documents will have to be produced by the student.
Courier Charges	Rs.300/- for domestic, Rs.4000/- for International	Online Payment through Parul University Website

By Order

Registrar

To,

1) Deans of Faculties
2) Principals/ Directors of Colleges/ Institutes
3) Dean, Doctoral Studies and Research
4) Campus Director
5) Managing Director (Global), Industrial Collaborations; Academic Strategies
6) Academic Directors
7) Dean, Students' Welfare
8) Controller of Examinations
9) Chief Technology Officer
10) Chief Librarian

EXAMINATION RULES & REGULATIONS FOR STUDENTS:

If any examinee is violating university rules, he/she is liable to be punished under the provisions of the University rules.

- A candidate has to check the bar-code to be pasted by the supervisor on the front page of his answer book.
- Enter your enrolment number, subject code, date of exam etc. in the given space.
- Students should ensure that any objectionable material leading to a UFM case is not lying around his/her seat prior to the start of the examination, and it will be the sole responsibility of the student to inform the supervisor.

- A disciplinary action will be taken against the candidate who disobeys the instructions of the supervisor or misbehaves or violates the code of conduct of examination of Parul University.
- All unused answer books and other University examination materials must be handed over to Supervisor and must **NOT** be taken from the room.
- Hollow sticker should be pasted after the warning bell on the appropriate space on the front page of the answer book, covering the information filled up by the candidate and the bar-code label.
- There will be a warning bell 10 minutes before the completion of the examination.
- If you need to do any rough work, use the back of the page(s) of your answer book(s) and either cross out the rough work or mark it clearly as such before submitting your script; you may not use paper of your own for rough work.
- No candidate will be allowed to leave the examination hall before 45 minutes after the commencement and during the last 15 minutes of the examination session.
- Do not leave your seat under any circumstances without the prior permission of the supervisor.
- Do not write anything in the space provided for marks.
- Write relevant answers of the questions in clear and legible handwriting on both sides of the answer book in respective Section-A and Section-B.
- Begin a new answer on a separate page.
- Please read the instructions at the top of your paper, and make sure you understand and follow them.
- Under no circumstances will the candidate be allowed to enter the examination hall after commencement of the examination.
- Please check that you have the correct examination paper and that your copy is complete.
- Do not write your name or number or sign in the answer book which reveals your identity.
- A candidate has to check the bar-code to be pasted by the supervisor on the front page of his answer book.
- Enter your enrolment number, subject code, date of exam, etc. in the given space.
- Students should ensure that any objectionable material leading to a UFM case is not lying around his/her seat prior to the start of the examination, and it will be the sole responsibility of the student to inform the supervisor.

PARUL UNIVERSITY
WAGHODIA, VADODARA

No: PU/EXAM/Functions/2015/5 **Date: 18/08/2015**

Instructions to Examinees/Candidates

1. All candidates must occupy their seats in the exam hall before half an hour of the commencement of the examination time.
2. Students should ensure that any objectionable material leading to a UFM case is not lying around his/her seat prior to the start of the examination, and it will be the sole responsibility of the student to inform the supervisor.
3. Enter your enrollment number, subject code, date of exam, etc. at the appropriate location.
4. A candidate has to check the barcode to be pasted by the supervisor on the front page of his answer book.
5. Do not write your name or number or any sign in the answer book which reveals your identity.
6. Please check that you have the correct examination paper and that your copy of it is complete.
7. Under no circumstances will the candidate be allowed to enter the examination hall after commencement of the examination.
8. Please read the instructions at the head of your paper and make sure you understand and follow them.
9. Begin a new answer on a separate page.
10. Write relevant answers to the questions in clear and legible handwriting on both sides of the answer book in respective Section-A and Section-B.
11. Do not write anything in the space provided for marks.
12. Do not leave your seat under any circumstance without the prior permission of the supervisor.
13. No candidate will be allowed to leave the examination hall after the commencement of the examination till 45 minutes and during the last 15 minutes of the examination session.
14. Students may not leave the examination room (except when they have finished their examination). If the medical condition appears serious and the student is not able to resume the exam, the Jr. Supervisor should call Security and request medical assistance. In this case, the student may be able to resume the examination.

15. If you need to do any rough work, use the back page(s) of your answer book(s) and either cross through the rough work or identify it clearly as such before handing in your script; you may not use paper of your own for rough work.
16. It is the students' responsibility to make sure their calculators are working and have fresh batteries. Students are neither permitted to share calculators nor to pass them between each other during an examination. Use of a non-permissible type of calculator or other electronic device will be regarded as cheating.
17. If any objectionable material related to the exam or mobile phone or programmable calculator is found with a candidate, or if found exchanging answer book/question paper with another candidate, he/she will be immediately expelled and final punishment will be imposed once the matter is taken up by the authorities.
18. There will be a warning bell 10 minutes before the completion of the examination.
19. Hollow sticker should be pasted after the warning bell at the appropriate place on the front page of the answer book, covering the information filled up by the candidate and the barcode label.
20. All unused answer books and other University examination materials must be handed over to the Supervisor and must **NOT** be taken from the room.
21. Disciplinary action will be taken against the candidate who disobeys the instructions of the supervisor or misbehaves or violates the code of conduct of examination of Parul University.
22. If any examinee is violating university rules, he/she is liable to be punished under the provisions of the University rules.

- Controller of Examination

Centre for Distance and Online Education (CDOE)

Backdrop

The Centre for Distance & Online Education (CDOE) at Parul University is committed to democratizing access to quality higher education by leveraging flexible, technology-enabled learning modes. Our mission is to empower learners—especially working professionals, remote learners, and lifelong learners—with UGC-approved, digitally delivered programmes that maintain the highest academic standards.

Key Features & Strengths

- **UGC-Recognized & Regulated:** The programs offered under CDOE comply with UGC (OL & ODL) regulations, ensuring equivalence with regular-mode degrees.
- **NAAC A++ Accredited University:** Parul University's institutional reputation adds strength and credibility to its distance and online program offerings.
- **Diverse Program Portfolio:** CDOE offers a wide range of undergraduate and postgraduate online and distance programs—spanning Arts, Commerce, Science, Management, Computer Applications, and more.
- **Flexible Learning Mode:** Examinations for online learners are conducted in online mode to suit remote learners.
- **Scholarship & Dual Degree Options:** Offered programs include scholarship options and dual degree opportunities to maximize value for students.
- **Global & Virtual Exposure:** Through virtual classrooms and global exchange programs, learners gain exposure to international perspectives and best practices.

Why Choose CDOE, Parul?

- **Accessibility & Convenience:** Learn from anywhere, anytime, without needing to relocate to campus.
- **Quality & Credibility:** Degrees awarded are recognized and equivalent to traditional formats under UGC norms.
- **Industry-Relevant Curriculum:** Programs are designed to meet evolving industry needs, with input from domain experts and faculty.
- **Learner Support & Resources:** Digital libraries, faculty mentors, technical support, and online study materials ensure smooth learning experiences.
- **Career Advancement & Upskilling:** Ideal for professionals aiming to progress in their careers without interrupting work commitments.

Program Offerings List

S.No	Type	Programme Name	Programme Code	Duration	Eligibility	Lump Sum Fee	Annual Fee	Semester Fee
1	OL	Bachelor of Arts	B.A.	3 Years	10+2 Exam or equivalent	70,000	30,000/yr	18,500/sem
2	OL	Bachelor of Business Administration	B.B.A.	3 Years	10+2 Exam or equivalent	70,000	30,000/yr	18,500/sem
3	OL	Bachelor of Computer Application	B.C.A.	3 Years	10+2 Exam or equivalent	70,000	30,000/yr	18,500/sem
4	OL	Master of Business Administration	M.B.A.	2 Years	Graduate min 50% (45% SC/ST/OBC)	90,000	55,000/yr	37,500/sem
5	OL	Master of Computer Application	M.C.A.	2 Years	Graduate min 50% (45% SC/ST/OBC)	80,000	50,000/yr	30,000/sem
6	OL	Master of Commerce	M.Com	2 Years	Graduate min 50% (45% SC/ST/OBC)	40,000	50,000/yr	15,000/sem
7	OL	M.A. Journalism & Mass Communication	M.A.-JMC	2 Years	Graduate min 50% (45% SC/ST/OBC)	40,000	50,000/yr	15,000/sem
8	OL	M.A. English Language Teaching	M.A.-ELT	2 Years	Graduate min 50% (45% SC/ST/OBC)	40,000	50,000/yr	15,000/sem
9	OL	Master of Social Work	M.S.W	2 Years	Graduate min 50% (45% SC/ST/OBC)	40,000	50,000/yr	15,000/sem
10	OL	M.Sc Applied Mathematics	M.Sc Applied Math	2 Years	Graduate min 50% with Maths	40,000	25,000/yr	15,000/sem

S.No	Type	Programme Name	Programme Code	Duration	Eligibility	Lump Sum Fee	Annual Fee	Semester Fee
11	ODL	B.A. Economics (Hons)	B.A Economics (Hons)	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
12	ODL	B.A. English (Hons)	B.A English (Hons)	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
13	ODL	B.A. Political Science (Hons)	B.A Political Science	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
14	ODL	B.A. Sociology (Hons)	B.A Sociology (Hons)	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
15	ODL	B.Sc Mathematics (Hons)	B.Sc Mathematics	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
16	ODL	BSW (Hons)	B.S.W (Hons)	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
17	ODL	BBA (Hons)	B.B.A (Hons)	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
18	ODL	BCA (Hons)	B.C.A (Hons)	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
19	ODL	B.Com (Hons)	B.Com (Hons)	4 Years	10+2 or equivalent	85,000	30,000/yr	18,500/sem
20	ODL	MBA	M.B.A	2 Years	Graduate min 50% (45% SC/ST)	90,000	55,000/yr	37,500/sem
21	ODL	MCA	M.C.A	2 Years	Graduate min 50% (45% SC/ST/OBC)	80,000	50,000/yr	30,000/sem
22	ODL	M.Com	M.Com	2 Years	Graduate min 50%	40,000	25,000/yr	15,000/sem

S.No	Type	Programme Name	Programme Code	Duration	Eligibility	Lump Sum Fee	Annual Fee	Semester Fee
23	ODL	M.A. ELT	M.A. ELT	2 Years	Graduate min 50%	40,000	25,000/yr	15,000/sem
24	ODL	M.A Economics	M.A Economics	2 Years	Graduate min 50%	40,000	25,000/yr	15,000/sem
25	ODL	M.A English Literature	M.A English Lit	2 Years	Graduate min 50%	40,000	25,000/yr	15,000/sem
26	ODL	M.A Geography	M.A Geography	2 Years	Graduate min 50%	40,000	25,000/yr	15,000/sem

Parul University Alumni Association

The Parul University Alumni Association (PUAA) proudly represents a global family of over 80,000 alumni across 38 constituent institutions, creating a lifelong network that connects graduates with their peers, mentors, and the University. PUAA serves as a bridge between the past and the present, fostering collaboration, mentorship, and professional growth while celebrating the achievements of our alumni community across the world.

Through its various initiatives, PUAA promotes knowledge sharing, career development, entrepreneurship, social impact, and global engagement, empowering every alumnus to contribute towards building a more innovative and inclusive future.

Our Chapters: Strengthening Global Bonds

PUAA has established 08 National and 25 International Alumni Chapters that act as local hubs of networking, mentorship, and collaboration, with many more upcoming chapters set to expand and enrich our global alumni network.

National Chapters:

- Vadodara Chapter,Ahmedabad Chapter, Rajkot Chapter, Surat Chapter, Udaipur Chapter, Navsari Chapter, Hyderabad Chapter, Pune Chapter, Mumbai Chapter, Dahod Chapter

International Chapters:

- New Jersey Chapter – USA, Texas Chapter – USA, London Chapter – UK, Berlin Chapter – Germany , Paris Chapter – Franc, Bhutan Chapter, Nepal Chapter, Zimbabwe Chapter, Uganda Chapter, Kenya Chapter, Zambia Chapter, Lithuania Chapter, Dubai Chapter – UAE, Mozambique Chapter, South Sudan Chapter, Madagascar Chapter, Gambia Chapter, Guinea Chapter, Nigeria Chapter, Niger Chapter, Tanzania Chapter, Ethiopia Chapter, Cameroon Chapter, Ghana Chapter, Ivory Coast Chapter

Each chapter promotes cultural exchange, professional opportunities, and university pride, reinforcing Parul University's global presence.

Student–Alumni Engagement Initiatives

PUAA actively bridges the gap between students and alumni through various programs focused on learning, growth, and career development.

- **Alumni Talk Series:** Alumni from diverse industries regularly visit the campus or connect virtually to share insights, experiences, and professional guidance with current students.
- **ASMP – Alumni Student Mentorship Program (Industry Connect):** A structured initiative pairing students with alumni mentors for personalized career guidance, higher

education support, and skill development.

- **HR Connect:** Alumni working in HR, recruitment, and corporate sectors collaborate with the University to provide placement opportunities and industry exposure.
- **Alumni Webinars & Industry Panels:** Interactive sessions where global alumni discuss emerging trends, innovation, and career paths.
- **Entrepreneur Connect:** Alumni entrepreneurs inspire students to explore startup ideas, innovation, and business leadership.

